

The Far Field of Gila: A Cell-Size Expansion for Every Separated Pair

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Abstract

Every entry of Gila's discretized vacuum Green operator at a separated relative offset is today a signed sum of thirty-six four-dimensional Gauss-Legendre quadratures of the Helmholtz kernel over pairs of cell faces. It is the bottleneck of the build — a 128^3 self-volume at $\lambda/32$ costs 1.94×10^{10} kernel evaluations and 1058 s of single-threaded fill — and it is not accurate: the assembly amplifies the per-pair error by 36 to 1300 on a cubic cell and by up to 1.67×10^5 on the slender cell Gila actually uses, and along that cell's short axis the fixed rule is wrong by a factor of order one. This document replaces the quadrature. The thirty-six face pairs are shown to be one integral of a smooth function against a non-negative product weight over the difference box of the pair; that integral is expanded by the spherical addition theorem into a frequency-independent geometry table, exact in the edge lengths, and a per-offset sum of elementary functions. Two proved theorems bound the truncation of exactly the sum the code performs; they choose the multipole order per offset, and where they cannot, the eight octants of the difference box are used instead, and where neither converges — one band of sixty-seven offsets on the 16:16:1 slender cell — the moment series of the companion document is. The result, `notes/farfield/farfield.jl`, agrees with a 220-bit reference to 8.6×10^{-15} in max-norm and 9.3×10^{-15} per entry over 427 tensors at four cell shapes, five frequencies and separations from two to 128 cells, losing at most 1.59 decimal digits to `Float64` rounding; over sixty random cases spanning aspect ratios 1 to 1020 the truncation is certified by the bound with no violation and the rounding error is at most 16.4ε times the largest tensor entry. The far field of a 128^3 self-volume at $\lambda/32$ is built in 0.200 s on twelve threads against 200.76 s, a factor of 1002, and the whole `GlaVacOprMem` build falls from 206.54 s to 6.73 s. What remains of that build is the contact and touching-shell integrals, whose order-9 rule is shown here to be wrong by 117% on the touching shell of a slender cell and to be the sole cause of the indefiniteness of that operator's anti-Hermitian part; replacing it by the exact moment series makes the operator positive definite.

A final section extends the method to the unequal cells of Gila's composite volumes, where a coarse region meets a fine one. The per-axis weight becomes a trapezoid, which is the difference of two triangles, so the geometry table, the parity argument and the truncation theorem carry over with new moments and nothing else (132 exact-rational identities, all zero); the sub-box theorem needs no change at all. The twenty-seven-piece split the task statement proposed for the near band was built, measured, and dropped: it does not converge where it is needed once the scale ratio reaches 4. The near band is served instead by the gcd average of equal-cell tensors written as a box sum over the fine block the same library already fills, whose cancellation was measured at 1.0–2.3. Over 438 220-bit references at fourteen unequal pairs, three frequencies and separations from one to thirty-two fine cells the worst error is 3.4×10^{-15} in max-norm and 4.7×10^{-15} per entry with no certificate violation; the 1.4×10^6 -offset block of a $\lambda/2$ coarse region against a $\lambda/32$ fine one is filled in 5.7 s on twelve threads, against about 1.8 h for Gila's two separated-pair rules, whose adaptive class is accurate to 10^{-6} and whose fixed rule is wrong by 1.8×10^{-4} on a rod approached from the side.

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1 How to use it

1.1 The files

Everything is under `notes/farfield/`.

<code>farfield.jl</code>	the library, 2009 lines, generic in the number type; lines 1537–2009 are the unequal-cell routes of Section 11
<code>verify.jl</code>	the nine verification parts (a)–(i); every table of Sections 2–9
<code>bench.jl</code>	the before/after build comparison of Section 10
<code>ref.jl</code>	the 220-bit face-pair reference (<code>refPairs</code> , <code>refSum</code> , <code>refTensor</code> , <code>volTensor</code>)
<code>refquad.jl</code>	the graded 220-bit volume reference, sharing no code with <code>ref.jl</code>
<code>mkrefcache.jl</code>	rebuilds the reference caches from the round-1 caches
<code>crosscheck/famG.jl</code>	the independent Cartesian Taylor route, used by part (f) only
<code>shapetab/</code>	the geometry tables, one per cell shape, 16–20 MB each (Section 1.3); a pair table of unequal cells is 0.4–2.5 MB and sits next to them under the pair's name
<code>refcache/</code>	1068 records of 220-bit reference values, append-only text
<code>ksrccache/</code>	the route-(iii) tensors, keyed on (shape, frequency), append-only text
<code>tables/</code>	every table this document quotes, as <code>verify.jl</code> and <code>bench.jl</code> print it
<code>tex/</code>	this document's section files and the generated table bodies

`farfield.jl` is standalone for routes (i) and (ii). Route (iii) — the k -series fallback of Section 8.6 — lazily includes `notes/moments/moments.jl` and errors with a clear message if it is absent; a build whose `ksrccache/` is populated never touches it, and the three cubic shapes never reach it at all.

1.2 The API

<code>farTensor(D, s, f; tol, fs, dir)</code>	the 3×3 tensor at one offset $D::NTuple\{3,Int\}$ of a cell of edges $s::NTuple\{3,<:Rational\}$ at frequency $f::Complex\{T\}$; errors when $\max D \leq 1$
<code>farBlock!(egoToe, s, f; tol, fs, dir)</code>	fills <code>egoToe[:, :, i1, i2, i3]</code> for every index with $\max(i) \geq 3$ (offset $i.-1$), leaving the eight contact indices untouched; threaded, bitwise identical at 1, 4 and 12 threads
<code>isFarInd(i)</code>	the predicate that partition uses; its complement is exactly Gila's <code>egoFunSng!</code> set
<code>farShape(s; dir)</code>	builds or loads the exact geometry table for one shape
<code>farSetup(s, f; tol, scl, dir)</code>	contracts it against one frequency and builds the threshold table; returns the <code>FrqSet</code> to pass as <code>fs</code>
<code>farRoute(fs, D; cst)</code>	the route, the order L and the eight sub-box orders for one offset
<code>farRouteStat(fs, dim)</code>	counts, the whole-box L histogram, and the (ii) and (iii) offset lists of a whole block
<code>farSetupX(sT, sS, f; tol, nBlk, offs, dir)</code>	the set of one <i>unequal</i> pair (Section 11): the trapezoid table keyed on the unordered pair, the thresholds of Theorem 11.2, the n -cut certified at the nearest whole-box offset of <code>offs</code> ; errors unless the edges are integer multiples per axis (Gila's <code>GlaExtInf</code> rule)
<code>farRouteX(fx, R)</code>	(<code>kind</code> , L) for one exact-rational offset R of the target centre, the source at the origin: 1 whole trapezoid box, 2 gcd box sum, 3 k -series on the thirty-six unequal panels; errors when the cells touch or overlap
<code>farTensorX(R, sT, sS, f; tol, fs, cert, dir)</code>	the 3×3 tensor of the pair at one offset; <code>cert = true</code> also returns the certified bound on $\max_{ab} T_{ab} - G_{ab} $
<code>farBlockX!(G, Rs, sT, sS, f; tol, fs, cert, dir, tms)</code>	fills <code>G[:, :, q]</code> for every offset <code>Rs[q]::NTuple\{3,Rational\{BigInt\}\}</code> , threaded, the near band as one box sum over a fine equal-cell block filled by <code>farBlock!</code> ; returns the route kinds (and the certificates); <code>tms</code> receives the phase times

The library's own bounds and special functions — `est`, `jbnd`, `hbnd`, `jt看il`, `bndTrm`, `bndCum`, `boundL`, `boundLoct`, `costWhl`, `costOct`, `tnsWhl!`, `tnsOct!`, `tnsKsr`, `srfsSum` — are all reachable, and `verify.jl` uses every one of them. The defaults are `TOL = 1e-14`, `LMAX = 56`, `NMAX = 12` (a floor, not a cap; see Remark 5.6), `NCAP = 40`, `TABPRC = 192`, `BLDPRC = 256`, `KSRPRC = 128`. The two cache directories are `Refs`, `TABDIR[]` and `KSRDIR[]`, and every entry point also takes a `dir` keyword, which is the safer way to point at a table directory: a missing table is not an error, it is a silent rebuild.

The intended use inside a build is one `farSetup` per (shape, frequency) followed by one `farBlock!`:

```
include("notes/farfield/farfield.jl")
s = (1//32, 1//32, 1//32); f = 1.0 + 0.0im
fs = farSetup(map(Rational{BigInt}, s), f; nBlk = 128) # once per (shape,f)
farBlock!(egoToe, s, f; fs = fs) # every far offset
```

and, for the external block between a fine and a coarse region, one `farSetupX` per (pair, frequency) followed by one `farBlockX!` over the block's non-touching offsets (`Rs` as exact rationals, target centre minus source centre):

```

sT = (1//32, 1//32, 1//32)           # the fine target cell
sS = (1//2, 1//2, 1//2)             # the coarse source cell
Q = Rational{BigInt}
fx = farSetupX(map(Q, sT), map(Q, sS), f; offs = Rs) # once per (pair,f)
farBlockX!(G, Rs, sT, sS, f; fs = fx) # G is 3 x 3 x length(Rs)

```

1.3 What it costs the first time

The exact geometry table is built once per cell shape and is frequency independent; every frequency after the first costs a contraction of tens of milliseconds. The table is 16–20 MB on disk (the four shipped in `shapetab/` are 19.5 MB each) and reloads in one to two seconds. Cold, single-threaded, one shape per process, sized for a 32^3 block:

shape	L_{whl}	L_{oct}	n_{max}	build + setup	peak RSS
$(1/32)^3$	48	51	12	14.3 s	0.66 GB
$(1/8)^3$	50	51	12	14.8 s	0.67 GB
$(1/4)^3$	52	52	12	15.7 s	0.70 GB
$(1/32, 1/32, 1/512)$	56	55	12	18.3 s	0.74 GB

The build allocates nothing per term. Its three inner loops — the integer solid-harmonic recurrence (17), the multiplication by $|\delta|^2$ and the moment accumulator — run in preallocated GMP and MPFR registers through the in-place `Base.GMP.MPZ` and `mpfr_*` calls, one distinct `BigInt` per polynomial coefficient. The allocating first version of the same loops peaked at 21 to 25 GB of resident memory and took 44.5 to 64.7 s per shape: 2.4 to 5.8 MB of `BigInt` garbage per (l, m) column, 15 GB over the 2601 columns of an $L = 50$ octant table, which neither a forced collection every sixteen columns nor a heap-size hint returned to the system. The in-place rewrite was checked bit-identical against it on 24 whole-box and octant tables (three shapes, $n_{\text{max}} \in \{6, 12\}$, $L \in \{16, 20\}$, segments from $l = 0$ and from $l > 0$), and the cold cache files above are byte-for-byte the sizes the allocating version wrote. The peak of 0.66 to 0.74 GB is now the Julia process itself.

A table already on disk removes even this: setup is then 0.7 to 1.3 s, and the per-offset cost of the fill is 665 to 1470 ns single-threaded (Section 10). The one remaining cold cost is the k -series fallback, which needs `notes/moments/moments.jl`: on the slender needle its 67 offsets take 128 s on eight threads from cold (633 s serially), against 0.14 s for the other 524 213, and are then cached to `ksrccache/` per (shape, frequency).

The table of an unequal pair costs what the equal-cell whole box costs at the same L : 3.3–5.7 s cold per pair for $L = 40$ –56 against 4.0 s for the equal-cell whole box to $L = 56$ under the same load, 0.2–0.3 GB over the loaded process, and 0.4–2.5 MB on disk, because a pair has no octant table (Section 11.6). The near band of such a block also needs the equal-cell set of the gcd cell, which is the shipped table when the fine cell is one of the four shapes.

2 Groundwork: the measurements that came first

Nothing in this document was designed before the following was measured. Each subsection names what was unknown, the script that measured it and the table file that holds the numbers; every table lives in `notes/farfield/tables/`.

2.1 Profile of the build being replaced

`scratch/work/root/bench.before.jl`, output `bench.before_1.txt` and `bench.before_12.txt`, on a quiet machine (Apple M3 Pro, 12 cores, 36 GB, Julia 1.12.6; load average 2.8–3.5 at the start).

The far fill is attributed band by band by timing `egoFunInn!` at one representative offset of each `quadOrd` order and multiplying by that band’s population; the kernel-evaluation count is 36 ord^4 per offset.

build	contact	far fill	full build	kernel evals	ns/eval
32^3 at $\lambda/32$, 1 thread	26.90 s	20.04 s	23.82 s	3.827×10^8	52.4
64^3 at $\lambda/32$, 1 thread	26.89 s	133.01 s	157.91 s	2.497×10^9	53.3
128^3 at $\lambda/32$, 1 thread	27.48 s	1058.37 s	1311.87 s	1.941×10^{10}	54.5
32^3 at $\lambda/8$, 1 thread	27.02 s	39.79 s	45.69 s	7.527×10^8	52.9
32^3 at $\lambda/4$, 1 thread	27.05 s	80.16 s	91.92 s	1.536×10^9	52.2
32^3 at $\lambda/32$, 12 threads	3.58 s	—	4.42 s		
64^3 at $\lambda/32$, 12 threads	3.52 s	—	29.81 s		
128^3 at $\lambda/32$, 12 threads	3.78 s	—	262.99 s		
32^3 at $\lambda/8$, 12 threads	3.54 s	—	10.42 s		
32^3 at $\lambda/4$, 12 threads	3.54 s	—	20.99 s		

The band attribution at 128^3 , $\lambda/32$:

quadOrd	offsets	$\mu\text{s}/\text{offset}$	band s	kernel evals
4	2 092 239	502.7	1051.8	1.928×10^{10}
5	4570	1174.9	5.37	1.028×10^8
6	218	2444.0	0.533	1.017×10^7
7	98	4595.7	0.450	8.471×10^6
9	19	12422.0	0.236	4.488×10^6

Three readings, and one correction to the task statement.

- **The estimate of 10^{11} kernel evaluations is too high by a factor of five.** The measured count at 128^3 , $\lambda/32$ is 1.941×10^{10} . The reason is item 2.2: Gila evaluates the N^3 offsets of the Toeplitz block, not the $(2N)^3$ of the doubled circulant, and 99.75 % of them fall in the cheapest band, order 4, at $4^4 = 256$ nodes per face pair.
- **The far fill is 81 % of the single-threaded build at 128^3 and 99.4 % of it is the order-4 band,** i.e. the cheapest rule applied two million times, not the expensive rules applied a few times. The whole order-5-to-9 tail is 6.6 s of 1058 s. A cheaper near band would buy nothing; only a cheaper far band matters.
- **The contact integrals are a fixed 27 s and are memoized,** so they do not grow with N and they set the floor of any build. After the far field is replaced they are the whole build (Section 10).

2.2 Which offsets Gila evaluates, and its use of reflections

Gila evaluates N^3 relative offsets — `egoToe`, indexed $i \leftrightarrow i - 1$ cells, all components non-negative — and fills the doubled $(2N)^3$ circulant grid from them by reflection with sign flips (`egoToeCrc!`, whose rule is $M[(c, a), (c', b)] = \text{egoToe}[a, b, |d| + 1] \sigma_a \sigma_b$ with $\sigma_m = \text{sign}(d_m)$). *The factor of eight at stake in the task statement is therefore already taken;* the count at 128^3 is 2 097 144 separated offsets, not 1.7×10^7 . The eight indices with all components ≤ 2 go through the contact path (`egoFunSng!`, `wekTrp`), and `egoSrfFxd!` returns `Inf` on them, so the far method’s domain is exactly `isFarInd`, $\max_i i_i \geq 3$. The reflection rule was checked end to end: the circulant arrays built from Gila’s own fill and from `farBlock!` agree to 9.81×10^{-15} relative at $f = 1$ and 8.68×10^{-15} at $f = 1 + 0.1i$ over a 20^3 grid, with zero sign-flipped entries out of 8748.

2.3 The far-field error today

`verify.jl` part (c), tables `c_gila_<shape>_<r|c>.txt`: Gila’s `egoSrfFxd!` at its own `quadOrd` order, assembled by `srfSum!`, against the 220-bit reference, at separations 2, 3, 4, 6, 8, 16, 32, 64 cells along an axis, a face diagonal and the body diagonal, at $f = 1$ and $f = 1 + 0.1i$. `pairErr` is the worst of the thirty-six face-pair integrals; `amp` is the assembly amplification $(\sum_{\text{fp}} |\text{srfMat}_{\text{fp}}|)/|G_{ab}|$ over the pairs summed into the entry that attains it; `tMx` and `tEn` are the assembled tensor error in max-norm and per entry.

shape	direction	worst pairErr	amp range	worst tMx	worst tEn
$(1/32)^3$	axis	8.9×10^{-14}	36–1300	3.2×10^{-13}	7.3×10^{-13}
$(1/32)^3$	face diagonal	5.9×10^{-15}	90–413	3.0×10^{-13}	3.6×10^{-13}
$(1/32)^3$	body diagonal	6.3×10^{-15}	311–312	4.7×10^{-13}	7.9×10^{-13}
$(1/8)^3$	all three	9.3×10^{-14}	15–326	1.9×10^{-13}	3.2×10^{-13}
$(1/4)^3$	all three	9.5×10^{-14}	4.8–163	8.6×10^{-14}	2.8×10^{-13}
slender, long axes	$(n, 0, 0), (0, n, 0)$	8.0×10^{-14}	4.3×10^3 – 1.67×10^5	2.0×10^{-11}	3.4×10^{-11}
slender, short axis	$(0, 0, n)$	1.6×10^{-2}	425–1850	0.789	1.60

Four things this fixes.

- **The face pairs themselves are good and the assembly is what is lost.** Per-pair errors are 10^{-15} to 9×10^{-14} everywhere; the assembled tensor is 10 to 1000 times worse. The amplification grows as $(|\mathbf{R}|/s)^2$ on an axis offset — it is a *fine*-cell effect, worst at $\lambda/32$ and mildest at $\lambda/4$ — and is flat at 312 on the body diagonal.
- **The slender cell’s long axes lose four to five digits to the assembly alone.** Face pairs good to 2.6×10^{-15} become a tensor good to 1.1×10^{-11} at 64 cells, because the amplification reaches 1.67×10^5 .
- **Along the slender cell’s short axis Gila’s fixed rule is wrong by $O(1)$:** relative tensor errors of 0.789, 0.227, 3.68×10^{-2} , 4.50×10^{-3} , 1.79×10^{-3} , 2.97×10^{-6} , 8.94×10^{-7} , 9.12×10^{-9} at $n = 2, 3, 4, 6, 8, 16, 32, 64$, and per entry up to 1.60. `quadOrd` keys the rule order on the separation in *cells*, so a $1/512$ boundary layer across a $1/32$ face is invisible to it. (The rows at $n = 3, 6, 32, 64$ use the graded volume reference of `refquad.jl`: `pairKer` did not return on those panels in 25 minutes.)
- **The target the expansion has to beat is therefore 10^{-13} per entry on the cubes and something much better than 10^{-11} on the slender cell.** The claim in the task statement that the far tensor is good to about 10^{-10} is pessimistic for cubes — 5×10^{-14} to 8.7×10^{-13} is what it is — and optimistic for the slender cell.

2.4 The volume identity in exact arithmetic

Part (a), table `a_div.txt`. On the polynomial kernels $F = |\mathbf{x} - \mathbf{y}|^{2j}$ both sides of the divergence identity are rational numbers in the edge lengths, so they can be compared with no arithmetic error at all. The thirty-six face-pair moments, from an exact multinomial integrator on the `facePair` panels and summed with the `srfSum!` signs, against the difference-box form with the exact weight moments, at $\mathbf{n} \in \{(2, 0, 0), (3, 1, 0), (2, 2, 2)\}$, $j = 1, 2, 3$, for the cubic and the slender cell: $\max |\text{difference}| = 0/1$ **exactly in all eighteen rows**. The theory is Section 3; the smallest case can be checked by hand and is written out there. The weight moments themselves were checked separately (`a_mu.txt`): the library’s closed form against the same integral done as two elementary halves, both in `Rational{BigInt}`, **zero disagreements** over five edge lengths and $n = 0, \dots, 24$.

2.5 The convergence ratios, and the fixed size of the slow band

Part (a), tables `a_rho.txt` and `a_band.txt`. With $r_d = |\mathbf{s}|$ the half-diagonal of the difference box, $\rho = r_d/|\mathbf{R}|$ is the convergence ratio of every expansion in this document, and ρ_{oct} is the largest of the eight octant ratios. For cubic cells both are scale invariant.

class	$n = 2$	3	4	6	8	16	32	64
axis, ρ	0.866025	0.577350	0.433013	0.288675	0.216506	0.108253	0.054127	0.027063
axis, ρ_{oct}	0.522233	0.333333	0.242536	0.156174	0.114960	0.055815	0.027486	0.013637
face diagonal, ρ	0.612372	0.408248	0.306186	0.204124	0.153093	0.076547	0.038273	0.019137
body diagonal, ρ	0.500000	0.333333	0.250000	0.166667	0.125000	0.062500	0.031250	0.015625
$(n, 1, 0)$, ρ	0.774597	0.547723	0.420084	0.284747	0.214834	0.108042	0.054100	0.027060
slender $(n, 0, 0)$	0.707797	0.471865	0.353898	0.235932	0.176949	0.088475	0.044237	0.022119
slender $(0, 0, n)$	11.32475	7.549834	5.662376	3.774917	2.831188	1.415594	0.707797	0.353898
slender $(0, 0, n)$, oct	0.992293	0.977398	0.956263	0.900237	0.834305	0.590143	0.338354	0.175578

The ratio ρ/ρ_{oct} rises from 1.48 at two cells to 1.98 at sixty-four: *the octant split buys a factor two in ρ only in the far field, and 1.5 where it is needed*. Along the slender cell's short axis $\rho = 22.63/n$, so the whole-box expansion *diverges* until $n = 23$ and reaches $\rho < 0.5$ only at $n = 46$; the octant version is below one from $n = 2$ but only below 0.5 from $n = 21$. This one direction is the reason there is a third route at all.

The offsets with large ρ are a ball of fixed radius in $\mathbf{n}$, so *their number does not grow with the volume*. Counting the `egoToe` octant offsets that the far field sees:

shape	offsets	$\rho > 0.5$	$\rho > 0.3$	$\rho > 0.2$	$\rho > 0.1$
cube, $N = 32$	32 760	30	134	429	3089
cube, $N = 64$	262 136	30	134	429	3089
cube, $N = 128$	2 097 144	30	134	429	3089
slender, $N = 32$	32 760	242	682	1417	5458
slender, $N = 64$	262 136	288	1135	2665	10 684
slender, $N = 128$	2 097 144	288	1192	3618	19 736

Thirty offsets of two million have $\rho > 0.5$, at every N . The expensive band is an $O(1)$ set and the average cost of a large build converges to the far-plateau cost. The slender cell is the exception the table makes explicit, and again for the one reason: $\rho > 1$ along $(0, 0, n)$ up to $n = 22$, so that column grows with N until N passes 23.

2.6 The conditioning of the derivative recursions

The Cartesian organization of the expansion needs $\partial^\alpha g(\mathbf{R})$ to total order 25–40, and whether that is stable in `Float64` decided the shape of the whole design. Three independent measurements, all against 220- or 256-bit references, all reporting $\log_{10}(|x_{64} - x_{\text{ref}}|/(\varepsilon|x_{\text{ref}}|))$.

Hobson's formula is unusable. Building the radial derivatives $g^{(n)}$ from $rg' = (ikr - 1)g$, then $T_m = (r^{-1}d/dr)^m g$, then the Cartesian derivatives from the ν -sum, loses 7.3 to 8.4 digits by $|\alpha| = 25$ at $k\mathbf{R} \leq 3$, and 10.4 digits at $|\alpha| = 30$, $k\mathbf{R} = 10$. The loss is the alternating cancellation inside the ν -sum, where $\mathbf{R}^{\alpha-2\nu}T_{p-|\nu|}$ terms of comparable size subtract. *Discarded*.

The coupled Cartesian recurrence is fine. With $u = g$, $w = rg$, the identities $r^2 \partial_j u + x_j u - ikx_j w = 0$ and $\partial_j w = ikx_j u$, differentiated α times and solved for the highest index with the pivot $j = \arg \max_i |R_i|$, lose 1.4 digits at $p = 25$, $k\mathbf{R} = 1$ on the axis and less off it (1.17 face diagonal, 0.85 body diagonal, 0.38 generic). The worst band is $k\mathbf{R} \approx 10$, where the loss rises to 4.6 digits at $p = 25$; below $k\mathbf{R} = 3$ it is flat at 1.5–2 and above $k\mathbf{R} = 30$ it falls back to about one, because the exponential then dominates and nothing cancels. *Answer to the design question: yes, the Float64 recursion is stable to order 25 at $k\mathbf{R} \approx 1$, provided it is this route.*

The pair recurrence on (G, S) is better still, and it is the one that was audited. With $G = g(\mathbf{R} + \delta)$ and $S = |\mathbf{R} + \delta| G$, the two systems $\nabla S = ik(\mathbf{R} + \delta)G$ and $P\nabla G = (\mathbf{R} + \delta)(ikS - G)$, $P = |\mathbf{R} + \delta|^2$, give a two-level recurrence whose Float64 digits lost, maximised over four directions and two frequencies, are

$k\mathbf{R}$	0.3	1.0	3.0	10	30	100	300
order 10	1.06	0.87	0.93	1.01	1.13	1.66	2.15
order 20	1.25	1.11	1.13	2.07	1.13	1.67	2.15
order 30	1.39	1.31	1.37	2.04	1.84	1.68	2.15
order 40	1.41	1.39	1.54	2.02	3.84	1.70	2.15

The flat floors at $k\mathbf{R} = 100$ and 300 are the *seed* $e^{ik\epsilon}$, which carries $k\rho\epsilon$ of relative error and never degrades with order; a double-double phase (twelve flops) removes it, recovering 2.5 of the 3.35 digits lost at $k\mathbf{R} = 3000$, and is used for the h_0 seed of the shipped method, where $k\mathbf{R}$ reaches about 350 on a 128^3 block at $\lambda/4$. The independent audit of this route put it at ≤ 2.15 digits through order 30 at every $k\mathbf{R}$ and ≤ 1.4 digits for $k\mathbf{R} \leq 3$, cross-checked against four other routes to 10^{-65} at 220 bits.

And the quantity that matters loses nothing. The individual degree sums $S_p = \sum_{|\alpha|=p} \partial^\alpha g \mu_\alpha / \alpha!$ lose 3.2 to 6.3 digits relative to *themselves*, but $|S_{30}|/|S_0|$ is 10^{-9} to 10^{-29} and $\sum_p |S_p| / |\sum_p S_p|$ is 1.00 to 1.74: over the whole physical grid the accumulated tensor loses between -1.06 and $+0.68$ digits. *The moment weights kill exactly the orders at which the recursion loses digits*, and the a priori amplification $1/(1 - \rho) = 7.7$ at $\rho = 0.87$ is itself four to seven times pessimistic. This is why no extended-precision path is needed in the per-offset evaluation — and why the one place that does need exact arithmetic is the geometry table, whose contraction cancels by 7.3×10^{11} (Section 5.8).

2.7 The exact moments

The only geometry the far field ever needs is the moment of the triangle weight,

$$\mu_n(s) = \int_{-s}^s (s - |t|) t^n dt = \begin{cases} \frac{2s^{n+2}}{(n+1)(n+2)}, & n \text{ even,} \\ 0, & n \text{ odd,} \end{cases} \quad \int_D w \delta^\alpha d\delta = \prod_{i=1}^3 \mu_{\alpha_i}(s_i), \quad (1)$$

an exact rational in the edge lengths, zero unless every α_i is even. Note $\mu_0(s) = s^2$, so $\int_D w = V_t^2$ and the cell average $(1/V_t) \int_D w$ is V_t . Sample values, all verified as exact rationals (`a.mu.txt`): at $s = 1/32$, $\mu_0 = 1/1024$, $\mu_2 = 1/6291456$, $\mu_4 = 1/16106127360$, $\mu_{12} = 1/107433837485284428611584$. The corresponding volume moments $\int_D w |\mathbf{R} + \delta|^{2j}$ were computed by two independent exact routes — the multinomial expansion against (1) and the face-pair route — and agree exactly; at $\mathbf{s} = (1/32)^3$,

n	$j = 1$	$j = 2$	$j = 3$
	9	691	82279
(2, 0, 0)	2199023255552	33776997205278720	726340547902313594880
	21	3511	866881
(3, 1, 0)	2199023255552	33776997205278720	726340547902313594880
	25	4931	285155
(2, 2, 2)	2199023255552	33776997205278720	145268109580462718976

and the even face-pair moments of `notes/moments/moments.jl` agree with the exact polynomial integrator to 154 digits at 512 bits, which is that library’s own floor. One practical note that cost time to find: `pairMoments` fails on `Rational{BigInt}` panels (a `sqrt` at `moments.jl:442`); pass `BigFloat` panels, which is exact for these lengths.

2.8 The joint error target

Three numbers set what the kernel must deliver so that nothing downstream is limited by it.

- **The FFT of the Toeplitz block.** At $N = 256$ on the doubled grid the round-off of the forward transform is 5.3×10^{-15} in `Float64` and 2.9×10^{-6} in `Float32`.
- **The anti-Hermitian positivity check** of Section 2.4 of *The Integrals of Gila*. On a 6^3 block at real frequency, where $\text{Im } M$ must be positive semi-definite, the measured floor is $\lambda_{\min} = -5 \times 10^{-15}$ against $\lambda_{\max} = 0.077$, i.e. 6.5×10^{-14} relative. A 10^{-13} kernel error sits 1.5 times above that floor, which is the reason this task targets 10^{-13} where the earlier work targeted 10^{-10} .
- **The solver does not need any of it.** Gila’s default storage type is `Float32` (`dfltPrc`) and the default relative tolerance of `GMRES`, `BiCGStab` and the mixed-precision refinement is $\sqrt{\epsilon}$: 1.49×10^{-8} in `Float64` and 3.45×10^{-4} in `Float32`. The solve discards six to eight digits regardless. The target is not for the solve; it is so that the operator is the operator it claims to be, so that the positivity check means something, and so that a future solve at higher precision is not blocked by the build.

Finally, the scale against which “ 10^{-13} ” is measured matters, because far entries are small. At $\lambda/32$, $f = 1$, the largest entry of the tensor on an axis offset is 2.08×10^{-2} at two cells, 5.77×10^{-4} at eight, 9.44×10^{-5} at thirty-two and 4.76×10^{-5} at 128, against a self term of order one (the identity $-1/f^2$). **Accuracy “relative to the largest entry of the tensor” is four orders looser than per-entry accuracy for a far offset, and per-entry relative accuracy in `Float64` is the honest target throughout this document.** Both are reported everywhere.

3 The volume formulation

Gila assembles every entry of the discretized vacuum Green operator from thirty-six face-pair integrals of the scalar kernel. That assembly is a signed sum of nearly equal numbers, and it is the reason the far field is only good to 10^{-10} today (Section 3.6 quantifies it). This section undoes it: it shows that the signed sum of the thirty-six face pairs *is* a single smooth integral over the difference box of the pair, derives every constant and every sign of that identity from the code rather than asserting them, checks the identity in exact rational arithmetic on the polynomial kernels $|\mathbf{x} - \mathbf{y}|^{2j}$, and pins it numerically against a 220-bit reference. Every expansion in the rest of this document expands that one integral.

3.1 Conventions, and what `srfMat` holds

Lengths are in wavelengths, the frequency is the single complex scalar f (the field `frqPhz`), $k = 2\pi f$, and Gila's scalar kernel is

$$g(r) = \frac{e^{2\pi i f r}}{4\pi f^2 r} = \frac{e^{i k r}}{4\pi f^2 r}, \quad r = |\mathbf{x} - \mathbf{y}|. \quad (2)$$

A cell has edge lengths $\mathbf{s} = (s_1, s_2, s_3)$, exact `Rationals` in `GlaVol.scl`; the source cell V_s is centred at the origin, the target cell V_t at $\mathbf{R} = (n_1 s_1, n_2 s_2, n_3 s_3)$ with integer $\mathbf{n}$, and $V_t = s_1 s_2 s_3$ denotes the cell volume as well as the target cell. The six faces are ordered `yzL yzU xzL xzU xyL xyU`; face F has axis $c(F) \in \{1, 2, 3\}$ and outward normal $\sigma_F \mathbf{e}_{c(F)}$ with $\sigma_F = -1$ for the lower (L) and $+1$ for the upper (U) face. The face-pair index is `fp` = $6(F - 1) + F'$ with F the *target* face and F' the *source* face (`facPar` writes `fPairs[1,k] = i` for the target and `fPairs[2,k] = j` for the source).

Lemma 3.1 (what the code stores). *Let $I_{FF'} = \int_F \int_{F'} g \, dS' \, dS$ be the raw face-pair integral. Then $\text{srfMat}[\text{fp}] = I_{FF'}/V_t$.*

Proof. `egoSrfFxd!` evaluates the integrand on the unit hypercube $[0, 1]^4$ with the bilinear maps of the two rectangles built by `cubFac` and multiplies the result by `srfScales[fp]`. The bilinear map of a rectangle F onto $[0, 1]^2$ has constant Jacobian A_F , the area of F , so the quadrature approximates $\int_{[0,1]^4} = I_{FF'}/(A_F A_{F'})$. `srfScl` sets `srfScl[(trgFid-1)*6+srcFid]` to $A_{F'}/s_{c(F)}$: the source face area over the target edge normal to the target face. Hence $\text{srfMat}[\text{fp}] = I_{FF'}/(A_F A_{F'}) \cdot A_{F'}/s_{c(F)} = I_{FF'}/(A_F s_{c(F)})$, and for a cuboid $A_F s_{c(F)} = V_t$ for every face F . $\square$

3.2 The divergence theorem, read backwards

Equation (7) of the companion document *The Integrals of Gila* turns the volume–volume integral of the strongly singular dyadic into surface–surface integrals of the weakly singular kernel. Read in the other direction it recovers a volume integral, and that is the direction the far field needs.

Lemma 3.2 (the divergence identity in both directions). *Let F be C^2 on a neighbourhood of the (closed) difference set $V_t - V_s$. Then for $a \neq b$*

$$\int_{V_t} \int_{V_s} (\partial_a \partial_b F)(\mathbf{x} - \mathbf{y}) \, d\mathbf{y} \, d\mathbf{x} = - \sum_{\substack{F: c(F)=a \\ F': c(F')=b}} \sigma_F \sigma_{F'} \int_F \int_{F'} F \, dS' \, dS, \quad (3)$$

and for the diagonal, using $(\partial_a \partial_a - \nabla^2)F = - \sum_{b \neq a} \partial_b \partial_b F$,

$$\int_{V_t} \int_{V_s} [(\partial_a \partial_a - \nabla^2)F](\mathbf{x} - \mathbf{y}) \, d\mathbf{y} \, d\mathbf{x} = + \sum_{b \neq a} \sum_{\substack{F, F': \\ c(F)=c(F')=b}} \sigma_F \sigma_{F'} \int_F \int_{F'} F \, dS' \, dS. \quad (4)$$

Proof. F depends on $\mathbf{x}$ and $\mathbf{y}$ only through $\mathbf{u} = \mathbf{x} - \mathbf{y}$, so $\partial_{u_a} = \partial_a^x = -\partial_a^y$ acting on $F(\mathbf{x} - \mathbf{y})$, whence $(\partial_a \partial_b F)(\mathbf{x} - \mathbf{y}) = -\partial_a^x \partial_b^y F(\mathbf{x} - \mathbf{y})$. Each derivative is moved onto its own domain by the divergence theorem, $\int_V \partial_a \Phi \, d\mathbf{x} = \oint_{\partial V} n_a \Phi \, dS$, giving $-\oint_{\partial V_t} \oint_{\partial V_s} n_a(\mathbf{x}) n'_b(\mathbf{y}) F \, dS' \, dS$. On a cuboid n_a vanishes on the four faces whose axis is not a and equals σ_F on the two whose axis is a , which is (3). The diagonal follows by applying (3) with $b = a$ to each of the two terms of $-\sum_{b \neq a} \partial_b \partial_b F$, the two minus signs cancelling. $\square$

Proposition 3.3 (`srfSum!` is exactly Lemma 3.2). With `srfMat[fp]` = $I_{FF'}/V_t$ of Lemma 3.1, the nine signed sums of `srfSum!` (`glaVacOprMemGen.jl`, lines 734–757) are

$$(G_{\text{gila}})_{ab}(\mathbf{R}) = \frac{1}{V_t} \int_{V_t} \int_{V_s} [(\partial_a \partial_b - \delta_{ab} \nabla^2) g](\mathbf{x} - \mathbf{y}) d\mathbf{y} d\mathbf{x}, \quad (5)$$

with the sign +1 on the diagonal and on the off-diagonals and with no further factor of k^2 , 4π , f^2 or V_t .

Proof. Read the coefficients off the code and compare with Lemma 3.2. For the off-diagonal entry (1, 2), `srfSum!` gives `-srfMat[3] + srfMat[4] + srfMat[9] - srfMat[10]`; the four indices are $\text{fp} = 6(F - 1) + F'$ for $(F, F') = (1, 3), (1, 4), (2, 3), (2, 4)$, i.e. the two x -faces of the target against the two y -faces of the source, and their normal signs are $\sigma_F \sigma_{F'} = (-)(-) = +1$, $(-)(+) = -1$, $(+)(-) = -1$, $(+)(+) = +1$. The code's coefficients are $-1, +1, +1, -1$, i.e. exactly $-\sigma_F \sigma_{F'}$, which is (3). For the diagonal entry (1, 1), `srfSum!` gives `srfMat[15] - srfMat[16] - srfMat[21] + srfMat[22] + srfMat[29] - srfMat[30] - srfMat[35] + srfMat[36]`; those eight indices are $(F, F') = (3, 3), (3, 4), (4, 3), (4, 4)$ (both faces on the y axis) and $(5, 5), (5, 6), (6, 5), (6, 6)$ (both on the z axis), with $\sigma_F \sigma_{F'} = +1, -1, -1, +1$ in each block, matching the code's coefficients with no extra sign, which is (4). The remaining seven entries are the same two patterns with the axes permuted. $\square$

3.3 Difference variables and the triangle weight

Lemma 3.4 (the difference box). For any integrable Φ ,

$$\int_{V_t} \int_{V_s} \Phi(\mathbf{x} - \mathbf{y}) d\mathbf{y} d\mathbf{x} = \int_D w(\boldsymbol{\delta}) \Phi(\mathbf{R} + \boldsymbol{\delta}) d\boldsymbol{\delta}, \quad D = \prod_{i=1}^3 [-s_i, s_i], \quad w(\boldsymbol{\delta}) = \prod_{i=1}^3 (s_i - |\delta_i|). \quad (6)$$

Proof. The integrand depends only on $\mathbf{u} = \mathbf{x} - \mathbf{y}$ and both cells are products of intervals, so the identity factors over the axes. On axis i the target interval is $R_i + [-s_i/2, s_i/2]$ and the source interval is $[-s_i/2, s_i/2]$; the push-forward of the product Lebesgue measure under $(x_i, y_i) \mapsto x_i - y_i$ is the convolution of the two indicators, a triangle of base $2s_i$ and apex height s_i centred at R_i . Writing $u_i = R_i + \delta_i$ gives the density $s_i - |\delta_i|$ on $[-s_i, s_i]$. $\square$

The moments of that weight are the only geometry the far field ever needs:

$$\mu_n(s) = \int_{-s}^s (s - |t|) t^n dt = \begin{cases} \frac{2s^{n+2}}{(n+1)(n+2)}, & n \text{ even,} \\ 0, & n \text{ odd,} \end{cases} \quad \int_D w \boldsymbol{\delta}^\alpha d\boldsymbol{\delta} = \prod_{i=1}^3 \mu_{\alpha_i}(s_i), \quad (7)$$

by $\mu_n = 2 \int_0^s (s - t) t^n dt$ for even n and oddness of the integrand otherwise. Note $\mu_0(s) = s^2$, so $\int_D w = V_t^2$ and the cell average $(1/V_t) \int_D w = V_t$.

Combining Proposition 3.3, Lemma 3.4 and the Helmholtz equation $\nabla^2 g = -k^2 g$, valid away from $r = 0$ and therefore on all of $\mathbf{R} + D$ as soon as the two cells do not touch, gives the object of this document.

Theorem 3.5 (the volume form of the far field). Let the two cells be separated, i.e. $\mathbf{0} \notin \mathbf{R} + D$. Then

$$\begin{aligned} (G_{\text{gila}})_{ab}(\mathbf{R}) &= \frac{1}{V_t} \int_D w(\boldsymbol{\delta}) [(\partial_a \partial_b + \delta_{ab} k^2) g](\mathbf{R} + \boldsymbol{\delta}) d\boldsymbol{\delta} \\ &= \frac{1}{V_t} \int_D w(\boldsymbol{\delta}) [(\partial_a \partial_b - \delta_{ab} \nabla^2) g](\mathbf{R} + \boldsymbol{\delta}) d\boldsymbol{\delta}. \end{aligned} \quad (8)$$

The integrand is analytic on a neighbourhood of D ; explicitly, with $\mathbf{z} = \mathbf{R} + \boldsymbol{\delta}$, $r = |\mathbf{z}|$,

$$g' = \left(ik - \frac{1}{r}\right)g, \quad g'' = \left(ik - \frac{1}{r}\right)g' + \frac{g}{r^2}, \quad \partial_a \partial_b g = (g'' - g'/r) \frac{z_a z_b}{r^2} + \delta_{ab} \frac{g'}{r}, \quad (9)$$

and $\nabla^2 g = g'' + 2g'/r = -k^2 g$.

The point of (8) is that the right-hand side carries no cancelling signed sum: $w \geq 0$ on D , and the thirty-six face-pair integrals that Gila subtracts have been replaced by one integral of a smooth function against a non-negative weight. What the far field must do is evaluate that integral without quadrature.

3.4 Relation to the physical dyadic

The free-space dyadic and the form Gila uses are

$$\mathbf{G}(\mathbf{r}) = \left(\mathbf{I} + \frac{\nabla \nabla}{k^2}\right) G_s(r) = \frac{1}{k^2} (\nabla \nabla - \mathbf{I} \nabla^2) G_s(r) - \frac{\mathbf{I}}{k^2} \delta(\mathbf{r}), \quad G_s(r) = \frac{e^{ikr}}{4\pi r}, \quad (10)$$

and the entry of the discretized operator is the double cell average $(1/V_t) \int_{V_t} \int_{V_s} G_{ab}$. Gila's operator is $4\pi^2$ times that average in units where the wavelength is one; the fixed constant is visible in `genEgoCrcSlf!`, where the identity term $-\delta_{ab}\delta_{ts}/k^2$ is implemented as `egoToe[i,i,1,1,1] -= 1/frqPhz^2`, i.e. as $-1/f^2 = -4\pi^2/k^2$. The two constants of (10) and of Gila's normalization combine once and for all:

$$\frac{4\pi^2}{k^2} = \frac{4\pi^2}{4\pi^2 f^2} = \frac{1}{f^2}, \quad \text{so} \quad \frac{4\pi^2}{k^2} G_s(r) = \frac{e^{ikr}}{4\pi f^2 r} = g(r). \quad (11)$$

The $1/k^2$ of the dyadic and the $4\pi^2$ of Gila's convention are exactly the $1/(4\pi f^2)$ already inside g , which is why no $1/k^2$ and no $4\pi^2$ survive anywhere in (8): they are not dropped, they are absorbed. Two consequences are worth recording because they are cheap end-to-end tests. First, every integral of $1/r$ is proportional to f^{-2} and nothing else, so g and hence G_{gila} is homogeneous, $G_{\text{gila}}(\alpha \mathbf{s}, \alpha \mathbf{n}, f/\alpha) = \alpha^2 G_{\text{gila}}(\mathbf{s}, \mathbf{n}, f)$. Second, the far entries are small: at $\lambda/32$, $f = 1$, the largest entry of G_{gila} on an axis offset is 2.08×10^{-2} at two cells, 5.77×10^{-4} at eight, 9.44×10^{-5} at thirty-two and 2.39×10^{-5} at 128 cells, against a self term of order one (the identity $-1/f^2$). Accuracy “relative to the largest entry of the tensor” is therefore four orders looser than per-entry accuracy for a far offset, and per-entry relative accuracy in `Float64` is the honest target throughout this document.

3.5 The identity in exact rational arithmetic

Theorem 3.5 contains two claims that can be separated and tested exactly: the divergence identity of Lemma 3.2 with the `srfsSum!` signs, and the difference-box reduction of Lemma 3.4. On a polynomial kernel both sides are rational numbers in the edge lengths and can be compared with no arithmetic error at all. Take $F(\mathbf{u}) = |\mathbf{u}|^{2j}$. Then $(\partial_a \partial_a - \nabla^2)|\mathbf{u}|^2 = -4$ identically, and more generally every side of (3)–(4) is a finite rational combination of the $\mu_n(s_i)$ and of the corresponding face moments.

The check was run for $\mathbf{n} \in \{(2, 0, 0), (3, 1, 0), (2, 2, 2)\}$ and $j = 1, 2, 3$ at $\mathbf{s} = (1/32)^3$ in `Rational` of `BigInt` arithmetic, and both identities hold *exactly* — not to a tolerance. The volume moments $\int_D w |\mathbf{R} + \boldsymbol{\delta}|^{2j} d\boldsymbol{\delta}$, computed by two independent exact routes (multinomial expansion against (7), and the face-pair route), agree exactly and are

n	$j = 1$	$j = 2$	$j = 3$
	9	691	82279
(2, 0, 0)	2199023255552	33776997205278720	726340547902313594880
	21	3511	866881
(3, 1, 0)	2199023255552	33776997205278720	726340547902313594880
	25	4931	285155
(2, 2, 2)	2199023255552	33776997205278720	145268109580462718976

The smallest case can be checked by hand and is worth writing out, because it fixes the sign of (4) with no computer at all: for $j = 1$, $(\partial_1 \partial_1 - \nabla^2)|\mathbf{u}|^2 = 2 - 6 = -4$, so the left-hand side of (4) is $-4 \int_D w = -4 \prod_i \mu_0(s_i) = -4s^6 = -4/32^6 = -1/268435456$ at $\mathbf{s} = (1/32)^3$, and the signed sum of the corresponding face-pair moments is that same rational number. The even moments of `notes/moments/moments.jl` agree with the exact polynomial integrator to 154 digits at 512 bits, which is that library’s own floor.

3.6 The identity pinned numerically, and the sign

The exact check of Section 3.5 tests the geometry but not the kernel. The kernel test compares two 220-bit evaluations of the same tensor that share no code: the *face-pair reference* (the thirty-six integrals $I_{FF'}$ of g from `pairKer` of `notes/gen/ref/mom.jl` at `ordN = 44`, divided by V_t and assembled with the `srfSum!` signs) and the *volume reference* (a graded `BigFloat` tensor Gauss–Legendre rule of order 32 applied to the right-hand side of (8), with every axis of D split at 0 where w has its kink, and panels graded geometrically towards the corner nearest the singularity at $\boldsymbol{\delta} = -\mathbf{R}$; nodes generated by Newton iteration on the Legendre recursion in `BigFloat`, so nothing in the rule is `Float64`). All four sign conventions were tested at every offset: e_{++} is (+diag, +off), e_{+-} is (+diag, –off), and so on, each entry being $\max_{ab} |G_{\text{conv}} - G_{\text{ref}}| / \max_{ab} |G_{\text{ref}}|$; ent_{++} is the worst per-entry relative error of the surviving convention. Shapes: $c_{32} = (1/32)^3$, $c_8 = (1/8)^3$, $c_4 = (1/4)^3$, $\text{sl} = (1/32, 1/32, 1/512)$; $f = 1$ (r) or $1 + 0.1i$ (c).

shape	f	n	e_{++}	ent_{++}	e_{+-}	e_{-+}	e_{--}
c_{32}	r	(2, 0, 0)	2.43e-58	2.81e-58	2.43e-58	2.0	2.0
c_{32}	r	(2, 2, 0)	9.72e-62	1.74e-61	2.0	1.115	2.0
c_{32}	r	(2, 2, 2)	2.00e-63	3.60e-63	2.0	0.567	2.0
c_{32}	r	(3, 1, 0)	1.09e-59	2.31e-59	0.943	2.0	2.0
c_{32}	r	(8, 0, 0)	1.16e-63	1.16e-63	1.16e-63	2.0	2.0
c_{32}	r	(64, 0, 0)	2.63e-63	—	2.63e-63	2.0	2.0
c_{32}	r	(64, 64, 64)	4.14e-63	8.26e-63	1.003	2.0	2.0
c_{32}	c	(2, 0, 0)	2.44e-58	2.77e-58	2.44e-58	2.0	2.0
c_{32}	c	(8, 8, 8)	6.16e-63	8.97e-63	1.373	2.0	2.0
c_8	r	(2, 0, 0)	1.49e-58	1.49e-58	1.49e-58	2.0	2.0
c_8	c	(4, 4, 4)	6.80e-64	1.22e-63	1.111	2.0	2.0
c_4	r	(2, 0, 0)	7.41e-59	1.05e-58	7.41e-59	2.0	2.0
c_4	r	(2, 2, 2)	1.53e-64	2.90e-64	1.054	2.0	2.0
c_4	c	(16, 0, 0)	3.71e-65	—	3.71e-65	2.0	2.0
sl	r	(2, 0, 0)	1.85e-58	4.50e-58	1.85e-58	2.0	2.0
sl	r	(0, 2, 0)	1.85e-58	4.50e-58	1.85e-58	2.0	2.0
sl	r	(0, 0, 2)	1.05e-49	1.07e-49	1.05e-49	2.0	2.0
sl	c	(0, 0, 16)	9.41e-58	9.99e-58	9.41e-58	2.0	2.0

The (+diag, +off) convention of Theorem 3.5 agrees with the face-pair reference to between 3.7×10^{-65} and 1.05×10^{-49} at all eighteen offsets, four shapes and both frequencies — nineteen to thirty-four orders better than the 10^{-30} that was asked for — and every other convention is wrong by 0.57

to 2.0. The sign is +1 on the diagonal and on the off-diagonals, with no extra factor. The rows where $e_{+-} = e_{++}$ are the offsets whose off-diagonal entries vanish by symmetry; the off-diagonal sign is pinned by the seven rows where e_{+-} lies between 0.943 and 2.0. The two suppressed e_{++} values are an artefact of the per-entry metric at offsets whose off-diagonals vanish identically, where `pairKer` leaves them at its own roundoff (7.2×10^{-71} and 2.3×10^{-65} absolute) and the quotient is $O(1)$. The single worst row, slender $(0, 0, 2)$ at 1.05×10^{-49} , is the volume rule’s own truncation and not a disagreement: there the difference box is sixteen times wider than it is deep and the singularity sits $1/512$ outside one of its faces, so the graded rule at order 32 is itself converged only to 1.8×10^{-24} , and 250 s are needed per evaluation. Everything else sits at or near the 220-bit floor of 1.3×10^{-64} .

Two further independent confirmations were obtained, deliberately without shared code. A second `BigFloat` Gauss–Legendre volume rule, written from scratch, reproduces the face-pair reference to 10^{-63} – 10^{-65} on the cubes and 1.4×10^{-62} on the slender cell. And an independent `NIntegrate` evaluation of the right-hand side of (8) in Mathematica at $\mathbf{n} = (5, 1, 0)$, $\mathbf{s} = (1/32)^3$, $f = 1$, working precision 30–40, the box split at 0: entry (1, 2) agrees with the 220-bit face-pair reference to 3.4×10^{-14} , which is Mathematica’s own error estimate, and entry (1, 1), recomputed with a fixed 30-point Gauss–Kronrod rule per sub-box after the global adaptive strategy stalled, reproduces $0.001570738221266370299274 + 0.000360155079207178606379i$ in every printed digit. Sign +, no extra factor, normalization confirmed by three routes.

3.7 Why the volume form and not the face pairs

The assembly of Proposition 3.3 subtracts nearly equal numbers, and the amplification is measurable. Define $\Lambda_{ab} = (\sum_{\text{fp}} |\text{srfMat}_{\text{fp}}|)/|G_{ab}|$ over the face pairs summed into entry (a, b) . Against the 220-bit reference, Gila’s present fixed Gauss–Legendre rule achieves a per-face-pair relative error of 10^{-15} to 9×10^{-14} at $\lambda/32$, but Λ runs from 36 to 1300 (it grows as n^2 on an axis offset and is flat at 312 on the body diagonal), so the assembled tensor error is 5.0×10^{-14} to 8.7×10^{-13} . At $\lambda/8$ the amplification is 14–326 and the tensor error 5×10^{-15} to 3×10^{-13} ; at $\lambda/4$, 5–163 and 1.6×10^{-15} to 2.8×10^{-13} . The amplification is a *fine-cell* effect, growing like $(|\mathbf{R}|/s)^2$. On the slender cell it is far worse: the long axes carry $\Lambda = 4.3 \times 10^3$ to 4.0×10^4 , giving 3.2×10^{-12} to 2×10^{-11} despite per-pair errors of 10^{-15} to 8×10^{-14} , and along the short axis the fixed rule is wrong by $O(1)$: relative tensor errors of 0.79 at $(0, 0, 2)$, 3.7×10^{-2} at $(0, 0, 4)$, 1.8×10^{-3} at $(0, 0, 8)$, 3×10^{-6} at $(0, 0, 16)$, 4.5×10^{-6} at $(0, 0, 20)$ and 1.9×10^{-6} at $(0, 0, 24)$, because `quadOrd` keys the rule order on the separation in *cells* and not on the physical geometry, and a $1/512$ boundary layer across a $1/32$ face is invisible to it.

Equation (8) removes Λ from the problem entirely. It is the formulation used by every family of the registry (Section 12).

4 The local expansion: addition theorem, exact moments, sub-boxes

This section expands the volume form (8) in the size of the cell. The mechanism is the spherical addition theorem: it separates the singular dependence on the centre separation $\mathbf{R}$, where the kernel is elementary, from the regular dependence on the difference variable $\boldsymbol{\delta}$, where the kernel is a homogeneous polynomial whose integral against the triangle weight is an exact rational in the edge lengths. The result is a two-index expansion — a multipole order l and a frequency order n — whose geometry data are frequency independent and computed once per cell shape, and whose

per-offset work is a sum of elementary functions.

4.1 The addition theorem, and the sign the usual statement hides

Write $r = |\mathbf{R} + \boldsymbol{\delta}|$, $\hat{\mathbf{R}} = \mathbf{R}/|\mathbf{R}|$, $\hat{\boldsymbol{\delta}} = \boldsymbol{\delta}/|\boldsymbol{\delta}|$.

Proposition 4.1 (the form actually needed). *For $|\boldsymbol{\delta}| < |\mathbf{R}|$,*

$$\frac{e^{ikr}}{r} = ik \sum_{l \geq 0} (2l+1) (-1)^l j_l(k|\boldsymbol{\delta}|) h_l^{(1)}(k|\mathbf{R}|) P_l(\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\delta}}) \quad (12)$$

$$= 4\pi ik \sum_{l \geq 0} \sum_{m=-l}^l (-1)^l h_l^{(1)}(k|\mathbf{R}|) Y_{lm}(\hat{\mathbf{R}}) j_l(k|\boldsymbol{\delta}|) Y_{lm}(\hat{\boldsymbol{\delta}}). \quad (13)$$

Remark 4.2 (the convention, stated once). The classical theorem is usually written for $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ with $P_l(\cos \gamma)$ and γ the angle between $\mathbf{r}_1$ and $\mathbf{r}_2$. Here $\mathbf{r} = \mathbf{R} - (-\boldsymbol{\delta})$, so $\mathbf{r}_2 = -\boldsymbol{\delta}$ and $\cos \gamma = -\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\delta}}$; since $P_l(-t) = (-1)^l P_l(t)$, the form written in terms of $P_l(\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\delta}})$ — the one whose angular factor is the real solid harmonic of $\boldsymbol{\delta}$ itself, which is what the moments need — carries an explicit $(-1)^l$. Omitting it is a first-order error, not a subtlety: at $\mathbf{R} = (1, 2, 3)$, $\boldsymbol{\delta} = (0.3, -0.2, 0.1)$, $f = 1$, in 300-bit arithmetic with $L = 40$, (13) reproduces e^{ikr}/r to 1.06×10^{-41} with the factor and to a relative error of 6.66×10^{-1} without it. The constant $4\pi ik$ is therefore verified to 1.1×10^{-41} at $f = 1$ and 1.1×10^{-40} at $f = 1 + 0.1i$. For the whole difference box the factor turns out to be moot, because Section 4.5 shows that every surviving l is even; on a sub-box it is mandatory, because subdivision destroys that parity (Section 4.7).

Multiplying (13) by $1/(4\pi f^2)$ gives the kernel,

$$g(|\mathbf{R} + \boldsymbol{\delta}|) = \frac{ik}{f^2} \sum_{l,m} (-1)^l h_l(k|\mathbf{R}|) Y_{lm}(\hat{\mathbf{R}}) j_l(k|\boldsymbol{\delta}|) Y_{lm}(\hat{\boldsymbol{\delta}}), \quad (14)$$

absolutely and uniformly convergent on compact subsets of $|\boldsymbol{\delta}| < |\mathbf{R}|$, so it may be integrated against w and differentiated in $\mathbf{R}$ term by term. Throughout, h_l means $h_l^{(1)} = j_l + iy_l$.

4.2 Real solid harmonics as integer polynomials

Take real harmonics without the Condon–Shortley phase,

$$Y_{l0} = N_{l0} P_l(\cos \theta), \quad Y_{lm} = \sqrt{2} N_{lm} P_l^m(\cos \theta) \cos m\varphi, \quad Y_{l,-m} = \sqrt{2} N_{lm} P_l^m(\cos \theta) \sin m\varphi, \quad (15)$$

$N_{lm} = \sqrt{(2l+1)(l-m)!/(4\pi(l+m)!)}$, so that $\sum_m Y_{lm}(\mathbf{a}) Y_{lm}(\mathbf{b}) = \frac{2l+1}{4\pi} P_l(\mathbf{a} \cdot \mathbf{b})$. The regular solid harmonic $|\boldsymbol{\delta}|^l Y_{lm}(\hat{\boldsymbol{\delta}})$ is then an *integer* polynomial in $(\delta_1, \delta_2, \delta_3)$ times a single square root:

$$|\boldsymbol{\delta}|^l Y_{lm}(\hat{\boldsymbol{\delta}}) = \nu_{lm} C_l^m(\delta_3, |\boldsymbol{\delta}|^2) A_m(\delta_1, \delta_2), \quad \nu_{lm} = \sqrt{\frac{(2l+1)(2-\delta_{m0})}{4\pi(l-m)!(l+m)!}}, \quad (16)$$

with $A_m = \text{Re}(\delta_1 + i\delta_2)^m$ for the cosine family and $A_m = \text{Im}(\delta_1 + i\delta_2)^m$ for the sine family, and with C_l^m generated by the integer recurrence

$$C_m^m = (2m-1)!!, \quad C_{m+1}^m = (2m+1)!! \delta_3, \quad C_l^m = (2l-1) \delta_3 C_{l-1}^m - (l+m-1)(l-m-1) |\boldsymbol{\delta}|^2 C_{l-2}^m. \quad (17)$$

Note the $|\delta|^2$ and not $\delta_1^2 + \delta_2^2$: C_l^m is $(l-m)!$ times $|\delta|^l P_l^m$ stripped of the factor $(\delta_1 + i\delta_2)^m$, and it is exactly the factorial $(l-m)!$ that makes (17) integral, cancelling afterwards against ν_{lm} . Stored in the parity-compact array $P[a+1, b+1]$ = the coefficient of $\delta_1^{2a+p_1} \delta_2^{2b+p_2} \delta_3^{\deg - \dots}$, multiplication by δ_3 is free (the degree label carries it) and multiplication by $|\delta|^2$ is $\text{dst}[a, b] = \text{src}[a, b] + \text{src}[a-1, b] + \text{src}[a, b-1]$. Checked against the normalised associated-Legendre recursion at a generic point for every (l, m) with $l \leq 20$ at 300 bits: maximum relative difference 2.8×10^{-87} .

The recurrence (17) alternates in sign and the geometry contraction that follows cancels badly — $\sum |\text{term}|/|\text{result}| = 2.1 \times 10^5$ at $L = 20$ and 5.8×10^8 at $L = 40$ for the cubes, 1.1×10^7 for the slender cell, i.e. up to 8.8 decimal digits. That is why the geometry table is built in exact rational arithmetic once per cell shape (**Rational** of **BigInt** edge lengths, **BigInt** polynomial coefficients) and rounded to **Float64** only at the end. The cost is a one-off: 14.65 s at $L = 40$, $n_{\max} = 12$.

4.3 The regular factor as a power series in k

j_l is entire and even in its argument up to the prefactor z^l :

$$j_l(k|\delta|) Y_{lm}(\hat{\delta}) = \sum_{n \geq 0} c_{ln}(k) |\delta|^{2n} [|\delta|^l Y_{lm}(\hat{\delta})], \quad c_{ln}(k) = \frac{(-1)^n k^{l+2n}}{2^n n! (2l+2n+1)!!}, \quad (18)$$

which follows from $j_l(z) = \frac{z^l}{(2l+1)!!} \sum_n \frac{(-z^2/2)^n}{n! \prod_{q=1}^n (2l+2q+1)}$ and $(2l+1)!! \prod_{q=1}^n (2l+2q+1) = (2l+2n+1)!!$. Every summand of (18) is a homogeneous polynomial in δ of degree $l+2n$ with rational coefficients, so its integral against w is exact. That is the whole point: *k enters only through the scalars c_{ln} and $h_l(k|\mathbf{R}|)$; the geometry is frequency independent.*

4.4 Where the derivatives go

Because g depends on $\mathbf{R}$ and δ only through their sum, $\partial/\partial R_a$ and $\partial/\partial \delta_a$ act identically on $g(\mathbf{R} + \delta)$, and the two derivatives of (8) may be put on either side. Both organizations are used below, for different purposes.

Derivatives on the regular side (the evaluation). Acting on the polynomial (18) the derivatives only re-index the moment product, since $\partial_a^2 \delta^\alpha = \alpha_a(\alpha_a - 1) \delta^{\alpha - 2\mathbf{e}_a}$ and $\partial_a \partial_b \delta^\alpha = \alpha_a \alpha_b \delta^{\alpha - \mathbf{e}_a - \mathbf{e}_b}$. Define the three exact-rational geometry tables

$$\begin{aligned} A^d[a, n, lm] &= \frac{1}{V_t} \int_D w \partial_a^2 [|\delta|^{2n+l} Y_{lm}] d\delta, & a = 1, 2, 3, \\ B^d[n, lm] &= \frac{1}{V_t} \int_D w |\delta|^{2n+l} Y_{lm} d\delta, \\ A^o[ab, n, lm] &= \frac{1}{V_t} \int_D w \partial_a \partial_b [|\delta|^{2n+l} Y_{lm}] d\delta, & ab = 12, 13, 23, \end{aligned} \quad (19)$$

each an exact rational in s_1, s_2, s_3 times ν_{lm} , by (7). Contract them with the frequency once per f ,

$$\tilde{Q}_{lm}^{aa}(k) = \sum_n c_{ln}(k) (A^d[a, n, lm] + k^2 B^d[n, lm]), \quad \tilde{Q}_{lm}^{ab}(k) = \sum_n c_{ln}(k) A^o[ab, n, lm], \quad (20)$$

and then *per offset* only the elementary sum remains,

$$\boxed{T_{ab}(\mathbf{R}) = \frac{ik}{f^2} \sum_{l \leq L} \sum_{m=-l}^l (-1)^l h_l(k|\mathbf{R}|) Y_{lm}(\hat{\mathbf{R}}) \tilde{Q}_{lm}^{ab}(k).} \quad (21)$$

The frequency contraction (20) costs 31 ms once per frequency; the per-offset cost of (21) is 391, 1250, 2731 and 4972 ns at $L = 10, 20, 30, 40$, i.e. $t(L) \approx 3.1L^2 + 80$ ns with no allocation.

Derivatives on the singular side (the bound). The same object is

$$T_{ab}(\mathbf{R}) = \frac{ik}{f^2 V_t} \sum_{l,m} (-1)^l \left[(\partial_a \partial_b + \delta_{ab} k^2) (h_l(k|\mathbf{R}|) Y_{lm}(\hat{\mathbf{R}})) \right] \mathcal{I}_{lm}, \quad \mathcal{I}_{lm} = \int_D w j_l(k|\boldsymbol{\delta}|) Y_{lm}(\hat{\boldsymbol{\delta}}) d\boldsymbol{\delta}, \quad (22)$$

and it is this form that Section 5 bounds, because the $\boldsymbol{\delta}$ dependence has been reduced to a single non-negative radial moment while the derivatives sit on functions of $\mathbf{R}$ alone, whose growth in l is known in closed form. Note that the two organizations select different (l, m) : in (22) the surviving classes are only those of the diagonal (Section 4.5), the six tensor entries being produced by the derivative; in (21) the derivative has already been absorbed into the tables and four classes survive.

4.5 What the triangle weight kills

Each monomial $\delta_1^i \delta_2^j \delta_3^p$ of $|\boldsymbol{\delta}|^{2n} |\boldsymbol{\delta}|^l Y_{lm}$ has a parity triple fixed by (l, m) and by the family:

$$\text{par}(\delta_1) = (m - t) \bmod 2, \quad \text{par}(\delta_2) = t \bmod 2, \quad \text{par}(\delta_3) = (l - m) \bmod 2, \quad (23)$$

with t even for the cosine family and odd for the sine family. Since w is even in every coordinate, $\int_D w \boldsymbol{\delta}^\alpha \neq 0$ only when every α_i is even, and $\partial_a \partial_b$ flips the parity of α_a and α_b . Exactly four classes survive:

entry	surviving class
11, 22, 33	cosine family, m even ≥ 0 , $l - m$ even $\Rightarrow l$ even
12	sine family, m even ≥ 2 , $l - m$ even $\Rightarrow l$ even
13	cosine family, m odd, $l - m$ odd $\Rightarrow l$ even
23	sine family, m odd, $l - m$ odd $\Rightarrow l$ even

Every surviving l is even, which is why the $(-1)^l$ of Remark 4.2 never appears for the whole box. The stored count is 144, 484, 1024, 1764 numbers per n at $L = 10, 20, 30, 40$ against 121, 441, 961, 1681 for one full (l, m) table — the four classes together cost about what a single unpruned table would, while carrying all three diagonal derivatives and both off-diagonal families.

Remark 4.3 (the cube loses $l = 2$ as well). For a *cubic* cell the entire $l = 2$ shell vanishes identically, so the surviving shells are $l \in \{0, 4, 6, 8, \dots\}$. In the organization (22) this is immediate: $|\boldsymbol{\delta}|^2 Y_{20} \propto 2\delta_3^2 - \delta_1^2 - \delta_2^2$ and $|\boldsymbol{\delta}|^2 Y_{22} \propto \delta_1^2 - \delta_2^2$, and $\int_D w \delta_i^2 = \mu_2(s_i) \prod_{j \neq i} \mu_0(s_j)$, which is the same number for $i = 1, 2, 3$ when $s_1 = s_2 = s_3$; the two combinations therefore cancel exactly. Measured: the shells with $\max_m |\mathcal{I}_{lm}|$ above 10^{-40} of the largest are $\{0, 4, 6, 8, \dots\}$ for $(1/32)^3$ and $(1/4)^3$ and $\{0, 2, 4, 6, \dots\}$ for $(1/32, 1/32, 1/512)$ (script `chkRem.jl`). This is a genuine saving on the shapes Gila uses most and is not exploited by the term-count tables of Section 5, which are stated for the generic shape.

4.6 Internal identities of the geometry table

Two identities test the tables (19) without any reference value. First, harmonicity: $|\boldsymbol{\delta}|^l Y_{lm}$ is harmonic, so $\sum_a A^d[a, 0, lm] = 0$, and in rational arithmetic this is exactly zero. Second, the

Helmholtz identity term by term: for $H_{lm} = |\delta|^l Y_{lm}$,

$$\begin{aligned} \nabla^2 [|\delta|^{2n} H_{lm}] &= 2n(2n+2l+1) |\delta|^{2n-2} H_{lm}, \\ \text{hence } \sum_a A^d[a, n, lm] &= 2n(2n+2l+1) B^d[n-1, lm]. \end{aligned} \quad (24)$$

On the exact-rational table with $l \leq 20$, $1 \leq n \leq 6$, the maximum relative violation is 1.73×10^{-89} at 300 bits of output precision, i.e. the identity holds exactly and the printing is the only error. Independently, the constant $4\pi i k$ and the $(-1)^l$ of (13) are verified to 1.7×10^{-89} on the full assembled identity.

4.7 Octant sub-boxes: affine weights and the moments ν_n

The expansion (21) converges like ρ^l with $\rho = r_d/|\mathbf{R}|$, $r_d = |\mathbf{s}|$, and at the nearest non-touching offsets ρ is close to one. Subdividing D shrinks it. Split each axis into pieces, so that D is the disjoint union of boxes D_j with centres $\mathbf{c}_j$ and half-widths $\mathbf{h}_j$; with $\delta' = \delta - \mathbf{c}_j$,

$$T_{ab}(\mathbf{R}) = \sum_j T_{ab}^{(j)}(\mathbf{R} + \mathbf{c}_j), \quad T_{ab}^{(j)}(\mathbf{V}) = \frac{1}{V_t} \int_{D_j} w(\mathbf{c}_j + \delta') [(\partial_a \partial_b + \delta_{ab} k^2) g](\mathbf{V} + \delta') d\delta', \quad (25)$$

each piece being the same object with local centre $\mathbf{V} = \mathbf{R} + \mathbf{c}_j$, local radius $r_j = |\mathbf{h}_j|$ and local ratio $\rho_j = r_j/|\mathbf{V}|$.

Lemma 4.4 (no piece may straddle the origin). *On a piece $[a, b]$ of axis i with $a \geq 0$ the factor $s_i - |\delta_i|$ is $\alpha_i + \beta_i t$ with $t = \delta'_i \in [-h_i, h_i]$, $\alpha_i = s_i - |c_i|$ and $\beta_i = -\text{sign}(c_i)$; likewise for $b \leq 0$. A piece containing 0 in its interior carries the kink of $|\delta_i|$ and has no affine form, and using one anyway is not a small error: for $m = 3$ equal pieces the middle piece $[-s/3, s/3]$ has exact zeroth moment $5s^2/9$ against $2s^2/3$ for the best affine model, a 20% error on that piece.*

Hence 0 must be a split point on every axis. With m_i equal pieces that is automatic for even m_i and forces an extra split for odd m_i , which then has $m_i + 1$ pieces: $m = 3$ costs exactly what $m = 4$ costs and converges more slowly, its outer pieces having half-width $s/3$ against $s/4$, so odd splits are dominated and never used. The minimal legal split is the eight octants, $m = (2, 2, 2)$. On a piece the moments are again exact rationals,

$$\nu_n(\alpha, \beta, h) = \int_{-h}^h (\alpha + \beta t) t^n dt = \begin{cases} \frac{2\alpha h^{n+1}}{n+1}, & n \text{ even}, \\ \frac{2\beta h^{n+2}}{n+2}, & n \text{ odd}, \end{cases} \quad \int_{D_j} w \delta'^\gamma d\delta' = \prod_i \nu_{\gamma_i}(\alpha_i, \beta_i, h_i), \quad (26)$$

and only three one-dimensional tables are needed, $\nu_n^{(0)} = \nu_n$, $\nu_n^{(1)} = n \nu_{n-1}$, $\nu_n^{(2)} = n(n-1) \nu_{n-2}$, exactly as for the whole box. Two exact consistency checks hold with zero residual in **Rational** of **BigInt**: $\sum_{\text{pieces}} \nu_0 = \mu_0(s)$ for $m = 1, 2, 3, 4, 5, 8$, and $\sum_j \sum_q \binom{n}{q} c_j^{n-q} \nu_q = \mu_n(s)$ for $n \leq 8$, $m = 2, 3, 4, 6$.

Subdivision destroys the parity argument of Section 4.5: the affine weight $\alpha + \beta t$ on an off-centre piece is not even, so *every* (l, m) with $0 \leq l \leq L$ contributes to every one of the six entries (seven kinds $\times (n_{\max} + 1) \times (L + 1)^2$ numbers per sub-box type, and about 7 complex multiplies per (l, m) against roughly 1.7 for the whole box), and *odd l survives*, so the $(-1)^l$ becomes mandatory. Omitting it is a first-order error: at $(1/32)^3$, $m = (2, 2, 2)$, $L = 24$, $f = 1$, the relative discrepancy against the unsubdivided value is 3.65×10^{-2} at $\mathbf{n} = (4, 0, 0)$, 5.63×10^{-2} at $(3, 1, 0)$, 6.45×10^{-3} at $(6, 6, 6)$ and 7.58×10^{-3} at $(5, 1, 2)$, against 1.51×10^{-13} , 2.63×10^{-11} , 2.85×10^{-16} and 2.59×10^{-16} with the factor restored — and the first two residuals are the unsubdivided value's own l -truncation at $L = 24$, not the subdivided one's.

4.8 The reflection rule: one octant of tables

Proposition 4.5 (reflection). *Let $\sigma \in \{\pm 1\}^3$ act on coordinates. Then*

$$T_{ab}^{(\sigma j)}(\mathbf{V}) = \sigma_a \sigma_b T_{ab}^{(j)}(\sigma \mathbf{V}), \quad \text{so} \quad T_{ab}(\mathbf{R}) = \sum_{t \in \text{one octant}} \sum_{\sigma \in \{\pm 1\}^3} \sigma_a \sigma_b T_{ab}^{(t)}(\sigma \mathbf{R} + \mathbf{c}_t). \quad (27)$$

Proof. w is even, so $w(\sigma \mathbf{x}) = w(\mathbf{x})$, and $G_{ab}(\mathbf{x}) = [(\partial_a \partial_b + \delta_{ab} k^2)g](\mathbf{x})$ satisfies $G_{ab}(\sigma \mathbf{y}) = \sigma_a \sigma_b G_{ab}(\mathbf{y})$ because g depends only on $|\mathbf{y}|$ and the two derivatives carry one sign each. Substituting $\delta' = \sigma \mathbf{u}$ in (25) gives the first statement; the sub-box set is closed under σ with $\mathbf{c}_{\sigma j} = \sigma \mathbf{c}_j$, which gives the second. $\square$

Diagonal entries take no sign and the off-diagonal (a, b) flips whenever exactly one of the axes a, b is flipped — the same rule the assembled tensor obeys under a lattice reflection. Consistently, the mirrored moments are $\nu_n(\alpha, -\beta, h) = (-1)^n \nu_n(\alpha, \beta, h)$: the mirrored table is the original with every odd- γ_i monomial negated. So only $\prod_i$ (pieces per axis on the positive side) geometry tables are ever built — one for the octant split. Verified against tables built independently for all eight octants at $(1/32)^3$, $m = (2, 2, 2)$, $L = 16$: 1.70×10^{-16} at $\mathbf{n} = (3, 1, 0)$, 1.35×10^{-16} at $(2, 1, 1)$, 9.61×10^{-17} at $(5, 1, 2)$. Per sub-box type and sign pattern, with $\mathbf{V} = \sigma \mathbf{R} + \mathbf{c}_t$,

$$T_{ab}^{(t)}(\mathbf{V}) = \frac{ik}{f^2} \sum_{l=0}^{L_{t,\sigma}} \sum_{m=-l}^l (-1)^l h_l(k|\mathbf{V}|) Y_{lm}(\hat{\mathbf{V}}) \tilde{Q}_{lm}^{ab}(t, k), \quad (28)$$

with $\tilde{Q}$ built from (26) exactly as (20) is built from (7), and with the $(-1)^l$ folded into $\tilde{Q}$ once.

4.9 Real and imaginary parts as separate real sums

For real f the coefficients $c_{lm}(k)$ of (18) are real, the geometry tables (19) are real, and hence $\tilde{Q}_{lm}^{ab}$ is real. With $h_l = j_l + iy_l$ and $ik/f^2 = 2\pi i/f$, (21) splits into two independent real sums,

$$\text{Re } T_{ab} = -\frac{k}{f^2} \sum_{l,m} y_l(k|\mathbf{R}|) Y_{lm}(\hat{\mathbf{R}}) \tilde{Q}_{lm}^{ab}, \quad \text{Im } T_{ab} = +\frac{k}{f^2} \sum_{l,m} j_l(k|\mathbf{R}|) Y_{lm}(\hat{\mathbf{R}}) \tilde{Q}_{lm}^{ab}. \quad (29)$$

This is a structural advantage over any organization that forms one complex sum, and it is worth stating because it is measurable. The radiative (imaginary) part of a far entry is small compared with the reactive part: when $|\text{Im } T_{ab}| / \max_{ab} |T_{ab}|$ falls below about 10^{-3} — which happens routinely at low frequency and on slender cells — a single complex sum delivers the imaginary part only to $\varepsilon \cdot |\text{Re } T|$ in absolute terms, i.e. to a *relative* accuracy of $10^{-16} / (|\text{Im } T| / \max |T|)$. Measured on a Cartesian-Taylor route that does form one complex sum: relative imaginary errors of 1.33×10^{-13} and 1.31×10^{-13} on the $\lambda/32$ cube at $f = 1 + 0.1i$ where the ratio is 3.6×10^{-3} and 2.0×10^{-3} , 2.37×10^{-12} on the slender cell at $f = 0.37$ where it is 1.6×10^{-6} , and 3.99×10^{-12} at ratio 6.0×10^{-5} . The split (29) removes that coupling; it costs nothing, and it makes the y_l and j_l recurrences separately responsible for their own part. The price is that $j_l(k|\mathbf{R}|)$ must then be obtained to *relative* accuracy for l above $k|\mathbf{R}|$, where the upward recurrence for h_l destroys it (the error of $\text{Re } h_l$ from the upward recurrence is 1.3×10^{13} relative at $z = 0.3$, $l = 10$ and 1.4×10^{24} at $z = 10$, $l = 40$); Miller's downward recurrence returns j_l to 6.8×10^{-15} at worst over $l \leq 40$, $|z| \in [0.3, 300]$, at a flat cost of 850 ns.

5 The a priori bounds

Every truncation in this document is controlled by a proved bound, and every bound is checked numerically in `BigFloat` against the quantity it bounds. This section states the bounds, proves them, says exactly what each one bounds and relative to what scale, and gives the measured ratio of bound to truth.

Which bound the shipped code uses. The l -truncation bound of this section, Theorem 5.3, is *not* the one `farfield.jl` evaluates. Section 6 replaces it by two sharper theorems, and it is those — Theorem A (Theorem 6.4, whole box) and Theorem B (Theorem 6.6, sub-box) — that select L in the library, at the tolerance $\text{tol} = 10^{-14}$ (Section 8.2). Theorem 5.3 is kept here in full because it is the bound every measurement of the first round was made against, because Theorem 6.7 is its sharpened form, and because the two bound *different* truncations: Theorem 5.3 bounds the sum with the derivatives on the singular side, and the library truncates the one with the derivatives on the regular side (Section 6.1). Everything else in this section — the elementary bounds on j_l and h_l , the n -truncation, the k -series bound and the two radii — is used unchanged.

The scripts are `chkElem.jl` (the elementary bounds (a), (b), the scalar addition-theorem remainder, the radial moments W_l , and the n -tail (d)), `chk17.jl` (the two structural constants of (c)), `chkRem.jl` (the l -truncation bound end to end), `chkKser.jl` (the k -series bound (e)) and `chkRay.jl` (the convergence radius of remark (f)); their raw output is `out_elem.txt`, `out_c17.txt`, `out_rem.txt`, `out_kser.txt`, `out_ray.txt`.

5.1 What is bounded, and the scale it is measured against

The bounds below are *absolute* bounds on the truncation error of T_{ab} . A relative statement needs a scale, and using $\max_{ab} |T_{ab}|$ would be circular, since the tensor is what is being computed. The scale used throughout is the a priori magnitude

$$\text{est}(\mathbf{R}) = \frac{V_t |k|^2 e^{-\text{Im}(k)|\mathbf{R}|}}{4\pi |f|^2 |\mathbf{R}|} \left(1 + \frac{3}{|k\mathbf{R}|} + \frac{3}{|k\mathbf{R}|^2} \right), \quad (30)$$

the pointwise dyadic scale $|(\partial_a \partial_b + \delta_{ab} k^2)g|(\mathbf{R})$ majorised term by term, multiplied by the total weight $(1/V_t) \int_D w = V_t$. It is computable before anything else is and it is never optimistic: measured against a 220-bit reference tensor, $\text{est}/\max_{ab} |T_{ab}| = 2.36, 2.26, 2.59, 3.77, 1.94$ and 2.29 at $(4, 0, 0)$ and $(3, 1, 0)$ of the $\lambda/32$ cube (both frequencies), $(8, 8, 8)$ of the $\lambda/32$ cube, $(4, 0, 0)$ of the $\lambda/4$ cube and $(4, 0, 0)$ of the slender cell (`chkRem.jl`), and 2.0 to 5.1 over the wider set of offsets and shapes on which it was measured elsewhere. A tolerance of 10^{-13} against est is therefore between 2 and 5 times stricter than 10^{-13} against the largest entry.

5.2 The two elementary bounds

Theorem 5.1 (regular factor). *For every $l \geq 0$ and every $z \in \mathbb{C}$,*

$$|j_l(z)| \leq \frac{|z|^l}{(2l+1)!!} e^{|z|^2/(4l+6)}. \quad (31)$$

Proof. $j_l(z) = \frac{z^l}{(2l+1)!!} \sum_{n \geq 0} \frac{(-z^2/2)^n}{n! \prod_{q=1}^n (2l+2q+1)}$. Since $2l+2q+1 \geq 2l+3$ for every $q \geq 1$, $\prod_{q=1}^n (2l+2q+1) \geq (2l+3)^n$, so by the triangle inequality

$$|j_l(z)| \leq \frac{|z|^l}{(2l+1)!!} \sum_{n \geq 0} \frac{(|z|^2/2)^n}{n! (2l+3)^n} = \frac{|z|^l}{(2l+1)!!} e^{|z|^2/(2(2l+3))},$$

and $2(2l+3) = 4l+6$. □

Theorem 5.2 (singular factor). *For every $l \geq 0$ and every $z \neq 0$,*

$$|h_l^{(1)}(z)| \leq \text{hb}(l, z) := \frac{e^{-\text{Im} z}}{|z|} \sum_{s=0}^l \frac{(l+s)!}{s! (l-s)! (2|z|)^s}. \quad (32)$$

Moreover $\text{hb}(l, z)$ is increasing in l at fixed z .

Proof. The classical finite representation is $h_l^{(1)}(z) = (-i)^{l+1} \frac{e^{iz}}{z} \sum_{s=0}^l i^s \frac{(l+s)!}{s! (l-s)! (2z)^s}$, and $|e^{iz}| = e^{-\text{Im} z}$; apply the triangle inequality. Monotonicity in l holds term by term because $(l+s)!/(s!(l-s)!)$ increases with l at fixed s and the sum gains a term. □

Measured (script chkElem.jl, 400-bit). Over 64 rows of (31) — $l \in \{0, 1, 2, 5, 10, 20, 40, 80\}$ and $z \in \{0.3, 1, 3, 10, 30, 3 + 0.3i, 10 + 3i, 30 + 30i\}$ — the bound is never violated and $\text{bound}/|j_l|$ ranges from 1.0011 to 1.5×10^{119} . The tightness is entirely a function of $|z|^2/(2l+3)$, which is the argument regime that matters: at $|z| = 0.3$ the ratio is 1.031, 1.0039, 1.0011, 1.00055 at $l = 0, 10, 40, 80$; at $|z| = 1$ it is 1.404, 1.044, 1.012, 1.0062; at $|z| = 3$ it is 95.3 (at $l = 0$, where j_0 sits near its zero at $z = \pi$), 1.48, 1.11, 1.057. Family (b) evaluates j_l only at $|z| = |k| |\delta| \leq |k| r_d$, which is 0.34 for the $\lambda/32$ cube, 1.36 at $\lambda/8$, 2.72 at $\lambda/4$ and 0.278 for the slender cell, so (31) is tight to within a factor 1.06–1.5 everywhere it is used. The enormous ratios in the table occur only for $|z|^2 \gg l$, where the bound's $e^{|z|^2/(4l+6)}$ is the wrong majorant — a regime this expansion never enters.

Over 72 rows of (32) the bound is never violated, with equality at $l = 0$ (both sides are $e^{-\text{Im} z}/|z|$), and $\text{bound}/|h_l|$ runs from 1.0 to 1.8×10^{13} . Again the loose rows are a single regime, $|z| \lesssim l$ (the terms $(l+s)!/(s!(l-s)!(2|z|)^s)$ grow before they decay, so the triangle inequality gives away the oscillation): the ratio is 1.35 at $|z| = 0.3$, $l = 80$, but 1.2×10^4 at $|z| = 10$, $l = 80$, and 3.0×10^3 at $|z| = 100$, $l = 40$. That loss is one of the two main reasons the composite bound of Theorem 5.3 over-selects L .

5.3 The l -truncation bound

Theorem 5.3 (l -truncation of (21)). *Let the pair be separated and $r_d = |\mathbf{s}|$. Write $T_{ab}^{(L)}$ for the sum (21) truncated at $l \leq L$. Then for every a, b*

$$|T_{ab}(\mathbf{R}) - T_{ab}^{(L)}(\mathbf{R})| \leq \frac{17|k|^3}{4\pi|f|^2 V_t} \sum_{\substack{l > L \\ l \text{ even}}} (2l+1) \sqrt{2l+5} \frac{|k|^l e^{(|k|r_d)^2/(4l+6)}}{(2l+1)!!} W_l \text{hb}(l+2, k|\mathbf{R}|), \quad (33)$$

where $W_l = \int_D w(\delta) |\delta|^l d\delta$ is, for even l , the exact positive sum

$$W_{2p} = \sum_{i+j+q=p} \frac{p!}{i! j! q!} \mu_{2i}(s_1) \mu_{2j}(s_2) \mu_{2q}(s_3), \quad (34)$$

and hb is the majorant (32).

Proof. Start from (22), the organization with the derivatives on the singular side. Three estimates are needed.

(i) *The derivative factor, and the constant 17.* Each Cartesian first derivative of a singular solid harmonic $h_l(k|\mathbf{R}|)Y_{lm}(\hat{\mathbf{R}})$ is k times a combination of at most four terms $h_{l\pm 1}Y_{l\pm 1, m}$ or $m\pm 1$ with coefficients of modulus at most one; these are the standard differentiation relations, in which the coefficients are of the form $a_{lm} = \sqrt{(l-m)(l+m)/((2l-1)(2l+1))} \leq 1$ and its companions. Two derivatives therefore give at most 16 terms with $l' \in \{l-2, l, l+2\}$, and the term $\delta_{ab}k^2 h_l Y_{lm}$ adds one more: at most 17 terms, each of modulus at most $|k|^2 \max_{l'} \text{hb}(l', |k\mathbf{R}|) \max_{m'} |Y_{l'm'}|$. By Theorem 5.2, hb is increasing in l' , so the maximum is at $l' = l+2$; and $|Y_{l'm'}| \leq \sqrt{(2l'+1)/(4\pi)} = \sqrt{(2l+5)/(4\pi)}$. Hence

$$\left| (\partial_a \partial_b + \delta_{ab} k^2) (h_l(k|\mathbf{R}|)Y_{lm}(\hat{\mathbf{R}})) \right| \leq 17 |k|^2 \text{hb}(l+2, k|\mathbf{R}|) \sqrt{\frac{2l+5}{4\pi}}. \quad (35)$$

(ii) *The sum over m .* By Cauchy–Schwarz and the addition theorem at coincident arguments, $\sum_m Y_{lm}(\mathbf{u})^2 = (2l+1)P_l(1)/(4\pi) = (2l+1)/(4\pi)$, so

$$\sum_{m=-l}^l |Y_{lm}(\mathbf{u})| \leq \sqrt{(2l+1) \sum_m Y_{lm}(\mathbf{u})^2} = \frac{2l+1}{\sqrt{4\pi}}. \quad (36)$$

(iii) *The regular factor and the radial moment.* By Theorem 5.1 pointwise in δ , and since $|\delta| \leq r_d$ on D ,

$$|j_l(k|\delta|)| \leq \frac{|k|^l |\delta|^l}{(2l+1)!!} e^{(|k|r_d)^2/(4l+6)},$$

so, applying (36) to $\hat{\delta}$ under the integral and using $w \geq 0$,

$$\sum_m |\mathcal{I}_{lm}| \leq \frac{2l+1}{\sqrt{4\pi}} \frac{|k|^l e^{(|k|r_d)^2/(4l+6)}}{(2l+1)!!} \int_D w |\delta|^l d\delta = \frac{2l+1}{\sqrt{4\pi}} \frac{|k|^l e^{(|k|r_d)^2/(4l+6)}}{(2l+1)!!} W_l.$$

Multiplying (i) by this, collecting the prefactor $|k|/(|f|^2 V_t)$ of (22) and the two factors $1/\sqrt{4\pi}$ into $1/(4\pi)$ gives (33); the sum runs over even l only because Section 4.5 shows $\mathcal{I}_{lm} = 0$ for odd l . Finally (34) is the multinomial expansion of $|\delta|^{2p} = (\delta_1^2 + \delta_2^2 + \delta_3^2)^p$ integrated term by term against (7); every term is positive, so W_l is computed without cancellation. $\square$

Remark 5.4 (why W_l and not $V_t^2 r_d^l$). Bounding $|\delta|^l$ by r_d^l under the integral is legal and gives $W_l \leq V_t^2 r_d^l$, but it throws away exactly the feature that makes the expansion converge faster than ρ^l : the corner of D , where $|\delta| = r_d$, is where the triangle weight vanishes to third order. With the crude majorant the measured bound-to-truth ratio is 10^5 to 10^{12} ; with the exact W_l it is 10^3 to 10^7 (below). The practical consequence is that the effective convergence ratio for a cube is 0.68 to 0.71 times ρ rather than ρ , so the empirical rule for the cube is $L(10^{-13}) \approx 13/\log_{10}(1/(0.7\rho))$.

The two constants, measured (script chk17.jl). (36) is never violated over $l \leq 60$ and four directions; it is an equality at $l = 0$ and the ratio settles to a direction-dependent constant, 0.487 along an axis, 0.717 on the body diagonal, 0.654 on a generic direction at $l = 60$, so it costs at most a factor 2 and not a factor growing with l . (35) is far looser: over 72 rows ($\mathbf{n} \in \{(2, 0, 0), (4, 0, 0), (3, 1, 0), (16, 16, 16)\}$ at $\lambda/32$ and $\{(4, 0, 0), (32, 0, 0)\}$ at $\lambda/4$, $f = 1$ and

$1 + 0.1i$, $l \leq 24$, all nine (a, b) and all m , derivatives taken by 320-bit central differences at step 10^{-25}) the worst ratio of the true derivative to the right-hand side of (35) is 3.14×10^{-2} and the best is 2.94×10^{-5} . The constant 17 is loose by a factor of 32 to 34000, which is where most of the over-selection of Theorem 5.3 comes from: it counts 17 terms that could all be maximal at once, whereas the differentiation relations distribute the weight across $l' = l - 2, l, l + 2$ and across m .

The bound end to end (script `chkRem.jl`, 320-bit). The truth is $\max_{ab} |T_{ab} - T_{ab}^{(L)}|$ with T_{ab} from an independent `BigFloat` tensor Gauss–Legendre evaluation of (8) on the eight octants of D at order 26 (self-consistency against order 20: 10^{-37} to 10^{-67} depending on the offset), and $T^{(L)}$ assembled from (22) with $\mathcal{I}_{lm}$ by an order-14 octant rule and the derivatives by central differences. Over 79 rows the bound is never violated:

shape	f	$\mathbf{n}$	ρ	$ k\mathbf{R} $	L	$\max_{ab} \text{Rem}_L $	bound/truth
$(1/32)^3$	1	(4, 0, 0)	0.433	0.785	8	1.62e-9	3.17e4
$(1/32)^3$	1	(4, 0, 0)	0.433	0.785	16	2.53e-13	5.50e4
$(1/32)^3$	1	(4, 0, 0)	0.433	0.785	24	2.42e-17	2.21e5
$(1/32)^3$	$1 + 0.1i$	(4, 0, 0)	0.433	0.789	16	2.50e-13	5.10e4
$(1/32)^3$	1	(3, 1, 0)	0.548	0.621	16	1.16e-11	1.57e5
$(1/32)^3$	1	(3, 1, 0)	0.548	0.621	24	7.09e-14	6.54e4
$(1/32)^3$	1	(8, 8, 8)	0.125	2.721	8	1.29e-16	2.04e5
$(1/32)^3$	1	(8, 8, 8)	0.125	2.721	16	4.23e-25	8.51e5
$(1/4)^3$	1	(4, 0, 0)	0.433	6.283	8	4.19e-9	1.57e6
$(1/4)^3$	1	(4, 0, 0)	0.433	6.283	16	4.17e-13	5.50e6
$(1/4)^3$	1	(4, 0, 0)	0.433	6.283	24	3.31e-17	2.99e7
sl	1	(4, 0, 0)	0.354	0.785	8	4.44e-11	1.91e4
sl	1	(4, 0, 0)	0.354	0.785	16	1.62e-15	5.93e4
sl	1	(4, 0, 0)	0.354	0.785	24	1.49e-19	9.47e4

The full range over the 79 rows is $\text{bound/truth} \in [1.3 \times 10^3, 3.0 \times 10^7]$, the largest values at $\lambda/4$ where $|k\mathbf{R}| = 6.3$ and the hb majorant of Theorem 5.2 is at its loosest. In terms of the quantity that matters, the selected L , an over-estimate of 10^3 to 10^7 in the bound costs a factor $\log_{10}(10^{3 \dots 7})/13 \approx 0.23$ to 0.54 in $L(10^{-13})$, i.e. the bound asks for 1.2 to 1.5 times the L that is actually needed. The bound is a usable selector, not merely a certificate, and Section 8 uses it as one.

5.4 The sub-box version

Theorem 5.3 applies verbatim to each sub-box of (25) with three changes: the sum runs over *all* l , because subdivision destroys the parity of Section 4.5; the expansion centre is $\mathbf{V}_j = \mathbf{R} + \mathbf{c}_j$; and the radial moment is the sub-box’s own,

$$|T_{ab}^{(j)} - T_{ab}^{(j), L_j}| \leq \frac{17|k|^3}{4\pi|f|^2V_t} \sum_{l > L_j} (2l+1)\sqrt{2l+5} \frac{|k|^l e^{(|k|r_j)^2/(4l+6)}}{(2l+1)!!} W_l^{(j)} \text{hb}(l+2, k|\mathbf{V}_j|), \quad (37)$$

$r_j = |\mathbf{h}_j|$, with $W_{2p}^{(j)} = \sum_{i+j'+q=p} \frac{p!}{i!j'!q!} \nu_{2i}(\alpha_1, \cdot, h_1) \nu_{2j'}(\alpha_2, \cdot, h_2) \nu_{2q}(\alpha_3, \cdot, h_3)$ — only even one-dimensional moments enter, so β drops out and every term is positive — and $W_l^{(j)} \leq \sqrt{W_{l-1}^{(j)} W_{l+1}^{(j)}}$ for odd l by Cauchy–Schwarz with the non-negative weight. Each of the n_{sub} sub-boxes is given $\text{tol} \cdot \text{est}(\mathbf{R})/n_{\text{sub}}$ of the budget and L_j is the first l at which its tail falls below it. The sub-box bound is *tighter in L* than the whole-box one, by 1.2–1.3 against 1.5–2.0, because $W_l^{(j)}$ describes the affine weight of a sub-box far better than W_l describes the corner suppression of the triangle weight.

5.5 The n -truncation of the k -series inside each l

Theorem 5.5 (n -truncation). *Fix l and truncate the inner sum (18) at $n \leq N$. With $y_l = (|k|r_d)^2/(2(2l+3))$, the omitted part of shell l satisfies*

$$\sum_{n>N} |c_{ln}(k)| W_{l+2n} \leq \frac{|k|^l W_l}{(2l+1)!!} \frac{y_l^{N+1}}{(N+1)!} e^{y_l}, \quad (38)$$

and the induced error in T_{ab} is (33) with the sum restricted to $l \leq L$ and with $e^{(|k|r_d)^2/(4l+6)}$ replaced by $y_l^{N+1} e^{y_l}/(N+1)!$.

Proof. $W_{l+2n} = \int_D w |\delta|^{l+2n} \leq r_d^{2n} W_l$ because $|\delta| \leq r_d$ on D and $w \geq 0$. From (18) and $(2l+2n+1)!! = (2l+1)!! \prod_{q=1}^n (2l+2q+1) \geq (2l+1)!! (2l+3)^n$,

$$|c_{ln}(k)| W_{l+2n} \leq \frac{|k|^l W_l}{(2l+1)!!} \cdot \frac{1}{n!} \left(\frac{|k|^2 r_d^2}{2(2l+3)} \right)^n = \frac{|k|^l W_l}{(2l+1)!!} \cdot \frac{y_l^n}{n!},$$

and $\sum_{n>N} y^n/n! = (y^{N+1}/(N+1)!) \sum_{j \geq 0} y^j (N+1)!/(N+1+j)! \leq (y^{N+1}/(N+1)!) e^y$ since $(N+1)!/(N+1+j)! \leq 1/j!$. The second statement follows by substituting this in place of the corresponding factor in step (iii) of the proof of Theorem 5.3, all other steps being unchanged. $\square$

Measured (script chkElem.jl). (38) was checked in its scalar form, $|j_l(z) - \sum_{n \leq N}| \leq \frac{|z|^l}{(2l+1)!!} \frac{y^{N+1}}{(N+1)!} e^y$, at the four physical values $|z| = |k|r_d \in \{0.34, 1.36, 2.72, 0.278\}$ (the $\lambda/32$, $\lambda/8$ and $\lambda/4$ cubes and the slender cell), $l \in \{0, 4, 12, 24\}$ and $N \in \{2, 4, 6\}$: 48 rows, no violation, bound/truth from 1.122 to 3268. The bound is tight where it is used — 1.12 to 2.4 for $l \geq 12$ at every $|z|$ — and loose only at $l = 0$ and large N , where j_0 is nearly its own leading term. It confirms the term counts $n_{\max} = 6$ (at $\lambda/32$ and for the slender cell), 10 (at $\lambda/8$) and 12 (at $\lambda/4$) for a relative truncation of 10^{-16} , so the n series is a non-issue: the relative size of the n -th term of $\tilde{Q}_{lm}$, maximised over $l \leq 24$, m and entry type, is

shape	$ k r_d$	$n = 0$	$n = 2$	$n = 4$	$n = 6$	$n = 8$	$n = 12$
cube $\lambda/32$	0.340	2.0	2.5e-3	1.9e-9	4.6e-16	3.9e-23	3.1e-38
cube $\lambda/8$	1.360	2.1	4.2e-2	8.4e-6	5.1e-10	1.1e-14	5.8e-25
cube $\lambda/4$	2.721	2.4	2.0e-1	6.8e-4	6.4e-7	2.2e-10	2.9e-18
slender	0.278	2.0	1.8e-3	1.1e-9	1.7e-16	9.6e-24	3.3e-39

with its own cancellation $\sum_n |\text{term}|/|\text{sum}|$ equal to 3.07, 3.29 and 4.14 at $\lambda/32$, $\lambda/8$ and $\lambda/4$: only the $n = 0$ entry exceeds one, so the series alternates with a first term about twice the total.

Remark 5.6 (N is adaptive, and the library refuses rather than truncates). The term counts above are functions of $|k|r_d$ alone — a *shape and frequency* quantity, not a constant — and the number they produce must be built into the geometry table, because the table stores the first $N+1$ coefficients c_{ln} of (18). An earlier version of the library carried a compile-time cap $n_{\max} = 12$ that the per- l cut could only lower, so a shape needing more got 12 and no warning; the adversarial audit found the error growing as $(|k|r_d)^{25.6}$ above $|k|r_d \approx 3.2$, reaching 9.6×10^{-3} per entry at $|k|r_d = 8.96$ with a reported bound 7.2×10^8 times smaller (Section 13, defect D9). In the shipped library the cut returned by Theorem 5.5 is honoured: N is taken as the largest per- l cut the bound asks for, the table is built at $\max(12, N)$, N is re-checked at every evaluation, and above $N_{\text{cap}} = 40$ the call raises an error instead of returning a wrong number. The measured relation is

$ k r_d$	0.03	0.28	0.68	1.58	2.72	3.16	4.46	5.80	8.96	10.9	21.9
N	4	6	8	11	12	16	18	19	25	24	39

(70 rows, evaluated table-free at $\text{tol} = 10^{-14}$; the non-monotone pairs are different shapes at the same $|k|r_d$, since N also depends weakly on the aspect ratio). A cell whose longest edge is at most $\lambda/4$ has $r_d \leq \sqrt{3}s_{\max}$ and hence $|k|r_d \leq 2.72$, so $N \leq 12$: *the old constant was exactly right for Gila's stated range and wrong for anything coarser*, which is why the cap was invisible until the audit went outside it. $N_{\text{cap}} = 40$ covers $|k|r_d \leq 22$, a cubic cell of one wavelength at $f = 2$. On the four shapes Gila uses, the largest per-shell cut the bound asks for is (`tables/e_ncut.txt`, $\text{tol} = 10^{-14}$)

shape	$(1/32)^3$	$(1/8)^3$	$(1/4)^3$	$(1/32, 1/32, 1/512)$
$ k r_d$	0.340	1.360	2.721	0.278
N needed	6	9	12	6
N stored	12	12	12	12

at both $f = 1$ and $f = 1 + 0.1i$, so the n -series is certified everywhere the library evaluates on these shapes, with the cut falling from 6 at $l = 0$ to 4 at $l = 14$ on the finest cell and from 12 to 7 on the coarsest.

5.6 The k -series bound, for the near band

The near field uses the moment series of the companion document, and the join needs its bound in the same currency.

Theorem 5.7 (k -series truncation). *For a panel pair (A, B) write $I_m = \int_A \int_B |\mathbf{x} - \mathbf{y}|^m dS' dS$, $D_{\max} = \max_{\mathbf{x} \in A, \mathbf{y} \in B} |\mathbf{x} - \mathbf{y}|$ and $x = |k|D_{\max}$, and let $S = \int_A \int_B g = \frac{1}{4\pi f^2} \sum_{n \geq 0} \frac{(ik)^n I_{n-1}}{n!}$. Then for every $N \geq 0$*

$$\left| S - \frac{1}{4\pi f^2} \sum_{n \leq N} \frac{(ik)^n I_{n-1}}{n!} \right| \leq \frac{I_{-1}}{4\pi |f|^2} \frac{x^{N+1}}{(N+1)!} B(x, N), \quad B(x, N) = \min \left(e^x, \frac{1}{1 - x/(N+2)} \right), \quad (39)$$

the second branch only when $x < N + 2$. Truncating every face pair at the same N and assembling,

$$\frac{\max_{ab} |G_{ab} - G_{ab}^{(N)}|}{\max_{ab} |G_{ab}|} \leq \Lambda \frac{x_{\max}^{N+1}}{(N+1)!} B(x_{\max}, N), \quad \Lambda = \frac{\sum_{\text{fp}} I_{-1}^{(\text{fp})}}{4\pi |f|^2 V_t \max_{ab} |G_{ab}|}. \quad (40)$$

Proof. Pointwise on $A \times B$, $r^{n-1} = r^{-1} r^n \leq r^{-1} D_{\max}^n$ and $r^{-1} > 0$, so $|I_{n-1}| \leq D_{\max}^n I_{-1}$ and $|t_n| \leq x^n I_{-1}/n!$ with $t_n = (ik)^n I_{n-1}/n!$. Then $\sum_{n > N} x^n/n! = \frac{x^{N+1}}{(N+1)!} \sum_{j \geq 0} x^j \frac{(N+1)!}{(N+1+j)!}$, and $(N+1)!/(N+1+j)! \leq 1/j!$ gives the e^x branch while $(N+1)!/(N+1+j)! = \prod_{i=2}^{j+1} (N+i)^{-1} \leq (N+2)^{-j}$ gives the geometric branch when $x < N + 2$. (40) follows because every entry of G is a signed sum of at most eight of the thirty-six scaled pairs. $\square$

Measured (script `chkKser.jl`, 300-bit). Both the pointwise moment inequality and the resulting remainder bound were checked on real face pairs of the lattice, with the moments from the closed forms of the companion document and the “truth” the tail of the same series summed to $m_{\max} = 60$. At $\mathbf{n} = (2, 0, 0)$ of the $\lambda/32$ cube the monotonicity $|I_{n-1}| \leq D_{\max}^n I_{-1}$ holds with a worst ratio of 0.0802, and bound/truth is 4.44, 8.05, 14.4, 25.2, 43.2 at $N = 8, 12, 16, 20, 24$ for $f = 1$ and 4.46, 8.07, 14.4, 25.3, 43.3 at $f = 1 + 0.1i$; at $(3, 1, 0)$ of the same cell the ratios are 4.07–20.5 over the same N and both frequencies, with monotonicity ratio 0.111. On the slender cell, which is the

band the k -series actually owns, the offsets $(0, 0, 2)$, $(0, 0, 8)$ and $(1, 1, 17)$ give $x = |k|D_{\max} = 0.280, 0.299, 0.593$, monotonicity ratios $0.0198, 0.0273, 0.0595$, and bound/truth between 7.48 and 266 over $N = 8, 12, 16, 20$ and both frequencies, with no violation in any of the 44 rows. The bound therefore over-states the term count by at most three terms, which matches the wider sweep on which it was originally measured ($N_{64} - N_{\text{bnd}} \in [-3, +3]$ over 233 rows). The assembly amplification Λ is the expensive part: 68.5 at $(2, 0, 0)$ of the $\lambda/32$ cube, up to 1405 on its body diagonal, 15.4 to 23.5 at $\lambda/4$, and 2.2×10^3 to 8.1×10^4 for the slender cell.

5.7 Two radii, and which statement is about what

The Cartesian organization of the same expansion — family (a) of the registry, which sums $\sum_{\alpha} (\partial^{\alpha + \mathbf{e}_a + \mathbf{e}_b} g)(\mathbf{R}) \mu_{\alpha} / \alpha!$ — comes with a Cauchy bound whose convergence criterion is $\sqrt{2} \rho < 1$, and it is important to be precise about what that criterion is a statement about, because the two thresholds have been conflated.

Proposition 5.8 (the radius of the series). *Fix $\delta \neq 0$ and let $\varphi(t) = F_{ab}(\mathbf{R} + t\delta)$ with $F_{ab} = (\partial_a \partial_b + \delta_{ab} k^2)g$ continued to complex argument through $u = \sqrt{\mathbf{z} \cdot \mathbf{z}}$. Then φ is analytic in the disc $|t| < |\mathbf{R}|/|\delta|$ and has singularities on its boundary. Consequently the degree-graded Taylor series of F_{ab} about $\mathbf{R}$ converges absolutely at every δ with $|\delta| < |\mathbf{R}|$, and the integrated series $\sum_q \int_D w c_q(\delta)$ converges absolutely whenever $\rho = r_d/|\mathbf{R}| < 1$.*

Proof. g is singular exactly where $\mathbf{z} \cdot \mathbf{z} = 0$. Along $\mathbf{z} = \mathbf{R} + t\delta$, $\mathbf{z} \cdot \mathbf{z} = |\mathbf{R}|^2 + 2t(\mathbf{R} \cdot \delta) + t^2|\delta|^2$ is a quadratic in t whose discriminant $4(\mathbf{R} \cdot \delta)^2 - 4|\mathbf{R}|^2|\delta|^2$ is non-positive by Cauchy–Schwarz, so its two roots are complex conjugates with product $|\mathbf{R}|^2/|\delta|^2$; hence $|t_+| = |t_-| = |\mathbf{R}|/|\delta|$ exactly. The last statement follows by taking β with $\rho < \beta < 1$, applying the Cauchy estimate on $|t| = \beta/\rho\delta$ with $\rho\delta = |\delta|/|\mathbf{R}| \leq \rho$, and bounding $\sum_q \int_D w |c_q| \leq (\sup_D M) V_t^2 \sum_q (\rho/\beta)^q$, the supremum being finite because the relevant circles stay at distance $(1 - \beta)|\mathbf{R}|$ from the singular set. $\square$

Remark 5.9 ($\rho < 1$ is about the series, $\sqrt{2}\rho < 1$ is about the bound). The Cauchy bound of family (a) estimates the Taylor coefficients on a *complex ball* of radius r_d centred at $\mathbf{R}$, because that is what a multivariate Cauchy estimate needs. The distance from $\mathbf{R}$ to the null cone $\{\mathbf{z} \cdot \mathbf{z} = 0\}$ in $\mathbb{C}^3$ is $|\mathbf{R}|/\sqrt{2}$, not $|\mathbf{R}|$: with $\mathbf{w} = \mathbf{u}_r + i\mathbf{v}$ and $\mathbf{a} = \mathbf{R} + \mathbf{u}_r$, $|\mathbf{z} \cdot \mathbf{z}|^2 = (|\mathbf{a}|^2 - |\mathbf{v}|^2)^2 + 4(\mathbf{a} \cdot \mathbf{v})^2$ vanishes first at $|\mathbf{w}| = |\mathbf{R}|/\sqrt{2}$. So the ball-based bound exists only for $\sqrt{2}\rho < 1$, and it is silent — not false, silent — in the range $1/\sqrt{2} \leq \rho < 1$. Proposition 5.8 shows the series converges there anyway, because the real difference box never leaves the real directions, along which the singularity sits at $|t| = 1/\rho > 1$.

Measured (script `chkRay.jl`, 300-bit): the two roots have $||t_{\pm}| - |\mathbf{R}|/|\delta|| / (|\mathbf{R}|/|\delta|) \leq 8.5 \times 10^{-91}$ at $(2, 0, 0)$, $(2, 2, 0)$ and $(3, 0, 0)$ of the $\lambda/32$ cube, at $(2, 0, 0)$ of the $\lambda/4$ cube, and at $(0, 0, 16)$ of the slender cell. At the worst real corner $\delta = \mathbf{s}$ of the cubic $(2, 0, 0)$, where $\rho = 0.866$ and $\sqrt{2}\rho = 1.2247 > 1$ so that the Cauchy criterion fails, the Taylor coefficients on the circle $|t| = 1.097$ obey $|a_q| \leq M\beta^{-q}$ with $M = 3.53 \times 10^5$ ($|a_q| = 1418, 1616, 230, 6.43, 3.66 \times 10^{-3}$ at $q = 8, 16, 32, 64, 128$), and the partial sums converge at $t = 1$ with relative errors $0.595, 8.97, 1.879, 5.05 \times 10^{-3}, 1.75 \times 10^{-5}$ at the same q — slowly, with an observed asymptotic ratio of 0.887 approaching $\rho = 0.866$, but convergently. At $(2, 2, 0)$, where $\sqrt{2}\rho = 0.866 < 1$, the same partial sums reach 4.70×10^{-13} at $q = 64$ and 6.91×10^{-25} at $q = 128$. At the slender $(0, 0, 16)$, $\rho = 1.416 > 1$, the radius is $0.706 < 1$ and the series genuinely diverges at the corner.

Resolution of a disagreement between the two earlier analyses. One of them records $(2, 0, 0)$ as a divergent, asymptotic series that “never becomes a convergent tail”, on the evidence of a non-monotone `BigFloat` error sequence (7.9×10^{-12} at $p = 72$ falling to 2.2×10^{-15} at $p = 120$);

the other records it as convergent with an effective ratio of about 0.73, consistent with the multipole form needing $L = 94$ for 6.6×10^{-14} . Proposition 5.8 and the measurement above settle it: the series *converges* at $(2, 0, 0)$, slowly and non-monotonically, and the observed plateau is the interaction of a slow convergent tail with Float64/BigFloat rounding, not divergence. What is genuinely unavailable there is a Cauchy *bound*, which is the practical reason that offset is handed to the octant split in Section 8 rather than to a longer whole-box sum.

5.8 What is not proven

Everything above bounds a truncation. Nothing above bounds a rounding error, and no bound in this document is a proof that the Float64 evaluation of the truncated sum attains the truncated accuracy. That gap is real and it is where the remaining risk of the method sits. What exists instead is measurement, and it should be read as measurement.

- **The geometry table is not evaluable in Float64 at all.** The contraction of the integer solid-harmonic coefficients (17) against the box moments has $\sum |\text{term}|/|\text{result}| = 2.1 \times 10^5$ at $L = 20$, 5.8×10^8 at $L = 40$ and 7.31×10^{11} at $L = 56$ for the cubes, and up to 1.90×10^{12} over the twelve audit shapes — almost 12 decimal digits. This is a *measured* amplification, not a bounded one. The response is to build the table in exact integer and rational arithmetic and to contract at 256 bits, which removes the question rather than answering it; the one-off cost is 14–18 s and 0.7 GB of peak resident memory per cell shape (Section 1.3).
- **The per-offset sums are measured to be benign, not proved to be.** The l -sum has $\sum_l |t_l|/|T_{ab}| \leq 8.4$ over the whole $\lambda/32$ table and ≤ 67 at $\lambda/4$ on the axial $(1, 1)$ entry; the growth is linear in $|k\mathbf{R}|$, $c_l \approx 0.67|k\mathbf{R}|$, so it reaches 1.8 digits at $|k\mathbf{R}| = 100$. The n -sum has 3.07–4.14. The sub-box sum has 1.0–1.55. All four are measurements over the offsets tested.
- **The special functions.** The upward recurrence for h_l is measured accurate to 3×10^{-15} relative for $l \leq 40$, $|z| \in [0.3, 300]$, $\text{Im } z \in [0, 30]$ — including $z = 300 + 30i$ where $j_0/h_0 = 6 \times 10^{25}$ and the recurrence still returns h_{40} to 7.8×10^{-17} , because the contamination injected at $l = 0$ propagates as j_l/j_0 , which decreases, while $|h_l|$ increases. That mechanism is understood but not proved here. Miller’s downward recurrence returns j_l to 6.8×10^{-15} at worst. The real spherical harmonics from the normalised Legendre recursion have absolute error never above 6.2×10^{-15} over $l \leq 40$ and five directions.
- **The phase seed.** For lattice offsets $\rho^2 = \sum (n_i s_i)^2$ is exactly representable, so the only Float64 error in $e^{2\pi i f \rho}$ is $2\pi f \rho \cdot \text{ulp}(\rho)$; measured, that costs 2.15 digits at $|k\mathbf{R}| = 300$, 2.99 at 1000 and 3.35 at 3000, and a double-double phase (twelve flops) removes it, recovering 2.5 of the 3.35 digits at $|k\mathbf{R}| = 3000$. At $\lambda/4$ on a 128^3 block $|k\mathbf{R}|$ reaches about 350, so this matters and is cheap.
- **The end-to-end statement is empirical.** Against the 220-bit reference, over the 427 cached tensors of Section 9.1 — four shapes, offsets from two to 128 cells, five frequencies — the shipped library is accurate to 8.60×10^{-15} in max-norm and 9.34×10^{-15} per entry, losing at most 1.59 decimal digits to rounding (Section 9.4). Over the audit’s 60 random (shape, offset, frequency) cases the rounding error is at most $16.4 \varepsilon \max_{ab} |T_{ab}|$, with an l -sum amplification of at most 3.14×10^3 (Section 7). Those are the numbers that justify the method; they are not theorems.

6 A sharper l -truncation bound

Theorem 5.3 over-selects L because of two estimates made on the *singular* solid harmonic: the constant 17 that counts the terms produced by $\partial_a \partial_b$, measured loose by a factor 32 to 34000, and

the Cauchy–Schwarz $\sum_m |Y_{lm}| \leq (2l+1)/\sqrt{4\pi}$. This section replaces both. Two things change, and each is an exact identity rather than a sharper inequality:

1. the two derivatives are moved onto the *weight* by integration by parts. The triangle weight is an explicit piecewise-linear function whose distributional second derivative is a bounded measure; the harmonics are then never differentiated at all, and the factor $\text{hb}(l+2)/\text{hb}(l) \sim (2l+1)(2l+3)/|k\mathbf{R}|^2$ that the old bound pays for two derivatives on h_l becomes a factor $4/(s_a s_b)$;
2. the sum over m is done by *one* Cauchy–Schwarz applied to both factors at once, $|\sum_m x_m y_m| \leq \|x\|_2 \|y\|_2$, instead of a supremum over m against an ℓ^1 norm. Since $\|y\|_2 \leq \sqrt{2l+1} \|y\|_\infty$, the joint form is never worse and is typically better by $\sqrt{2l+1}$; and $\sum_m Y_{lm}(\mathbf{u})^2 = (2l+1)/4\pi$ is an equality, so no constant is given away.

The scripts are `chk0.jl` (the three structural identities), `chkA.jl` (the whole box, end to end), `chk0b.jl` and `chkB.jl` (the sub-box); the deliverable is `bound2.jl`. Raw output in `out_chk0.txt`, `out_chkA_main.txt`, `out_chkA_far.txt`, `out_chk0b.txt`, `out_chkB_main.txt`.

6.1 Which truncation is being bounded

Sections 4.4 and 5.3 use two organizations of the same sum. They are *not* the same term by term. With $\psi_{lm}(\mathbf{R}) = h_l(k|\mathbf{R}|)Y_{lm}(\hat{\mathbf{R}})$ and $\varphi_{lm}(\boldsymbol{\delta}) = j_l(k|\boldsymbol{\delta}|)Y_{lm}(\hat{\boldsymbol{\delta}})$, shell l is

$$\Sigma_l^{\text{sng}} = \sum_m [(\partial_a \partial_b + \delta_{ab} k^2) \psi_{lm}](\mathbf{R}) \int_D w \varphi_{lm}, \quad \Sigma_l^{\text{reg}} = \sum_m \psi_{lm}(\mathbf{R}) \int_D w (\partial_a \partial_b + \delta_{ab} k^2) \varphi_{lm}, \quad (41)$$

and $\Sigma_l^{\text{sng}} \neq \Sigma_l^{\text{reg}}$: the first differentiates $S_l(\mathbf{R}, \boldsymbol{\delta})$ in $\mathbf{R}$, the second in $\boldsymbol{\delta}$, and S_l is not a function of $\mathbf{R} + \boldsymbol{\delta}$. Both series converge to T_{ab} . *The library truncates the second*, because its geometry tables carry the derivatives ((19)), while Theorem 5.3 bounds the first. Measured over 116 rows (`chkA.jl`), $\max_{ab} |T - T^{(L), \text{reg}}|$ is a median 17.5 times $\max_{ab} |T - T^{(L), \text{sng}}|$ at the same L , and the L needed for 10^{-13} est is a median one shell larger for the regular organization ($L_{\text{meas}}^{\text{sng}}/L_{\text{meas}}^{\text{reg}}$ has median 0.909, range 0.75–1). The old bound is loose enough to cover the difference, so nothing computed so far is wrong; but a bound meant to *select* L should bound the sum that is actually truncated. Theorem 6.4 does; Theorem 6.7 is the sharpened form of Theorem 5.3 for the other organization.

6.2 Three exact identities

Lemma 6.1 (integration by parts onto the triangle weight). *Extend $w(\boldsymbol{\delta}) = \prod_i (s_i - |\delta_i|)_+$ by zero to $\mathbb{R}^3$. For every ϕ of class C^2 on a neighbourhood of $\bar{D}$ and every $a \neq b$, with c the remaining axis,*

$$\int_D w \partial_a \partial_b \phi \, d\boldsymbol{\delta} = \int_D \text{sign}(\delta_a) \text{sign}(\delta_b) (s_c - |\delta_c|) \phi \, d\boldsymbol{\delta}, \quad (42)$$

and for every a , writing Q_a for the rectangle in the two axes b, c ,

$$\int_D w \partial_a^2 \phi \, d\boldsymbol{\delta} = \int_{Q_a} (s_b - |\delta_b|)(s_c - |\delta_c|) [\phi|_{\delta_a=s_a} + \phi|_{\delta_a=-s_a} - 2\phi|_{\delta_a=0}] \, d\delta_b \, d\delta_c. \quad (43)$$

Proof. w is Lipschitz, vanishes on ∂D , and is a product of one-dimensional tent functions $w_i(t) = (s_i - |t|)_+$ with $w'_i = -\text{sign}(t)$ a.e. and $w''_i = \delta_{-s_i} - 2\delta_0 + \delta_{s_i}$ in $\mathcal{D}'(\mathbb{R})$. For $a \neq b$ integrate by parts once in δ_a and once in δ_b ; both boundary terms vanish because $w = 0$ on ∂D , and $w'_a w'_b = \text{sign}(\delta_a) \text{sign}(\delta_b)$ gives (42). For $a = b$ integrate by parts once in δ_a : the boundary term vanishes and $-\int w'_a \partial_a \phi = \int \text{sign}(\delta_a) \partial_a \phi = -\int_{-s_a}^0 \partial_a \phi + \int_0^{s_a} \partial_a \phi = \phi(s_a) + \phi(-s_a) - 2\phi(0)$, which is (43); equivalently it is $\langle w''_a, \phi \rangle$. $\square$

Lemma 6.2 (the two m -sums). *For every unit $\mathbf{u}$, $\sum_m Y_{lm}(\mathbf{u})^2 = (2l+1)/(4\pi)$. For every solution f of the spherical Bessel equation of order l and every $\mathbf{x} \neq 0$, with $r = |\mathbf{x}|$,*

$$\sum_m |\nabla(f_l(kr)Y_{lm}(\hat{\mathbf{x}}))|^2 = \frac{|k|^2}{4\pi} \left[l |f_{l-1}(kr)|^2 + (l+1) |f_{l+1}(kr)|^2 \right] =: \frac{|k|^2}{4\pi} \mathcal{G}_f(l). \quad (44)$$

Proof. The first is the addition theorem at coincident arguments. For the second, the gradient formula for a scalar multipole field is

$$\nabla(f_l Y_{lm}) = k \left[\sqrt{\frac{l}{2l+1}} f_{l-1} \mathbf{Y}_{l,l-1,m} - \sqrt{\frac{l+1}{2l+1}} f_{l+1} \mathbf{Y}_{l,l+1,m} \right],$$

$\mathbf{Y}_{Jl'm}$ the vector spherical harmonics, which uses only $f'_l - (l/z)f_l = -f_{l+1}$ and $f'_l + ((l+1)/z)f_l = f_{l-1}$. The matrix $M_{l'l''}(\hat{\mathbf{x}}) = \sum_m \mathbf{Y}_{Jl'm}(\hat{\mathbf{x}}) \otimes \overline{\mathbf{Y}_{Jl''m}(\hat{\mathbf{x}})}$ satisfies $M(Q\hat{\mathbf{x}}) = QM(\hat{\mathbf{x}})Q^\top$ for every rotation Q , by unitarity of the Wigner matrices; hence its trace is a rotation-invariant function of $\hat{\mathbf{x}}$, i.e. a constant, and integrating over the sphere with the orthonormality of the $\mathbf{Y}_{Jl'm}$ gives $\sum_m \mathbf{Y}_{Jl'm} \overline{\mathbf{Y}_{Jl''m}} = \delta_{l'l''} (2J+1)/(4\pi)$. The cross term therefore vanishes and $l/(2l+1) + (l+1)/(2l+1)$ recombines into (44). Passing from complex to real harmonics is a unitary change of basis inside the shell and leaves every $\sum_m |\cdot|^2$ unchanged. $\square$

Lemma 6.3 (the Hessian, in ℓ^2 over m — the replacement of the constant 17). *With f, l, r as in Lemma 6.2,*

$$\left(\sum_m \sum_{a,b} |\partial_a \partial_b (f_l(kr)Y_{lm})|^2 \right)^{1/2} \leq \frac{|k|^2}{\sqrt{4\pi}} \left[\sqrt{\frac{l \mathcal{G}_f(l-1)}{2l-1}} + \sqrt{\frac{(l+1) \mathcal{G}_f(l+1)}{2l+3}} \right] =: \Theta_f(l, r), \quad (45)$$

with equality at $l = 0$. Only f_{l-2}, f_l, f_{l+2} enter.

Proof. Apply the gradient formula once: $\partial_b(f_l Y_{lm}) = k[a_l f_{l-1}(\mathbf{Y}_{l,l-1,m})_b - b_l f_{l+1}(\mathbf{Y}_{l,l+1,m})_b]$ with $a_l = \sqrt{l/(2l+1)}$, $b_l = \sqrt{(l+1)/(2l+1)}$, and use Minkowski's inequality in $\ell^2(m, a, b)$ to separate the two channels. In the first channel, the assignment $m \mapsto ((\mathbf{Y}_{l,l-1,m})_b)_{b=1,2,3}$ is an equivariant map from the irreducible representation D^l into $D^1 \otimes D^{l-1} = D^{l-2} \oplus D^{l-1} \oplus D^l$, in which D^l occurs once; writing A for its matrix in the orthonormal bases $\{Y_{lm}\}$ and $\{\mathbf{e}_b Y_{l-1,m'}\}$, Schur's lemma gives $AA^\dagger = \lambda I$ with $\lambda(2l+1) = \text{tr}(AA^\dagger) = \sum_m \int |\mathbf{Y}_{l,l-1,m}|^2 = 2l+1$, so A is a coisometry and $A^\dagger A$ is the orthogonal projector onto that D^l component. Its partial trace over the vector index is equivariant on D^{l-1} , hence a multiple of the identity, and taking traces the multiple is $(2l+1)/(2l-1)$. Therefore

$$\sum_{m,b} |\nabla(f_{l-1}(\mathbf{Y}_{l,l-1,m})_b)|^2 = \frac{2l+1}{2l-1} \sum_{m'} |\nabla(f_{l-1} Y_{l-1,m'})|^2 = \frac{2l+1}{2l-1} \cdot \frac{|k|^2}{4\pi} \mathcal{G}_f(l-1),$$

and $a_l^2(2l+1)/(2l-1) = l/(2l-1)$. The second channel is identical with $b_l^2(2l+1)/(2l+3) = (l+1)/(2l+3)$. At $l = 0$ there is one channel, so Minkowski is an equality; direct computation of the Hessian of a radial function confirms $\Theta_f(0, r)^2 = (|k|^4/4\pi)(|f_0|^2 + 2|f_2|^2)/3$ on both sides. $\square$

Measured (chk0.j1, 400-bit, central differences at step 10^{-30}). (44) holds as an identity to the last printed digit (ratio 1.000000) in 48 rows, $f \in \{j, h\}$, four points, $l \leq 20$. (45) is never violated and the ratio true/bound is

$ k r$	$l = 1$	2	5	12	20
0.785 (h)	0.958	0.984	0.996	0.9991	0.9997
0.785 (j)	0.708	0.945	0.993	0.9989	0.9996
9.425 (h)	0.719	0.712	0.710	0.849	0.949

i.e. the replacement of the constant 17 is loose by at most 1.41 and typically by less than 1.01, against 32–34000 for the constant it replaces. (42)–(43) were checked against direct differentiation of φ_{lm} under the integral: relative difference 2.5×10^{-50} to 2.0×10^{-44} (quadrature limited) for $(1/32)^3$ and $(1/32, 1/32, 1/512)$, all six (a, b) , $l \leq 8$.

6.3 The whole box

Theorem 6.4 (*l*-truncation, derivatives on the regular side). *Let the pair be separated, $\rho < 1$, and let $T_{ab}^{(L)}$ be (21) truncated at $l \leq L$. Then for every a, b*

$$|T_{ab} - T_{ab}^{(L)}| \leq \frac{|k|}{4\pi|f|^2 V_t} \sum_{\substack{l > L \\ l \text{ even}}} (2l+1) |h_l^{(1)}(k|\mathbf{R}|)| \frac{|k|^l e^{(|k|r_d)^2/(4l+6)}}{(2l+1)!!} V_l^{ab}, \quad (46)$$

where V_l^{ab} is the exact positive geometry moment, for $l = 2p$,

$$V_l^{ab} = \int_D (s_c - |\delta_c|) |\delta|^l d\delta = \sum_{i+j+q=p} \frac{p!}{i!j!q!} \lambda_{2i}(s_a) \lambda_{2j}(s_b) \mu_{2q}(s_c), \quad a \neq b, \quad (47)$$

$$V_l^{aa} = 2 \sum_{i+j+q=p} \frac{p!}{i!j!q!} s_a^{2i} \mu_{2j}(s_b) \mu_{2q}(s_c) + 2 \sum_{j+q=p} \frac{p!}{j!q!} \mu_{2j}(s_b) \mu_{2q}(s_c) + |k|^2 W_l, \quad (48)$$

with $\lambda_n(s) = \int_{-s}^s t^n dt = 2s^{n+1}/(n+1)$, μ_n as in (7) and W_l as in (34). Every term is positive.

Proof. Because g depends on $\mathbf{R} + \delta$ only, $\partial_{R_a} \partial_{R_b} g = \partial_{\delta_a} \partial_{\delta_b} g$; inserting (14), which converges uniformly with all δ -derivatives on $\bar{D}$ when $\rho < 1$, gives $T_{ab} = \frac{ik}{f^2 V_t} \sum_{l,m} (-1)^l \psi_{lm}(\mathbf{R}) \mathcal{M}_{lm}^{ab}$ with $\mathcal{M}_{lm}^{ab} = \int_D w(\partial_a \partial_b + \delta_{ab} k^2) \varphi_{lm}$, which is $V_t \tilde{Q}_{lm}^{ab}$ of (20). Hence

$$|T_{ab} - T_{ab}^{(L)}| \leq \frac{|k|}{|f|^2 V_t} \sum_{l > L} \left(\sum_m |\psi_{lm}(\mathbf{R})|^2 \right)^{1/2} \left(\sum_m |\mathcal{M}_{lm}^{ab}|^2 \right)^{1/2},$$

and the first factor is $\sqrt{(2l+1)/4\pi} |h_l(k|\mathbf{R}|)|$ exactly by Lemma 6.2. For the second, apply Lemma 6.1 to $\phi = \varphi_{lm}$ (entire in δ), then Minkowski's integral inequality in $\ell^2(m)$, i.e. $\| \int F d\nu \|_{\ell^2(m)} \leq \int \|F\|_{\ell^2(m)} d\nu$ for the positive measures ν produced by the lemma, together with $\|\varphi_{lm}(\delta)\|_{\ell^2(m)} = \sqrt{(2l+1)/4\pi} |j_l(k|\delta|)|$:

$$\left(\sum_m |\mathcal{M}_{lm}^{ab}|^2 \right)^{1/2} \leq \sqrt{\frac{2l+1}{4\pi}} U_l^{ab},$$

$U_l^{ab} = \int_D (s_c - |\delta_c|) |j_l(k|\delta|)|$ for $a \neq b$ and, for $a = b$, $2 \int_{Q_a} w_b w_c [|j_l(\varrho_+) + |j_l(\varrho_0)|] + |k|^2 \int_D w |j_l|$ with $\varrho_+^2 = s_a^2 + \delta_b^2 + \delta_c^2$ and $\varrho_0^2 = \delta_b^2 + \delta_c^2$. Finally $|j_l(kx)| \leq (|k|x)^l e^{(|k|r_d)^2/(4l+6)}/(2l+1)!!$ pointwise (Theorem 5.1, using $x \leq r_d$ on $\bar{D}$ and on both sections), and integrating $x^l = |\delta|^{2p}$ against those measures multinomially gives (47)–(48). Odd l is omitted because $\mathcal{M}_{lm}^{ab} = 0$ there (Section 4.5). $\square$

Remark 6.5 (why this is so much smaller). Term by term the ratio of (46) to (33) is

$$\frac{|h_l(k\mathbf{R})| V_l^{ab}}{17 |k|^2 \sqrt{2l+5} W_l \text{hb}(l+2, k\mathbf{R})} \approx \frac{4 \sigma (|\mathbf{R}|/s)^2}{23 \sqrt{2l+5} (2l+1)(2l+3)},$$

using $V_0^{ab}/W_0 = 4/(s_a s_b)$ and $|h_l|/\text{hb}(l+2) \approx |k\mathbf{R}|^2/((2l+1)(2l+3) \cdot 1.35)$ at small argument, σ an $O(1)$ moment spread. Two derivatives cost $\sim (2l+1)^2/|k\mathbf{R}|^2$ on the singular side and $\sim 4/(s_a s_b)$ on the regular side; the regular side wins whenever $l \gtrsim |\mathbf{R}|/s$, which is the whole range in which the expansion is used, since $L \approx 13/\log_{10}(1/(0.7\rho))$ while $|\mathbf{R}|/s = \sqrt{3}/\rho$.

6.4 One affine-weight sub-box

On a sub-box the weight does not vanish on the boundary, so integration by parts leaves boundary terms; on the diagonal it leaves one first derivative, which Lemma 6.2 controls exactly.

Theorem 6.6 (*l-truncation on a sub-box*). *Let $D_j = \prod_i [-h_i, h_i]$ carry the weight $u(\delta') = \prod_i u_i(\delta'_i)$, $u_i(t) = \alpha_i + \beta_i t \geq 0$, and let $\mathbf{V} = \mathbf{R} + \mathbf{c}_j$, $r_j = |\mathbf{h}|$, $u_i^\pm = u_i(\pm h_i) \geq 0$. Then, with all l contributing,*

$$|T_{ab}^{(j)} - T_{ab}^{(j),L}| \leq \frac{|k|}{4\pi|f|^2V_t} \sum_{l>L} (2l+1) |h_l^{(1)}(k|\mathbf{V})| \Xi_l^{ab}, \quad (49)$$

where, with $j(q) = |k|^q e^{(|k|r_j)^2/(4q+6)}/(2q+1)!!$ and $N[\cdot]$ the $|\delta'|^q$ -moment of the indicated product measure (odd q by Cauchy-Schwarz, $N_q \leq \sqrt{N_{q-1}N_{q+1}}$),

$$\Xi_l^{ab} = j(l) N_l[\nu_a \otimes \nu_b \otimes u_c], \quad a \neq b, \quad (50)$$

$$\begin{aligned} \Xi_l^{aa} = & \sqrt{\frac{l}{2l+1}} j(l-1) |k| N_{l-1}[\pi_a \otimes u_b \otimes u_c] + \sqrt{\frac{l+1}{2l+1}} j(l+1) |k| N_{l+1}[\pi_a \otimes u_b \otimes u_c] \\ & + j(l) N_l[\beta_a \otimes u_b \otimes u_c] + |k|^2 j(l) N_l[u_a \otimes u_b \otimes u_c], \end{aligned} \quad (51)$$

and the one-dimensional positive measures are $\nu_i = u_i^+ \delta_{h_i} + u_i^- \delta_{-h_i} + |\beta_i| dt$, $\pi_i = u_i^+ \delta_{h_i} + u_i^- \delta_{-h_i}$, $\beta_i = |\beta_i|(\delta_{h_i} + \delta_{-h_i})$.

Proof. The one-dimensional identities are $\int_{-h}^h u X' = u^+ X(h) - u^- X(-h) - \beta \int_{-h}^h X$ and $\int_{-h}^h u X'' = u^+ X'(h) - u^- X'(-h) - \beta(X(h) - X(-h))$. For $a \neq b$ apply the first in axis a and in axis b and keep u_c in axis c ; taking absolute values term by term replaces each one-dimensional functional by the positive measure ν_i , and Minkowski in $\ell^2(m)$ with $\|\varphi_{lm}\|_{\ell^2(m)} = \sqrt{(2l+1)/4\pi}|j_l|$ gives the first line. For $a = b$ apply the second identity in axis a : the two surviving derivative evaluations are bounded by $\|\nabla \varphi_{lm}\|$, and Lemma 6.2 with $|j_{l\mp 1}| \leq j(l \mp 1)|\delta'|^{l\mp 1}$ and $\sqrt{A+B} \leq \sqrt{A} + \sqrt{B}$ gives the first two terms; the β_a term and the $\delta_{ab} k^2$ term give the last two. Every measure is positive, so every moment is an exact positive multinomial sum in the one-dimensional moments $\int |t|^{2i} d\nu$, and $\rho_j < 1$ makes the interchange of sum and integral legitimate. For the octant split $u_i^+ = 0$ and $u_i^- = s_i$, so only the inner face carries a derivative. $\square$

6.5 The singular organization, with the constant 17 removed

Theorem 6.7 (sharpened form of Theorem 5.3). *With $T^{(L),\text{sng}}$ the truncation of (22) and $\Theta_h(l, |\mathbf{R}|)$ of Lemma 6.3 built on $f = h^{(1)}$,*

$$|T_{ab} - T_{ab}^{(L),\text{sng}}| \leq \frac{|k|}{|f|^2V_t} \sum_{\substack{l>L \\ l \text{ even}}} \left[\Theta_h(l, |\mathbf{R}|) + \delta_{ab} |k|^2 \sqrt{\frac{2l+1}{4\pi}} |h_l| \right] \sqrt{\frac{2l+1}{4\pi}} \frac{|k|^l e^{(|k|r_d)^2/(4l+6)}}{(2l+1)!!} W_l. \quad (52)$$

Proof. As in Theorem 6.4 but with the roles exchanged: Cauchy-Schwarz over m between $((\partial_a \partial_b + \delta_{ab} k^2) \psi_{lm})_m$ and $(\mathcal{I}_{lm})_m$; the first ℓ^2 norm is at most $\Theta_h + \delta_{ab} |k|^2 (\sum_m |\psi_{lm}|^2)^{1/2}$ by Minkowski and Lemma 6.3, the second is at most $\sqrt{(2l+1)/4\pi} \int_D w |j_l|$ by Minkowski and Lemma 6.2. $\square$

Against (33) the term-by-term ratio is $\Theta_h \sqrt{(2l+1)/4\pi} / (17|k|^2 \text{hb}(l+2) \sqrt{(2l+5)/4\pi} (2l+1)/\sqrt{4\pi}) \approx 1/(34\sqrt{2l} \cdot 1.35)$, i.e. the improvement grows like $\sqrt{l}$: the old bound loses a factor $\sqrt{2l+5}$ that the exact m -sum does not.

6.6 Two routes that do not help, and one that helps a little

A Cauchy estimate in one complex variable. Along $\mathbf{R} + t\delta$ the kernel is analytic for $|t| < |\mathbf{R}|/|\delta|$ (Proposition 5.8) and a Cauchy estimate bounds the tail of its *Taylor* series in t . That is the degree-graded truncation of family (a), not the l -truncation: shell l contains the degrees $l, l+2, l+4, \dots$ and degree q contains the shells $q, q-2, \dots$, so neither tail majorises the other. The estimate therefore cannot be substituted here, and where it does apply it is 2.0–7.7 times pessimistic in the selected degree.

Exact Bessel magnitudes instead of majorants. Both $|h_l|$ and $|j_l|$ are computable to full precision at negligible cost, and using them in place of the majorants of Theorems 5.1 and 5.2 keeps the bound rigorous up to the rounding of an elementary function. Measured (`chkA.jl`, 29 cases, tails at $L = 32$): replacing $|h_l|$ by $\text{hb}(l, k\mathbf{R})$ inflates the bound by 1.48 at $|k\mathbf{R}| = 0.39$, 2.17 at 0.79, 4.6 at 1.6, 13.6 at 2.7, 20 at 3.1, 297 at 6.3, 2.7×10^4 at 12.6 and 2.3×10^6 at 21.8 — the regime $|k\mathbf{R}| \gtrsim l$, where the triangle inequality on the finite Hankel sum gives away the oscillation — but costs only a median 8% (at most 38%) in the selected L , because L is set by ρ and a constant factor moves it by a few shells. Replacing the exponential majorant of j_l by the termwise-absolute power series $\sum_n |k|^{l+2n} V_{l+2n} / (2^n n! (2l+2n+1)!!)$ gains between 0.002% and 2.2% and never changes the selected L : the j_l majorant is not where the looseness is.

6.7 Measured

Truth: $\max_{ab} |T_{ab} - T_{ab}^{(L)}|$ for the organization named, with T_{ab} from an independent **BigFloat** volume Gauss–Legendre rule on the eight octants of D (order 22; agreement with the shell sums at $L = 32$ is 10^{-18} to 10^{-57} relative) and the shell data from a 24-point positive-octant rule with exact parity factors. 116 rows: four cell shapes, offsets from 2 to 64 cells, $f \in \{1, 1 + 0.1i, 1 + i\}$, $L \in \{8, 16, 24, 32\}$; 320-bit, and 512-bit with a 10^{-38} difference step for the far offsets, where the central-difference evaluation of $(\partial_a \partial_b + \delta_{ab} k^2) \psi_{lm}$ has a floor. No violation.

bound / truth	min	median	max
(46), regular organization	8.79	49.1	2.93×10^3
(52), singular organization	1.49	140	792
(33), the first-round bound	1.35×10^3	1.53×10^5	1.73×10^{11}
gain (33)/(46)	31.3	3.23×10^3	4.61×10^9
gain (33)/(52)	162	661	7.21×10^8

Cost: V_l^{ab} depends on the cell shape alone once the $|k|^2 W_l$ piece is split off, so it is tabulated once per shape (31 ms to $l = 140$) and the per-offset work is the h_l recurrence and one length-70 dot product, measured at 3.8, 3.9 and $9.7 \mu\text{s}$ at $L = 8, 16, 32$ on a machine that was not quiet.

The largest ratios of (46) are not looseness of the bound but dips of the truth: the true remainder oscillates in L (the l -sum alternates with the phase of h_l), and no monotone majorant can follow a dip. At the L that matters — the one the bound selects — the ratio is 9–100.

shape	$\mathbf{n}$	ρ	$ k\mathbf{R} $	L_{meas}	L from (46)	L from (33)	cost ratio
$(1/32)^3$	(2, 0, 0)	0.866	0.39	> 32	126	> 136	–
$(1/32)^3$	(3, 0, 0)	0.577	0.59	32	40	50	1.56
$(1/32)^3$	(4, 0, 0)	0.433	0.79	24	28	34	1.47
$(1/32)^3$	(3, 1, 0)	0.548	0.62	32	36	46	1.63
$(1/32)^3$	(8, 0, 0)	0.217	1.57	16	16	20	1.56
$(1/32)^3$	(8, 8, 8)	0.125	2.72	12	12	14	1.36
$(1/32)^3$	(2, 2, 0)	0.612	0.56	> 32	44	56	1.62
$(1/8)^3$	(4, 0, 0)	0.433	3.14	22	26	36	1.92
$(1/8)^3$	(3, 1, 0)	0.548	2.50	28	36	48	1.78
$(1/4)^3$	(4, 0, 0)	0.433	6.28	22	26	38	2.14
$(1/4)^3$	(8, 0, 0)	0.217	12.57	14	16	24	2.25
$(1/4)^3$	(8, 8, 8)	0.125	21.77	14	16	20	1.56
sl	(4, 0, 0)	0.354	0.79	22	26	30	1.33
sl	(0, 0, 64)	0.354	0.79	24	26	30	1.33
sl	(0, 0, 32)	0.708	0.39	> 32	70	90	1.65

L_{meas} is the smallest even L with $\max_{ab} |T - T^{(L), \text{reg}}| \leq 10^{-13} \text{est}(\mathbf{R})$; “cost ratio” is $(L_{\text{old}}/L_{\text{new}})^2$, the saving in the $O(L^2)$ per-offset work. Over 25 cases with all three defined, $L_{\text{old}}/L_{\text{new}}$ is 1.00–1.50 with median 1.21 (cost 1.00–2.25, median 1.47) and the new bound over-selects by $L_{\text{new}}/L_{\text{meas}} = 1.00$ –1.29, median 1.14, against 1.17–1.73, median 1.40, for the bound it replaces. At (2, 0, 0) the old bound gives no certificate below $L = 136$ while (46) certifies $L = 126$.

6.8 What is still loose

The remaining factor of 9–100 is, in order of size: (i) the Cauchy–Schwarz over m , which is an equality only when $\mathcal{M}_{lm}$ is proportional to $Y_{lm}(\hat{\mathbf{R}})$ — for the whole box only one parity class per entry is non-zero, so a factor between 1 and $\sqrt{\#\{m \text{ surviving}\}}$ is given away and is not recovered by any argument that does not know the geometry table; (ii) Minkowski under the δ integral, which discards the cancellation of Y_{lm} over the box — the same loss the W_l moment already minimises for the radial part; (iii) the absolute values across l , worth the measured $\sum_l |t_l|/|T| \leq 8.4$ at $\lambda/32$ and ≤ 67 at $\lambda/4$; (iv) 1.00–1.41 for Lemma 6.3 and 1.0006–1.5 for the j_l majorant. Nothing here bounds a rounding error, and (46) uses $|h_l|$ from the upward recurrence, which is a measurement (3×10^{-15} for $l \leq 40$) and not a theorem; the variant with $\text{hb}(l, k\mathbf{R})$ is fully certified and costs a median 8% in L .

7 What the bound certifies, and what it does not

Theorem A (Theorem 6.4) and Theorem B (Theorem 6.6) bound a truncation. They do not bound a rounding error, and no bound in this document does. This section separates the two experimentally, on the hardest sample available, and states the one certificate that holds with no hypothesis at all.

7.1 Sixty random cases, and the classification of every violation

The sample is the adversarial audit’s: sixty cases drawn with a fixed seed, the shape uniform over twelve shapes of aspect ratio 1 to 1020 plus $(1/32)^3$ and $(1/4)^3$ (thirteen of the fourteen were drawn), the offset uniform in $[-20, 20]^3$ with $\max |\mathbf{D}| \geq 2$, and the frequency uniform over $\{1, 1 + 0.1i, 0.37, 2 + 0.2i, 1 + i\}$. The reference is an independent 220-bit graded Gauss–Legendre volume rule — no addition theorem, no moments, no face pairs, no shared code with the library — whose own convergence over these shapes is a median 6.0×10^{-50} . *bound* is the library’s own

tail at the L it actually selected, *actual* is $\max_{ab} |G - G_{\text{ref}}|$, *flr* is $\varepsilon \max_{ab} |G_{\text{ref}}|$, and *amp* is $\max_{ab} \sum_l |\text{term}|/|\text{sum}|$ of the l -sum the library performs.

over the 60 cases	min	median	max	violations (< 1)
bound / actual, shipped library	0.118	4.02	54.5	15
bound / actual, the audited version	9.96×10^{-10}	7.48	419	19
actual / $\varepsilon \max G_{\text{ref}} $	0.636	2.3	16.4	
$\max_{ab} G - G_{\text{ref}} / \max G_{\text{ref}} $			3.65×10^{-15}	
l -sum amplification $\sum \text{term} / \text{sum} $	1.27	3.7	3.14×10^3	

All fifteen remaining violations are the rounding floor. None is a truncation violation.

In every one of them the Theorem A bound lies *below* the **Float64** evaluation floor of the answer — bound 1.8×10^{-22} to 1.8×10^{-19} against $\varepsilon \max |G_{\text{ref}}| = 2 \times 10^{-22}$ to 3×10^{-19} — so what it fails to cover is rounding, not truncation. Their errors run from 0.964 to 15.6 times $\varepsilon \max |G_{\text{ref}}|$ and their l -sum amplifications from 1.78 to 388:

shape	D	f	L	bnd/act	act/flr	amp
r_3	$(-14, 13, -5)$	1	28	0.717	15.6	388
r_3	$(12, 9, -10)$	$1 + i$	32	0.122	10.5	45.1
r_2	$(17, 19, 15)$	$1 + i$	24	0.840	6.09	36.2
pl_8	$(-17, -11, -7)$	1	16	0.293	5.56	3.50
sl_X	$(-2, 13, 1)$	$1 + i$	16	0.635	4.53	2.79
pl_8	$(20, 14, -8)$	0.37	14	0.118	2.82	2.70
r_1	$(7, -19, 10)$	$2 + 0.2i$	16	0.119	2.76	14.5
<i>(eight further rows at act/flr ≤ 2.8, amp ≤ 6.9)</i>						

For comparison, the same sixty cases on the version this one replaces had nine *genuine* failures — the answer wrong, not the bound loose — with *bound / actual* down to 9.96×10^{-10} and errors up to 1.38×10^{-4} in max-norm, and four more where the answer was right to 10^{-13} but the rounding error reached 396ε . Both classes are now empty: r_3 at $(18, -14, -10)$ went from 2.43×10^{-3} to 54.5, r_3 at $(-14, 13, -5)$ from 9.60×10^{-3} to 0.717, the $\lambda/128$ cube at $(-18, 7, -13)$ from 3.86×10^{-2} to 11.5.

7.2 The statement this document carries

The truncation error is certified by the bound: over the sixty cases no violation is a truncation violation, and the largest truncation-side ratio is bound/actual = 54.5. The rounding error is measured, not bounded, at no more than $X\varepsilon$ times the largest tensor entry with

$$X = 16.4,$$

attained on an aspect-87.7 cell at $f = 2 + 0.2i$ whose l -sum amplification is 3.14×10^3 .

The corresponding a posteriori rule is $\text{bound} + 20\varepsilon \max_{ab} |T_{ab}|$, which no row of the sixty exceeds. (The audited version needed bound + several hundred ε and still failed on thirteen rows.)

7.3 The certificate that needs no hypothesis

Everything above compares the bound against $\text{tol} \cdot \text{est}(\mathbf{R})$, and est is an a priori scale, not a bound on $\max_{ab} |T_{ab}|$ (Section 14). After the tensor has been evaluated that hypothesis can be discharged,

because both the bound and the answer are in hand:

$$\frac{\max_{ab} |T_{ab} - T_{ab}^{(L)}|}{\max_{ab} |T_{ab}|} \leq \frac{\text{bnd}}{\max_{ab} |T_{ab}^{(L)}| - \text{bnd}}, \quad (53)$$

which follows from the triangle inequality alone and costs one already-computed number. Over the 427 cached references (53) evaluates to at most 4.5×10^{-14} . **That is the honest guarantee: a certified max-norm relative truncation error of 4.5×10^{-14} , against a measured one of 8.6×10^{-15} , with the rounding allowance of the previous subsection on top.**

A second, proven lower scale exists and is one keyword away. Away from the source $\nabla^2 g = -k^2 g$, so $\text{tr } T = 2k^2 S$ with $S = (1/V_t) \int_D w g(\mathbf{R} + \boldsymbol{\delta}) d\boldsymbol{\delta}$, hence $\max_{ab} |T_{ab}| \geq |\text{tr } T|/3$, and $|S|$ is bounded below by its $l = 0$ shell minus the Theorem-A tail with V_l^{ab} replaced by W_l — everything already in the shape table, $O(L)$ per offset, and a function of $|\mathbf{R}|$ alone, so the threshold table survives. Measured, $\text{est}_{\text{lo}}/\max |G_{\text{ref}}|$ reaches 0.98, 0.63 and 0.71 on the $\lambda/32$, $\lambda/8$ and slender shapes, against $\text{est}/\max |G_{\text{ref}}| \in [1.10, 6.34]$ above; it costs 4 to 13 % in terms where it applies. It is *vacuous* on a $\lambda/4$ cell, where the $l \geq 2$ tail of the scalar series exceeds its $l = 0$ shell (the ratio $|k|^2 r_d^2 / (18 j_0(|k| r_d))$ is 0.0065, 0.14 and 0.0009 at $\lambda/32$, $\lambda/8$ and the slender cell, and 2.86 at $\lambda/4$): the trace is direction-free precisely because it discards the anisotropy, and at small $k\mathbf{R}$ the anisotropic part is the whole of T . So a proven, direction-free, $O(1)$ -per-offset scale exists for cells up to about $\lambda/8$ across the diagonal and does not exist by this argument at $\lambda/4$. The default remains *est*, with (53) as the universal fallback.

8 Band selection: which method where, from the bounds

The bounds of Sections 5 and 6 are enough to decide, for every relative offset of every cell shape, which method computes that offset and with how many terms, before any value is computed. This section states the rule, gives the cost model it minimises, and tabulates the band edges the bounds produce. Nothing here is fitted to a measurement; measurement enters only through the cost model and through the term $\text{cap } L_{\text{max}}$, and both are labelled where they do.

Two sets of band-edge numbers appear below and it is worth separating them once. Section 8.3 tabulates the output of `bands.jl`, which evaluates the *first-round* bounds (33) and (37) at $\text{tol} = 10^{-13}$ with a cap $L_{\text{max}} = 64$; it is what established, before the library existed, that L is governed by ρ and not by $k\mathbf{R}$ and that exactly twelve offsets per octant need subdivision. Section 8.4 then gives the edges the *shipped* selector produces, from Theorem A and Theorem B of Section 6 at $\text{tol} = 10^{-14}$ with $L_{\text{max}} = 56$. The two agree on every qualitative conclusion and differ in L by -17% to $+17\%$.

8.1 The three methods and their costs

W — whole box. Equation (21) with $L = L_{\text{wb}}$ from Theorem 5.3. Measured cost 391, 1250, 2731, 4972 ns at $L = 10, 20, 30, 40$, i.e.

$$c_W(L) \approx 3.1 L^2 + 80 \text{ ns}, \quad (54)$$

about 3 ns per (l, m) slot at roughly 1.7 complex multiplies each, with no allocation.

O — octant split. Equation (28) over eight sub-boxes with L_j from (37). Seven complex multiplies per (l, m) instead of 1.7, so at the same measured 3 ns per (l, m) per 1.7 multiplies,

$$c_O(\{L_j\}) \approx 12.2 \sum_{j=1}^8 (L_j + 1)^2 + 640 \text{ ns}. \quad (55)$$

K — k -series. The moment series of the companion document on the thirty-six face pairs, with N from Theorem 5.7. Cost is dominated by the moments and is three to four orders larger per offset: 32 ms at two cells, $\lambda/32$, in **Float64**, and about 0.2 s per offset in 128-bit arithmetic, which is what the amplification Λ forces on the slender cell. The moments are frequency independent, so on a fixed shape this is a one-off per offset.

For reference, the rule being replaced — thirty-six face-pair integrals by the fixed Gauss–Legendre schedule of **quad0rd** — costs 13.9 ms per offset at two cells, $\lambda/32$, and 0.55 ms at thirty-two cells, and 23.3 s for a 32^3 block.

Caveat on all timings. Every microsecond figure quoted here was measured while several other processes were running on the same machine. Relative comparisons within one run are sound; absolute per-offset costs must be re-measured on a quiet machine before they are used for anything but ordering the methods.

Since $c_W(L) < c_O(\{L_j\})$ whenever $L \lesssim 2 \max_j L_j$, and c_K exceeds both by three orders, the cost ordering is $W < O \ll K$ at every offset where more than one of them applies. The selection rule is therefore a cascade.

8.2 The rule

Take the cheapest method whose *proven* bound reaches $\text{tol} \cdot \text{est}(\mathbf{R})$ within the term cap $L_{\max}$:

- (1) $L_{\text{wb}}(\mathbf{R}) \leq L_{\max} \implies W$ at order L_{wb} ;
- (2) else $\max_j L_j(\mathbf{R}) \leq L_{\max} \implies O$ with per-sub-box orders L_j ;
- (3) else K at order N from Theorem 5.7,

(56)

with N per shell from Theorem 5.5 in cases (1) and (2) (Remark 5.6). In the shipped library L_{wb} comes from Theorem A (Theorem 6.4) and L_j from Theorem B (Theorem 6.6), with

$\text{tol} = 10^{-14}$	TOL in farfield.jl
$L_{\max} = 56$	LMAX
$N_{\text{cap}} = 40$	NCAP; the call errors above it

Why $\text{tol} = 10^{-14}$ and not 10^{-13} . The criterion is $\text{bound} \leq \text{tol} \cdot \text{est}$, and est over-states $\max_{ab} |T_{ab}|$ by a measured factor 1.10 to 6.34 on the cubes (Section 14), so what the rule certifies is a relative error of $\text{tol} \cdot \text{est} / \max |T|$. With the first-round bound, loose by a median 1.5×10^5 , that slack was hidden; with Theorem A, loose by a median 49, it is not. Measured on the same references, $\text{tol} = 10^{-13}$ delivers 1.6×10^{-13} per entry and $\text{tol} = 10^{-14}$ delivers 9.3×10^{-15} , for +0.7% in terms. The tolerance was tightened because the bound was sharpened.

$L_{\max}$ is the one place where measurement, not proof, enters the rule, and the justification is threefold, each part a measurement: the exact geometry table costs 14–18 s per cell shape at $L \approx 50$ and grows steeply beyond; the whole-box form was measured to reach 6.6×10^{-14} at $L = 94$ at the worst offset and *not* to improve to 10^{-15} by $L = 100$, so a whole-box sum of order ~ 100 has a **Float64** floor above the target; and the octant form reaches 4.4×10^{-15} with $L_j \leq 52$ at exactly those offsets. A cap of 56 sits above every L_j the sub-box bound requires on the shapes Gila uses and below the region where the whole-box route stops paying.

8.3 Band edges for the four shapes, from the first-round bounds

L_{wb} is the whole-box cap from (33), even l only; $L_{\text{oct}} = \max_j L_j$ from (37) over the eight octant sub-boxes; costs from (54) and (55) in ns; $f = 1$. Method by (56) at $\text{tol} = 10^{-13}$ with $L_{\text{max}} = 64$. The shipped selector's own numbers, from Theorems A and B at 10^{-14} , are in Section 8.4.

shape	direction	n	ρ	$ k\mathbf{R} $	L_{wb}	c_W	L_{oct}	c_O	method
$(1/32)^3$	axis	2	0.8660	0.393	174	93936	55	203648	O
$(1/32)^3$	axis	3	0.5774	0.589	50	7830	32	84283	W
$(1/32)^3$	axis	4	0.4330	0.785	34	3664	25	57248	W
$(1/32)^3$	axis	6	0.2887	1.178	24	1866	19	35971	W
$(1/32)^3$	axis	8	0.2165	1.571	20	1320	16	27236	W
$(1/32)^3$	axis	16	0.1083	3.142	14	688	12	17134	W
$(1/32)^3$	axis	32	0.0541	6.283	12	526	10	12450	W
$(1/32)^3$	axis	64	0.0271	12.566	10	390	9	10400	W
$(1/32)^3$	face diag	2	0.6124	0.555	56	9802	39	95971	W
$(1/32)^3$	face diag	4	0.3062	1.111	24	1866	20	36923	W
$(1/32)^3$	face diag	8	0.1531	2.221	16	874	14	21892	W
$(1/32)^3$	body diag	2	0.5000	0.680	40	5040	32	67594	W
$(1/32)^3$	body diag	4	0.2500	1.360	22	1580	18	31006	W
$(1/32)^3$	body diag	8	0.1250	2.721	14	688	13	19440	W
$(1/8)^3$	axis	2	0.8660	1.571	176	96106	55	206820	O
$(1/8)^3$	axis	3	0.5774	2.356	52	8462	32	86772	W
$(1/8)^3$	axis	4	0.4330	3.142	36	4098	25	57248	W
$(1/8)^3$	axis	8	0.2165	6.283	22	1580	17	30554	W
$(1/8)^3$	axis	32	0.0541	25.13	14	688	12	17134	W
$(1/8)^3$	face diag	2	0.6124	2.221	58	10508	39	100094	W
$(1/8)^3$	body diag	2	0.5000	2.721	42	5548	33	72766	W
$(1/4)^3$	axis	2	0.8660	3.142	182	102764	56	215604	O
$(1/4)^3$	axis	3	0.5774	4.712	54	9120	33	92628	W
$(1/4)^3$	axis	4	0.4330	6.283	38	4556	26	64324	W
$(1/4)^3$	axis	8	0.2165	12.57	24	1866	19	37777	W
$(1/4)^3$	axis	32	0.0541	50.27	18	1084	15	25626	W
$(1/4)^3$	axis	64	0.0271	100.5	16	874	14	22600	W
$(1/4)^3$	face diag	2	0.6124	4.443	60	11240	40	106341	W
$(1/4)^3$	body diag	2	0.5000	5.441	44	6082	34	78134	W
slender	long axis	2	0.7078	0.393	90	25190	47	154116	O
slender	long axis	3	0.4719	0.589	42	5548	29	70375	W
slender	long axis	4	0.3539	0.785	30	2870	23	48269	W
slender	long axis	8	0.1769	1.571	18	1084	15	24113	W
slender	long axis	32	0.0442	6.283	10	390	10	12450	W
slender	face diag	2	0.5005	0.555	46	6640	34	76426	W
slender	body diag	2	0.5000	0.556	46	6640	34	76426	W
slender	short axis	2	11.325	0.025	–	–	–	–	K
slender	short axis	4	5.6624	0.049	–	–	–	–	K
slender	short axis	8	2.8312	0.098	–	–	207	3489254	K
slender	short axis	16	1.4156	0.196	–	–	70	459214	K
slender	short axis	32	0.7078	0.393	90	25190	34	116833	O
slender	short axis	64	0.3539	0.785	30	2870	21	45780	W

A dash means the bound provides no certificate at all: the term sequence of (33) does not decrease, which for the short axis of the slender cell is not a defect of the bound but the truth, since $\rho > 1$ there and the expansion diverges. Three readings of the table are worth stating.

- **L is set by ρ , not by $k\mathbf{R}$.** At a fixed offset in cells the selected L moves by at most 4 from $\lambda/32$ to $\lambda/4$ even though $|k\mathbf{R}|$ changes by a factor eight. Frequency enters the *cost* only through

$n_{\max}$ and through the cancellation of the l -sum, not through L .

- **The far plateau is cheap and flat.** Beyond about eight cells L settles at 10–16 for the cubes and the whole-box evaluation costs under 2 μ s per offset.
- **Only the whole box ever wins on cost**, at every offset where its bound applies. Subdivision is not a cost optimisation; it is the repair for the offsets where the whole-box bound needs an L that Float64 cannot deliver.

8.4 The band edges the shipped selector produces

The same quantities from Theorem A and Theorem B at $\text{tol} = 10^{-14}$, $L_{\max} = 56$, as `verify.jl` part (e) prints them (`tables/e_terms.txt`, 192 rows; “cost” is the number of complex multiply-adds the route issues, -1 means the bound is not met at $L \leq 56$). `route` is the method the library actually takes.

shape	dir	n	ρ	$ k\mathbf{R} $	L_{wb}	cost	L_{oct}	cost	route
c32	axis	2	0.866	0.393	−1	−1	50	98028	(ii)
c32	axis	3	0.5774	0.589	44	3491	30	43036	(i)
c32	axis	4	0.433	0.785	32	1892	24	29848	(i)
c32	axis	6	0.2887	1.178	22	931	19	20272	(i)
c32	axis	8	0.2165	1.571	18	641	16	16184	(i)
c32	axis	16	0.1083	3.142	14	405	13	10976	(i)
c32	axis	32	0.0541	6.283	12	307	11	8064	(i)
c32	axis	64	0.0271	12.566	10	223	10	6776	(i)
c32	face	2	0.6124	0.555	48	4132	36	49182	(i)
c32	face	3	0.4082	0.833	30	1673	24	27902	(i)
c32	face	4	0.3062	1.111	24	1096	20	20818	(i)
c32	face	6	0.2041	1.666	18	641	16	15722	(i)
c32	face	8	0.1531	2.221	16	516	14	12600	(i)
c32	face	16	0.0765	4.443	12	307	12	9464	(i)
c32	face	32	0.0383	8.886	10	223	11	8064	(i)
c32	face	64	0.0191	17.772	10	223	10	6776	(i)
c32	body	2	0.5	0.68	36	2371	31	36848	(i)
c32	body	3	0.3333	1.02	24	1096	21	23576	(i)
c32	body	4	0.25	1.36	20	779	18	18158	(i)
c32	body	6	0.1667	2.041	16	516	15	13468	(i)
c32	body	8	0.125	2.721	14	405	14	11179	(i)
c32	body	16	0.0625	5.441	12	307	11	8064	(i)
c32	body	32	0.0312	10.883	10	223	10	6776	(i)
c32	body	64	0.0156	21.766	10	223	10	6776	(i)
c8	axis	2	0.866	1.571	−1	−1	50	99736	(ii)
c8	axis	3	0.5774	2.356	46	3805	31	46172	(i)
c8	axis	4	0.433	3.142	32	1892	24	31052	(i)
c8	axis	6	0.2887	4.712	24	1096	20	22456	(i)
c8	axis	8	0.2165	6.283	20	779	17	18144	(i)
c8	axis	16	0.1083	12.566	16	516	15	13468	(i)
c8	axis	32	0.0541	25.133	14	405	13	10976	(i)
c8	axis	64	0.0271	50.265	14	405	13	10976	(i)
c8	face	2	0.6124	2.221	50	4473	36	51464	(i)
c8	face	3	0.4082	3.332	30	1673	25	30450	(i)
c8	face	4	0.3062	4.443	24	1096	21	23030	(i)
c8	face	6	0.2041	6.664	20	779	17	17654	(i)
c8	face	8	0.1531	8.886	18	641	16	14798	(i)
c8	face	16	0.0765	17.772	16	516	14	12600	(i)
c8	face	32	0.0383	35.543	14	405	13	10976	(i)

shape	dir	n	ρ	$ k\mathbf{R} $	L_{wb}	cost	L_{oct}	cost	route
c8	face	64	0.0191	71.086	14	405	13	10976	(i)
c8	body	2	0.5	2.721	38	2631	31	38276	(i)
c8	body	3	0.3333	4.081	26	1275	22	24164	(i)
c8	body	4	0.25	5.441	22	931	19	20230	(i)
c8	body	6	0.1667	8.162	18	641	16	15953	(i)
c8	body	8	0.125	10.883	16	516	15	14336	(i)
c8	body	16	0.0625	21.766	14	405	14	11179	(i)
c8	body	32	0.0312	43.531	14	405	13	10976	(i)
c8	body	64	0.0156	87.062	14	405	13	10976	(i)
c4	axis	2	0.866	3.142	-1	-1	51	104384	(ii)
c4	axis	3	0.5774	4.712	48	4132	32	49420	(i)
c4	axis	4	0.433	6.283	34	2125	26	35224	(i)
c4	axis	6	0.2887	9.425	26	1275	21	25900	(i)
c4	axis	8	0.2165	12.566	22	931	19	21308	(i)
c4	axis	16	0.1083	25.133	18	641	17	17164	(i)
c4	axis	32	0.0541	50.265	18	641	16	15260	(i)
c4	axis	64	0.0271	100.531	16	516	15	14336	(i)
c4	face	2	0.6124	4.443	52	4827	37	54880	(i)
c4	face	3	0.4082	6.664	32	1892	26	33740	(i)
c4	face	4	0.3062	8.886	26	1275	22	27132	(i)
c4	face	6	0.2041	13.329	22	931	19	20762	(i)
c4	face	8	0.1531	17.772	20	779	18	18662	(i)
c4	face	16	0.0765	35.543	18	641	16	16184	(i)
c4	face	32	0.0383	71.086	18	641	15	14336	(i)
c4	face	64	0.0191	142.172	16	516	15	14336	(i)
c4	body	2	0.5	5.441	40	2904	32	41244	(i)
c4	body	3	0.3333	8.162	28	1467	23	28392	(i)
c4	body	4	0.25	10.883	24	1096	20	23548	(i)
c4	body	6	0.1667	16.324	20	779	18	19180	(i)
c4	body	8	0.125	21.766	20	779	17	18144	(i)
c4	body	16	0.0625	43.531	18	641	16	16184	(i)
c4	body	32	0.0312	87.062	16	516	15	14336	(i)
c4	body	64	0.0156	174.125	16	516	15	14336	(i)
s1	ax1	2	0.7078	0.393	-1	-1	44	78652	(ii)
s1	ax1	3	0.4719	0.589	38	2631	28	38360	(i)
s1	ax1	4	0.3539	0.785	30	1673	23	28476	(i)
s1	ax1	6	0.2359	1.178	22	931	18	19180	(i)
s1	ax1	8	0.1769	1.571	18	641	16	15260	(i)
s1	ax1	16	0.0885	3.142	14	405	13	10976	(i)
s1	ax1	32	0.0442	6.283	12	307	11	8064	(i)
s1	ax1	64	0.0221	12.566	10	223	10	6776	(i)
s1	ax2	2	0.7078	0.393	-1	-1	44	78652	(ii)
s1	ax2	3	0.4719	0.589	38	2631	28	38360	(i)
s1	ax2	4	0.3539	0.785	30	1673	23	28476	(i)
s1	ax2	6	0.2359	1.178	22	931	18	19180	(i)
s1	ax2	8	0.1769	1.571	18	641	16	15260	(i)
s1	ax2	16	0.0885	3.142	14	405	13	10976	(i)
s1	ax2	32	0.0442	6.283	12	307	11	8064	(i)
s1	ax2	64	0.0221	12.566	10	223	10	6776	(i)
s1	ax3	2	11.3248	0.025	-1	-1	-1	-1	(iii)
s1	ax3	3	7.5498	0.037	-1	-1	-1	-1	(iii)
s1	ax3	4	5.6624	0.049	-1	-1	-1	-1	(iii)
s1	ax3	6	3.7749	0.074	-1	-1	-1	-1	(iii)
s1	ax3	8	2.8312	0.098	-1	-1	-1	-1	(iii)
s1	ax3	16	1.4156	0.196	-1	-1	-1	-1	(iii)
s1	ax3	32	0.7078	0.393	-1	-1	32	59164	(ii)

s1	ax3	64	0.3539	0.785	30	1673	21	27104	(i)
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Three things change against Section 8.3 and nothing else does. The whole-box L falls by two to six shells at two to four cells (Theorem A is sharper than (33) by a median factor 3.2×10^3) and rises by two shells in the far plateau, where the old $L = 8$ was never a certificate of the sum the code performs; the octant L_j falls from 55 to 50 at $(2, 0, 0)$; and the cap of 56 is reached by the whole-box bound at the two-cell offsets, which is what routes them to (ii).

Routing counts over a whole block. The number of offsets each route takes, over the `egoToe` octant that Gila evaluates (indices with $\max_i i_i \geq 3$, i.e. max-norm separation ≥ 2 cells), from `tables/e.route.txt`:

shape	f	block	(i)	(ii)	(iii)	setup s
c32	$1.0 + 0.0i$	$32 \times 32 \times 32$	32748	12	0	0.7
c32	$1.0 + 0.0i$	$64 \times 64 \times 64$	262124	12	0	0.7
c32	$1.0 + 0.0i$	$128 \times 128 \times 128$	2097132	12	0	0.8
c32	$1.0 + 0.1i$	$32 \times 32 \times 32$	32748	12	0	1.0
c32	$1.0 + 0.1i$	$64 \times 64 \times 64$	262124	12	0	0.9
c32	$1.0 + 0.1i$	$128 \times 128 \times 128$	2097132	12	0	0.9
c8	$1.0 + 0.0i$	$32 \times 32 \times 32$	32748	12	0	0.7
c8	$1.0 + 0.0i$	$64 \times 64 \times 64$	262124	12	0	0.8
c8	$1.0 + 0.0i$	$128 \times 128 \times 128$	2097132	12	0	0.8
c8	$1.0 + 0.1i$	$32 \times 32 \times 32$	32748	12	0	0.9
c8	$1.0 + 0.1i$	$64 \times 64 \times 64$	262124	12	0	0.9
c8	$1.0 + 0.1i$	$128 \times 128 \times 128$	2097132	12	0	1.1
c4	$1.0 + 0.0i$	$32 \times 32 \times 32$	32748	12	0	0.8
c4	$1.0 + 0.0i$	$64 \times 64 \times 64$	262124	12	0	0.8
c4	$1.0 + 0.0i$	$128 \times 128 \times 128$	2097132	12	0	0.9
c4	$1.0 + 0.1i$	$32 \times 32 \times 32$	32748	12	0	1.1
c4	$1.0 + 0.1i$	$64 \times 64 \times 64$	262124	12	0	1.0
c4	$1.0 + 0.1i$	$128 \times 128 \times 128$	2097132	12	0	1.1
s1	$1.0 + 0.0i$	$16 \times 16 \times 32$	8013	104	67	0.7
s1	$1.0 + 0.0i$	$32 \times 32 \times 64$	65350	111	67	0.7
s1	$1.0 + 0.0i$	$64 \times 64 \times 128$	524102	111	67	0.8
s1	$1.0 + 0.1i$	$16 \times 16 \times 32$	8013	104	67	0.9
s1	$1.0 + 0.1i$	$32 \times 32 \times 64$	65350	111	67	0.9
s1	$1.0 + 0.1i$	$64 \times 64 \times 128$	524102	111	67	0.9

On the three cubic shapes the k -series is never needed, and subdivision is needed at exactly twelve offsets out of two million. The whole-box L histogram of the largest block bottoms out at a flat floor — $L = 10$ at $\lambda/32$, 14 at $\lambda/8$, 16 at $\lambda/4$ — which 97.8%, 99.6% and 97.5% of the offsets attain, and this is what makes the average per-offset cost of a large build converge to a few hundred nanoseconds.

8.5 The twelve two-cell offsets of a cubic grid

On a cubic grid the offsets with $\max_i n_i \geq 2$ and $|\mathbf{n}| \leq 2.56$ are exactly twelve per octant, for every N :

$$(2, 0, 0), (0, 2, 0), (0, 0, 2); \quad (2, 1, 1), (1, 2, 1), (1, 1, 2); \\ (2, 1, 0), (2, 0, 1), (1, 2, 0), (0, 2, 1), (1, 0, 2), (0, 1, 2).$$

These are precisely the offsets the rule sends to the octant split, at all three cubic scales, and the reason is visible in the L columns:

shape	$\mathbf{n}$ class	ρ	L_{wb}	L_j over the eight sub-boxes
$(1/32)^3$	(2, 0, 0)	0.8660	174	31, 31, 31, 31, 55, 55, 55, 55
$(1/32)^3$	(2, 1, 0)	0.7746	104	28, 28, 31, 31, 38, 38, 55, 55
$(1/32)^3$	(2, 1, 1)	0.7071	78	26, 28, 28, 32, 32, 38, 38, 56
$(1/8)^3$	(2, 0, 0)	0.8660	176	32, 32, 32, 32, 55, 55, 55, 55
$(1/8)^3$	(2, 1, 0)	0.7746	104	28, 28, 32, 32, 38, 38, 55, 55
$(1/8)^3$	(2, 1, 1)	0.7071	78	26, 29, 29, 32, 32, 39, 39, 56
$(1/4)^3$	(2, 0, 0)	0.8660	182	33, 33, 33, 33, 56, 56, 56, 56
$(1/4)^3$	(2, 1, 0)	0.7746	108	30, 30, 33, 33, 39, 39, 56, 56
$(1/4)^3$	(2, 1, 1)	0.7071	82	27, 30, 30, 33, 33, 39, 39, 56

Every one of the twelve has $L_{\text{wb}} \geq 78 > L_{\text{max}}$ and $\max_j L_j \leq 56 \leq L_{\text{max}}$, at all three cubic scales; the next offset outwards, (2, 2, 0) with $\rho = 0.6124$, has $L_{\text{wb}} = 56, 58, 60$ at $\lambda/32, \lambda/8, \lambda/4$ and is handled by the whole box. *The rule therefore reproduces, from the bounds alone, the split that was found empirically:* octant for the twelve, whole box from (2, 2, 0) outwards. Since twelve is independent of N , subdividing them costs twelve tensor evaluations in a build of two million.

The shipped selector agrees, with room to spare: at $\text{tol} = 10^{-14}$ Theorem A gives no certificate at $L \leq 56$ for any of the twelve, and Theorem B gives $\max_j L_j = 50, 50$ and 51 at (2, 0, 0) and $51, 51$ and 52 at (2, 1, 1) for the $\lambda/32, \lambda/8$ and $\lambda/4$ cubes — below the cap $L_{\text{max}} = 56$ in every case, and delivering 4.4×10^{-15} or better against the reference. The routing counts of Section 8.4 are exactly twelve route-(ii) offsets per block, at all three cubic scales, at $32^3, 64^3$ and 128^3 alike, and at both frequencies.

Remark 8.1 (a discrepancy in the sub-box L , and its size). The earlier sub-box analysis records the bound as selecting $L_j \leq 48$ at these offsets, against $L_j \leq 56$ here, both from the same written bound (37), the same geometry ($\rho_j = 0.522$ at (2, 0, 0) in both) and the same budget $\text{tol} \cdot \text{est}/8$. The difference is a factor of about 100 in the value of the tail, i.e. ρ_j^7 ; it does not change any band edge, since both numbers are below $L_{\text{max}} = 64$ and both are above the measured requirement of $L_{13} = 38\text{--}42$. I have not been able to reproduce the smaller value from the bound as written, and I record the larger one because it is the one this document's script computes from the stated formula. The corresponding over-selection factor against the measured requirement is $56/42 = 1.33$ here and $48/42 = 1.14$ there.

8.6 The slender cell: which offsets the expansion cannot reach

For the slender cell $(1/32, 1/32, 1/512)$ the isotropic ratio is $\rho = r_d/|\mathbf{R}|$ with $r_d = 0.04421$, and along the short axis $|\mathbf{R}| = n/512$, so $\rho = 22.63/n$: the expansion *diverges* until $n = 23$ and reaches $\rho < 0.5$ only at $n = 46$. This is not slow convergence, and no constant repairs it: the difference box still spans $\pm 1/32$ in x and y while the offset advances in steps of $1/512$. Subdivision helps but not enough — the octant bound needs $L = 207$ at (0, 0, 8) and $L = 70$ at (0, 0, 16).

Enumerating every offset with $n_1, n_2 \leq 2$ and $n_3 \leq 64$ that is not a touching offset — 577 of them — and asking of each whether the whole-box bound (33) or the octant bound (37) reaches 10^{-13}est within a given cap:

cap	count	the set
$L > 48$	83	$(0, 0, n)$ for $2 \leq n \leq 21$; $(0, 1, n), (1, 0, n), (1, 1, n)$ for $2 \leq n \leq 22$
$L > 64$	64	$(0, 0, n), (0, 1, n), (1, 0, n), (1, 1, n)$ for $2 \leq n \leq 17$

Both sets are stable under the reflections and under N . Every offset with $n_1 = 2$ or $n_2 = 2$ is already covered by the whole box or the octant split, and so is every (n_1, n_2, n_3) with n_3 beyond the band edge. *This is the only band, on any of the four shapes, where neither form of the expansion has a certificate.* The shipped selector, with Theorem A/B at $\text{tol} = 10^{-14}$ and $L_{\max} = 56$, hands 67 **offsets** to the k -series — the four classes $(0, 0, n_3)$, $(0, 1, n_3)$, $(1, 0, n_3)$, $(1, 1, n_3)$ with n_3 up to 18 — and 111 to the octant split, out of 524 280 offsets of a $64 \times 64 \times 128$ block; the counts do not change between $32 \times 32 \times 64$ and $64 \times 64 \times 128$, nor between $f = 1$ and $f = 1 + 0.1i$.

The k -series owns it comfortably, because the cell is short in the only direction that matters to k . Recomputed on exactly the 67 offsets the router sends there (`verify.jl` part (f), `tables/f_bands.txt`), with $D_{\max}$ the largest $|\mathbf{x} - \mathbf{y}|$ over the thirty-six face pairs, $x = |k|D_{\max}$, $\Lambda = (\sum_{\text{fp}} I_{-1}^{(\text{fp})}) / (4\pi |f|^2 V_{\text{t est}}(\mathbf{R}))$ the assembly amplification, and N the order Theorem 5.7 needs for $10^{-16}/\Lambda$:

$$x \in [0.280, 0.602], \quad N \in [12, 17], \quad \Lambda \leq 1002, \quad \varepsilon\Lambda \leq 2.23 \times 10^{-13},$$

identically at $f = 1$ and $f = 1 + 0.1i$. *That last number is why route (iii) is done in 128-bit arithmetic:* a `Float64` face-pair k -series on this set would be 2.2×10^{-13} wrong from rounding alone, before any truncation. The three cubic shapes have no route-(iii) offsets at all, and the table says so explicitly. The price is the cost: 1.9 to 9.5 s per offset on eight threads, 128 s for all 67 from cold, once per (shape, frequency), cached thereafter (`ksrccache/`), against 0.14 s for the other 524 213 offsets of the block.

8.7 The geometry that makes this affordable

Two facts about the lattice decide the cost of the whole scheme.

The convergence ratio is scale free and tabulated. For cubic cells $\rho = \sqrt{3} s / |\mathbf{R}|$ depends only on $\mathbf{n}$, and ρ_m , the ratio after splitting each axis into m pieces, is $\min_j |\mathbf{h}| / |\mathbf{R} + \mathbf{c}_j|$:

class	n	ρ	ρ_2	ρ_4	ρ_8
axis	2	0.866025	0.522233	0.333333	0.190117
axis	3	0.577350	0.333333	0.190117	0.101535
axis	4	0.433013	0.242536	0.132453	0.069171
axis	8	0.216506	0.114960	0.059655	0.030378
axis	16	0.108253	0.055815	0.028387	0.014313
axis	32	0.054127	0.027486	0.013856	0.006956
axis	64	0.027063	0.013637	0.006846	0.003430
face diag	2	0.612372	0.397360	0.242536	0.135665
face diag	8	0.153093	0.081559	0.042220	0.021485
body diag	2	0.500000	0.333333	0.200000	0.111111
body diag	8	0.125000	0.066667	0.034483	0.017544
$(n, 1, 0)$	2	0.774597	0.522233	0.333333	0.190117
$(n, 2, 1)$	2	0.577350	0.397360	0.242536	0.135665

Refinement divides ρ by roughly $m(n - 1) + 1$ rather than by m , which is why the octant split ($m = 2$) is worth its factor of about 3.6 in cost at two cells and nothing at all beyond three.

The slow band has a fixed size. The offsets with large ρ form a ball of fixed radius in $\mathbf{n}$, $n_1^2 + n_2^2 + n_3^2 < 3/\rho^2$, whose population does not grow with the volume: in the octant of offsets Gila actually evaluates there are 37 offsets with $\rho > 0.5$, 141 with $\rho > 0.3$, 436 with $\rho > 0.2$ and 3096

with $\rho > 0.1$, *identically* for $N = 32, 64$ and 128 . So the expensive offsets are an $O(1)$ set and the average per-offset cost of a large build converges to the far-plateau cost of the table in Section 8.3, a few hundred nanoseconds. The slender cell is the same statement with a different constant: its k -series band contains 67 offsets whatever N is, and its octant band 111.

9 Verification

Every number in this section comes from `notes/farfield/verify.jl`, single-threaded, against 220-bit references held in `refcache/reftensors.txt`. Three independent references exist and were used: the *face-pair* reference (`pairKer` of `notes/gen/ref/mom.jl` on each of the thirty-six pairs, assembled with the `srfSum!` signs), the *volume* reference (a graded `BigFloat` tensor Gauss–Legendre rule on the difference box, split at the kink of w and graded to the corner nearest the singularity), and a second volume rule written independently for family (d). They agree with each other to 3.7×10^{-65} – 1.05×10^{-49} (Section 3.6) and to 1.8×10^{-61} on the cases where all three were run. `pairKer` stalls beyond twenty-five minutes on slender panels, which is why the slender short-axis rows use the volume rule.

9.1 Against the reference: all bands, shapes, directions, frequencies

`verify.jl` part (d), table `d_ref.txt`: every cached reference, 427 rows. $e_{\max} = \max_{ab} |G - G_{\text{ref}}| / \max_{ab} |G_{\text{ref}}|$; “per entry” is the worst $|G_{ab} - G_{\text{ref},ab}| / |G_{\text{ref},ab}|$ over entries carrying more than 10^{-8} of the largest; `Re` and `Im` are the same for the two parts separately, each relative to that part’s own value.

shape	f	rows	$e_{\max}$	per entry	Re	Im
c32	0.37 + 0.0i	14	7.08×10^{-16}	8.60×10^{-16}	5.73×10^{-15}	9.43×10^{-14}
c32	1.0 + 0.0i	81	4.40×10^{-15}	4.40×10^{-15}	9.80×10^{-15}	3.20×10^{-14}
c32	1.0 + 0.1i	79	1.73×10^{-15}	2.92×10^{-15}	8.16×10^{-14}	3.48×10^{-14}
c32	1.0 + 1.0i	13	7.87×10^{-16}	1.71×10^{-15}	9.26×10^{-15}	3.18×10^{-15}
c32	3.0 + 0.3i	13	5.38×10^{-16}	1.32×10^{-15}	1.57×10^{-14}	2.43×10^{-15}
c4	1.0 + 0.0i	29	8.60×10^{-15}	9.34×10^{-15}	9.69×10^{-13}	8.80×10^{-13}
c4	1.0 + 0.1i	29	2.50×10^{-15}	3.61×10^{-15}	1.71×10^{-14}	1.89×10^{-14}
c8	1.0 + 0.0i	28	5.50×10^{-15}	5.53×10^{-15}	4.97×10^{-14}	1.55×10^{-13}
c8	1.0 + 0.1i	28	1.91×10^{-15}	3.35×10^{-15}	3.66×10^{-14}	7.32×10^{-15}
s1	0.37 + 0.0i	8	8.80×10^{-16}	1.82×10^{-15}	1.82×10^{-15}	4.38×10^{-16}
s1	1.0 + 0.0i	45	2.50×10^{-15}	8.69×10^{-15}	9.13×10^{-15}	1.21×10^{-14}
s1	1.0 + 0.1i	44	2.44×10^{-15}	7.83×10^{-15}	6.19×10^{-14}	2.32×10^{-14}
s1	1.0 + 1.0i	8	1.35×10^{-15}	1.35×10^{-15}	1.01×10^{-14}	1.35×10^{-15}
s1	3.0 + 0.3i	8	2.48×10^{-15}	2.48×10^{-15}	3.01×10^{-15}	6.72×10^{-15}

Over 427 tensors at four shapes, five frequencies, separations from two to 128 cells and every offset direction, the worst max-norm error is 8.60×10^{-15} and the worst per-entry error is 9.34×10^{-15} . The *Re* and *Im* columns above 10^{-13} are the known formulation floor and nothing else: in every such row the offending part carries less than about 1% of that entry’s own modulus, and its *absolute* error is still at $\varepsilon \max |G|$. Wherever both parts are $O(1)$ fractions of the entry, both per-part errors are at 1.5×10^{-14} .

The bands are covered on both sides of every routing boundary. The boundaries bite only on the two flat shapes, $(1/32, 1/32, 1/512)$ and $(1/8, 1/8, 1/128)$, and there both sides of all five class boundaries — $(0, 0, n_3)$, $(1, 0, n_3)$, $(1, 1, n_3)$, $(2, 0, n_3)$, $(2, 1, n_3)$ — were measured against the volume reference at both frequencies: over those forty rows the worst per-entry error is 1.0×10^{-14} and

the worst per-part 1.17×10^{-14} , with the k -series side at 2.4×10^{-17} to 9.3×10^{-17} and the octant side at 1.6×10^{-15} to 1.0×10^{-14} . (That sweep was run on the audited version at $\text{tol} = 10^{-13}$; the shipped selector moves each of those boundaries by at most one lattice step, and the offsets on either side of the new position are covered by the rows above.) *The accuracy is continuous across every boundary; the cost is not* (Section 13, defect D3).

9.2 Aspect ratios, and the parameter that governs everything

The requirement is aspect ratios 10^{-3} to 10^3 . Twelve shapes were built for this, with aspect ratios 1, 4, 11.4, 16 (in all three axis permutations), 87.7, 295, 678 and 1020, six shape-adapted offsets each — two and four cells along the *shortest* edge, two along the longest, (3, 3, 3), (7, -3, 2), (16, 16, 16) — and four frequencies, giving 288 tensors against the independent volume reference.

The whole sweep collapses onto one parameter, $|k|r_d$ with $r_d = \sqrt{s_1^2 + s_2^2 + s_3^2}$. Not the longest edge: a needle whose long edge is 0.503λ passes, and a cell whose long edge is 0.459λ fails at one frequency and passes at another, because $|k|r_d$ is 3.17 in the first case and 5.80 in the second. On the audited version the break was at $|k|r_d \approx 3.2$ and the error above it grew as $(|k|r_d)^{25.6}$, the exponent $2n_{\max} + 2 = 26$ of the frozen k -series (defect D9). With the adaptive N of Remark 5.6 the four failing rows become

shape	f	$ k r_d$	N	$e_{\max}$	per entry	per entry, before
r_3 (aspect 678)	$2 + 0.2i$	8.96	25	4.91×10^{-15}	1.71×10^{-13}	9.6×10^{-3}
r_4 (aspect 87.7)	$2 + 0.2i$	5.80	19	7.29×10^{-15}	1.02×10^{-13}	6.4×10^{-7}
r_2 (aspect 295)	$2 + 0.2i$	4.78	18	3.47×10^{-15}	2.10×10^{-14}	1.3×10^{-9}
r_3	$1 + 0.1i$	4.48	18	3.81×10^{-15}	3.29×10^{-14}	1.8×10^{-10}
r_3	1	4.46	18	5.28×10^{-15}	2.71×10^{-14}	1.7×10^{-10}
$(1/2)^3$ cube	1	5.44	16	3.79×10^{-15}	3.79×10^{-15}	7.2×10^{-13}

i.e. a gain of 6.3×10^3 to 5.6×10^{10} , and every row now at or below 1.71×10^{-13} per entry with 7.3×10^{-15} in max-norm. The two rows that miss 10^{-13} per entry are aspect-87.7 and aspect-678 cells with longest edges of 0.92 and 1.43 wavelengths — three to six times outside Gila’s stated range — and what is left there is the small-part floor, visible in their imaginary-part errors of 9.3×10^{-12} on parts carrying less than 10^{-3} of their entry, not the k -series. Inside Gila’s range every shape has $|k|r_d \leq 2.72$ and passes with margin.

The two finest cells, which failed for the opposite reason. A $\lambda/128$ cube — the finest cell Gila claims — was wrong by a flat 4.65×10^{-13} per entry at *every* separation from three to 128 cells and at every $k\mathbf{R}$ from 0.255 to 5.44, because the per-shell n -cut was being chosen against a budget evaluated at a radius where the expansion does not converge, and cut the j_l series at two to four of its thirteen stored terms. With the cut re-evaluated at the true radius the same sixteen rows run from 2.6×10^{-16} to 4.3×10^{-15} , a factor 110 to 1800, and the separation-independence that diagnosed the bug is gone. The aspect-4 shape (1/16, 1/32, 1/64) went from 1.01×10^{-12} at (128, 0, 0) to 2.96×10^{-15} . A third, independent mechanism — the order L under-selected at very large separation — was fixed by the same tolerance change: (48, 48, 48) and (64, 64, 64) on an aspect-11.4 shape went from 1.23×10^{-13} and 1.19×10^{-13} to 3.22×10^{-15} and 3.83×10^{-15} , with *two* extra shells rather than the twelve the diagnosis needed, because Theorem A bounds the sum the code actually performs.

9.3 Symmetries and homogeneity

Part (g), table `g_sym.txt`: fifty random offsets per (shape, frequency) drawn from $[-40, 40]^3$ with a fixed seed.

shape	f	$\mathbf{D} \rightarrow -\mathbf{D}$	transpose	reflections	permutation	homogeneity
$(1/32)^3$	1	0	0	0	7.24×10^{-16}	1.29×10^{-15}
$(1/32)^3$	$1 + 0.1i$	0	0	0	7.75×10^{-16}	9.33×10^{-16}
$(1/4)^3$	1	0	0	0	1.63×10^{-15}	4.77×10^{-15}
$(1/4)^3$	$1 + 0.1i$	0	0	0	1.04×10^{-15}	6.55×10^{-15}
slender	1	0	0	0	—	1.96×10^{-15}
slender	$1 + 0.1i$	0	0	0	—	8.72×10^{-16}

The three zeros prove bookkeeping, not accuracy, and should be read that way. $\mathbf{D} \rightarrow -\mathbf{D}$, the transpose and the axis reflections are exact *by construction* in this expansion: the whole-box table keeps only even l and $Y_{lm}(-\mathbf{u}) = (-1)^l Y_{lm}(\mathbf{u})$, so route (i) cannot fail them. The honest test is the axis permutation, which mixes different (l, m) columns of the table and comes out at 1.6×10^{-15} ; it is not available on the slender cell, whose three axes are inequivalent. Homogeneity, $T(\lambda f, \mathbf{s}/\lambda) = \lambda^{-2} T(f, \mathbf{s})$ at $\lambda = 3.7$, is $\leq 6.6 \times 10^{-15}$. On route (ii), which reflects sub-boxes and keeps odd l , the reflection test is *not* vacuous and gives 1.6 to 2.7×10^{-16} , with the reflected offsets matching their own independent references to 3.9×10^{-16} – 1.6×10^{-15} .

Number type: `ComplexF32` in gives `ComplexF32` out at 6.20×10^{-8} of the `Float64` value ($\varepsilon(\text{Float32}) = 1.19 \times 10^{-7}$), and `Complex{BigFloat}` in gives `Complex{BigFloat}` out at 5.14×10^{-16} . Changing the working precision from 128 to 256 bits moves the answer in the 38th digit, not the 16th, so no untyped constant survives in the generic path. The ceiling is 192 bits, the precision at which the geometry table is stored (Section 13, D5).

9.4 Digits lost

Part (e), table `e_digits.txt`: the same route and the same L evaluated in `Float64` and in `BigFloat(192)`, so the truncation cancels exactly and only rounding is left. digits = $\log_{10}(\max_{ab} |G_{64} - G_B| / (\varepsilon \max_{ab} |G_B|))$.

shape	$\mathbf{n}$	ρ	$ k\mathbf{R} $	route	L	rel. err	digits lost
c32	(2, 0, 0)	0.866	0.393	2	50	4.25×10^{-15}	1.28
c32	(3, 0, 0)	0.5774	0.589	1	44	3.78×10^{-16}	0.23
c32	(4, 0, 0)	0.433	0.785	1	32	3.98×10^{-16}	0.25
c32	(6, 0, 0)	0.2887	1.178	1	22	2.49×10^{-16}	0.05
c32	(8, 0, 0)	0.2165	1.571	1	18	3.06×10^{-16}	0.14
c32	(16, 0, 0)	0.1083	3.142	1	14	4.05×10^{-16}	0.26
c32	(32, 0, 0)	0.0541	6.283	1	12	2.35×10^{-16}	0.02
c32	(64, 0, 0)	0.0271	12.566	1	10	1.01×10^{-15}	0.66
c32	(2, 2, 0)	0.6124	0.555	1	48	7.43×10^{-16}	0.52
c32	(3, 3, 0)	0.4082	0.833	1	30	1.35×10^{-15}	0.78
c32	(4, 4, 0)	0.3062	1.111	1	24	4.69×10^{-16}	0.33
c32	(6, 6, 0)	0.2041	1.666	1	18	3.73×10^{-16}	0.22
c32	(8, 8, 0)	0.1531	2.221	1	16	1.47×10^{-16}	−0.18
c32	(16, 16, 0)	0.0765	4.443	1	12	3.03×10^{-16}	0.14
c32	(32, 32, 0)	0.0383	8.886	1	10	3.48×10^{-16}	0.19
c32	(64, 64, 0)	0.0191	17.772	1	10	3.54×10^{-16}	0.2
c32	(2, 2, 2)	0.5	0.68	1	36	2.08×10^{-15}	0.97

shape	n	ρ	$ k\mathbf{R} $	route	L	rel. err	digits lost
c32	(3, 3, 3)	0.3333	1.02	1	24	3.26×10^{-16}	0.17
c32	(4, 4, 4)	0.25	1.36	1	20	4.86×10^{-16}	0.34
c32	(6, 6, 6)	0.1667	2.041	1	16	2.62×10^{-16}	0.07
c32	(8, 8, 8)	0.125	2.721	1	14	4.79×10^{-16}	0.33
c32	(16, 16, 16)	0.0625	5.441	1	12	7.58×10^{-16}	0.53
c32	(32, 32, 32)	0.0312	10.883	1	10	3.19×10^{-16}	0.16
c32	(64, 64, 64)	0.0156	21.766	1	10	1.75×10^{-15}	0.9
c4	(2, 0, 0)	0.866	3.142	2	51	4.61×10^{-16}	0.32
c4	(3, 0, 0)	0.5774	4.712	1	48	3.71×10^{-16}	0.22
c4	(4, 0, 0)	0.433	6.283	1	34	3.82×10^{-16}	0.24
c4	(6, 0, 0)	0.2887	9.425	1	26	4.52×10^{-16}	0.31
c4	(8, 0, 0)	0.2165	12.566	1	22	9.56×10^{-16}	0.63
c4	(16, 0, 0)	0.1083	25.133	1	18	8.22×10^{-16}	0.57
c4	(32, 0, 0)	0.0541	50.265	1	18	2.72×10^{-15}	1.09
c4	(64, 0, 0)	0.0271	100.531	1	16	8.59×10^{-15}	1.59
c4	(2, 2, 0)	0.6124	4.443	1	52	2.36×10^{-16}	0.03
c4	(3, 3, 0)	0.4082	6.664	1	32	8.14×10^{-16}	0.56
c4	(4, 4, 0)	0.3062	8.886	1	26	4.92×10^{-16}	0.35
c4	(6, 6, 0)	0.2041	13.329	1	22	7.00×10^{-16}	0.5
c4	(8, 8, 0)	0.1531	17.772	1	20	3.06×10^{-16}	0.14
c4	(16, 16, 0)	0.0765	35.543	1	18	1.83×10^{-16}	-0.08
c4	(32, 32, 0)	0.0383	71.086	1	18	4.90×10^{-16}	0.34
c4	(64, 64, 0)	0.0191	142.172	1	16	2.23×10^{-16}	0.0
c4	(2, 2, 2)	0.5	5.441	1	40	1.13×10^{-15}	0.71
c4	(3, 3, 3)	0.3333	8.162	1	28	5.05×10^{-16}	0.36
c4	(4, 4, 4)	0.25	10.883	1	24	3.44×10^{-16}	0.19
c4	(6, 6, 6)	0.1667	16.324	1	20	5.06×10^{-16}	0.36
c4	(8, 8, 8)	0.125	21.766	1	20	2.20×10^{-15}	1.0
c4	(16, 16, 16)	0.0625	43.531	1	18	4.66×10^{-15}	1.32
c4	(32, 32, 32)	0.0312	87.062	1	16	3.45×10^{-16}	0.19
c4	(64, 64, 64)	0.0156	174.125	1	16	4.38×10^{-16}	0.3
s1	(2, 0, 0)	0.7078	0.393	2	44	1.17×10^{-15}	0.72
s1	(3, 0, 0)	0.4719	0.589	1	38	5.13×10^{-16}	0.36
s1	(4, 0, 0)	0.3539	0.785	1	30	1.78×10^{-15}	0.9
s1	(6, 0, 0)	0.2359	1.178	1	22	4.94×10^{-16}	0.35
s1	(8, 0, 0)	0.1769	1.571	1	18	2.57×10^{-16}	0.06
s1	(16, 0, 0)	0.0885	3.142	1	14	2.51×10^{-16}	0.05
s1	(32, 0, 0)	0.0442	6.283	1	12	1.51×10^{-16}	-0.17
s1	(64, 0, 0)	0.0221	12.566	1	10	9.65×10^{-16}	0.64
s1	(0, 2, 0)	0.7078	0.393	2	44	1.31×10^{-15}	0.77
s1	(0, 3, 0)	0.4719	0.589	1	38	5.13×10^{-16}	0.36
s1	(0, 4, 0)	0.3539	0.785	1	30	1.78×10^{-15}	0.9
s1	(0, 6, 0)	0.2359	1.178	1	22	4.94×10^{-16}	0.35
s1	(0, 8, 0)	0.1769	1.571	1	18	2.57×10^{-16}	0.06
s1	(0, 16, 0)	0.0885	3.142	1	14	2.51×10^{-16}	0.05
s1	(0, 32, 0)	0.0442	6.283	1	12	1.51×10^{-16}	-0.17
s1	(0, 64, 0)	0.0221	12.566	1	10	9.65×10^{-16}	0.64
s1	(0, 0, 32)	0.7078	0.393	2	32	1.07×10^{-15}	0.68
s1	(0, 0, 64)	0.3539	0.785	1	30	1.96×10^{-16}	-0.05

The worst over the whole grid is 1.59 digits, at the $\lambda/4$ cube at (64, 0, 0) where $k\mathbf{R} = 100.5$; the two-cell octant route loses 1.28; most rows are between 0 and 0.9, and several are negative. This is the measurement that says the expansion is rounding-limited rather than truncation-limited at

the L the bound selects, and it is the reason no extended-precision path exists in the per-offset evaluation.

9.5 The anti-Hermitian positivity check

Part (h), table `h_pos.txt`. Each block is built twice: once by Gila (`wekTrp` at `intOrd` 48, `egoFunInn!` on every Toeplitz index, `egoFunSng!` on the eight contact indices), and once with every separated offset replaced by `farBlock!`, so the eight contact blocks are identical by construction. The dense matrix uses `egoToeCrc!`'s rule and is complex symmetric, so the anti-Hermitian part is $\text{Im } M$; at real frequency the operator is lossless and $\text{Im } M$ must be positive semi-definite, and the meaningful number is the most negative eigenvalue.

block	f	$\lambda_{\min}$ (Gila)	$\lambda_{\min}$ (farfield)	$\lambda_{\max}$	$\varepsilon\lambda_{\max}$	negatives
$(1/32)^3, 6^3$	1	-4.4681×10^{-15}	-2.4488×10^{-15}	7.729×10^{-2}	1.7×10^{-17}	248 $\rightarrow$ 201
$(1/4)^3, 6^3$	1	-6.3987×10^{-15}	-4.8810×10^{-15}	2.3897	5.3×10^{-16}	58 $\rightarrow$ 49
slender, $6 \times 6 \times 12$	1	-6.553×10^{-14}	-1.921×10^{-14}	1.0117×10^{-2}	2.3×10^{-18}	596 $\rightarrow$ 576
$(1/32)^3, 6^3$	$1 + 0.1i$	$+2.855 \times 10^{-3}$	$+2.855 \times 10^{-3}$	0.18815		0/648
$(1/4)^3, 6^3$	$1 + 0.1i$	$+7.124 \times 10^{-3}$	$+7.124 \times 10^{-3}$	1.7452		0/648
slender, $6 \times 6 \times 12$	$1 + 0.1i$	-9.939×10^{-3}	-1.465×10^{-2}	0.19539		108 $\rightarrow$ 76

At real frequency `farBlock!` moves $\lambda_{\min}$ 1.8 times (cube at $\lambda/32$), 1.3 times ($\lambda/4$) and 3.4 times (slender) closer to zero and removes 9 to 47 negative eigenvalues; both builds stay above the `Float64` floor, because the residual sits in the contact blocks, which are Gila's in both. At complex frequency the two cubes are positive definite and the two builds agree to thirteen digits.

The slender block at $f = 1 + 0.1i$, and what it turned out to be. There $\text{Im } M$ is indefinite at the percent level in both builds, and *farBlock! makes it worse*: $\lambda_{\min}$ goes from -9.94×10^{-3} to -1.465×10^{-2} against $\lambda_{\max} = 0.195$, from 5.1 % to 7.5 % of the spectral radius, twelve orders above any rounding floor, while the number of negative eigenvalues *falls* from 108 to 76. Two earlier passes concluded from this that the indefiniteness is a property of the 1:1:16 discretization and that no far-field accuracy fixes it. That conclusion is wrong, and the test that shows it is to replace the eight *near* offsets — the contact cell and the seven touching-shell offsets with all $|n_i| \leq 1$ — by the exact k -series of `notes/moments/moments.jl` (`momentSeries` on the thirty-six `facePair` panels at 192 bits, $m_{\max} = 30$, assembled by the `srfsSum!` signs, then the identity term $-1/f^2$ on the diagonal of $(0, 0, 0)$; the series' worst last-term ratio over the 288 panel pairs is 6.6×10^{-36}). How far Gila's order-9 rule is from the exact value there:

offset	(0, 0, 0)	(1, 0, 0)	(0, 1, 0)	(1, 1, 0)	(0, 0, 1)	(1, 0, 1)	(0, 1, 1)	(1, 1, 1)
rel. difference	0.127	0.0244	0.0244	7.4×10^{-5}	1.17	0.0365	0.0365	8.2×10^{-5}

117 % **on the touching shell across the 1/512 face, and 12.7 % on the contact cell itself.** With those eight offsets exact and `farBlock!` everywhere else:

$(6 \times 6 \times 12)$ slender, $\text{Im } M$	$\lambda_{\min}$	$\lambda_{\max}$	negatives	$\lambda_{\min}/\lambda_{\max}$
$f = 1 + 0.1i$, Gila everywhere	-9.93932×10^{-3}	0.1953792	108/1296	-5.09×10^{-2}
Gila contact + <code>farBlock!</code>	-1.46500×10^{-2}	0.1953886	76/1296	-7.50×10^{-2}
exact contact + <code>farBlock!</code>	$+3.25013 \times 10^{-4}$	0.1953884	0/1296	$+1.66 \times 10^{-3}$
$f = 1$, Gila everywhere	-6.55324×10^{-14}	1.011658×10^{-2}	596/1296	-6.5×10^{-12}
Gila contact + <code>farBlock!</code>	-1.92133×10^{-14}	1.011658×10^{-2}	576/1296	-1.9×10^{-12}
exact contact + <code>farBlock!</code>	-4.63544×10^{-18}	1.011658×10^{-2}	605/1296	-4.6×10^{-16}

With exact contact integrals the anti-Hermitian part of the slender operator is positive definite at $f = 1 + 0.1i$ and positive semi-definite to the Float64 floor at $f = 1$ ($\lambda_{\min} = -2.1 \varepsilon \lambda_{\max}$, against $-29\,000 \varepsilon \lambda_{\max}$ for Gila’s own build). The -1.465×10^{-2} is Gila’s order-9 touching-shell and contact quadrature, it is 45 times larger than the whole anti-Hermitian spectrum’s true distance from zero, and the far-field replacement is what *exposes* it — it moves $\lambda_{\min}$ from -9.9×10^{-3} to -1.5×10^{-2} by removing the compensating error at separation two. This is a diagnosis, not a deliverable: the exact contact rebuild costs 114–117 s per (shape, frequency) for eight offsets and is not wired into anything.

Finally, $M_{ab}(\mathbf{R}) = M_{ba}(-\mathbf{R})$ holds to 8.3×10^{-16} – 3.1×10^{-15} in every case, and the `farBlock!` build is never worse than Gila’s.

9.6 The overlap: two independent expansions compared where both converge

Part (f), table `f_overlap.txt`: the shipped method against the Cartesian (G, S) Taylor recurrence of family (g), at that route’s own order rule, over every offset of a 16^3 octant with max-norm separation ≥ 2 — 4085 offsets per case. The two share no geometry table, no special function and no truncation rule; they share only the volume form and the `srfsSum!` sign convention, so their agreement is an end-to-end check of the bookkeeping, the normalization and both truncations at once.

shape	f	band	offsets	worst per entry	worst offset
$(1/32)^3$	1	2 cells	7	1.02×10^{-14}	(2, 2, 2)
$(1/32)^3$	1	3–4	98	6.86×10^{-15}	(3, 1, 1)
$(1/32)^3$	1	5–8	604	3.09×10^{-15}	(2, 2, 8)
$(1/32)^3$	1	9–15	3367	2.73×10^{-15}	(10, 9, 9)
$(1/32)^3$	$1 + 0.1i$	2–15	4076	9.79×10^{-15}	(2, 2, 2)
$(1/4)^3$	1	2–15	4076	8.21×10^{-15}	(13, 13, 13)
$(1/4)^3$	$1 + 0.1i$	2–15	4076	1.02×10^{-14}	(2, 2, 2), (3, 0, 0)

Two independent expansions agree to 1.02×10^{-14} per entry over 16 340 offsets at $\lambda/32$ and $\lambda/4$, both frequencies. The per-part columns of the same table reach 8.9×10^{-12} , and they are the Taylor route’s floor rather than this method’s: that route forms one complex sum, so its imaginary part is accurate only to $\varepsilon |\operatorname{Re} T|$, whereas for real f the multipole form splits into two independent real sums, $\operatorname{Re} T$ from y_l and $\operatorname{Im} T$ from j_l (Section 4.9).

Where the k -series fallback is used the overlap is checked the other way, against the graded volume reference (table `f_ksr.txt`): route (iii) lands at 2.3×10^{-17} to 9.8×10^{-17} in max-norm and $\leq 9.8 \times 10^{-17}$ per entry on all ten of its rows, at both frequencies. Two offsets just outside the routed set, (0, 0, 20) and (0, 1, 30), were computed both ways: the octant route the library actually takes gives 5.3×10^{-16} to 2.5×10^{-15} and the forced k -series 2.5×10^{-17} to 6.7×10^{-17} , so the two routes agree with each other and with the reference across the seam.

10 The build, before and after

`notes/farfield/bench.jl` drives the real `GlaVacOprMem` build stage by stage through `GilaElectromagnetics.GilaVacuum`, on a quiet machine, and nothing under `src/` is touched.

Before is the stages of `genEgoCrcSlf!` called one by one: `wekTrp` (the contact integrals, `intOrd` = 48), the threaded `egoFunInn!` loop over every Toeplitz offset, `egoFunSng!` on the eight contact offsets, the identity term, `egoToeCrc!` into the doubled circulant, `gthEgoCmp!`, `genEgoFur`. Since

`wekTrp` is memoized on (scale, order, frequency) it is timed once per shape and that time is carried into both totals. *After* is the same with `farBlock!(egoToe, s, f)` in place of the `egoFunInn!` loop, `farSetup` timed separately as the one-time table, everything else identical. Both end in `GlaVacOprMem(cmpInf, egoFur, vol, vol)`, so both are usable operators, and `bench.jl` checks that the staged *before* path reproduces `GlaVacOprMem`'s own `egoFur` **bit for bit** (0.000×10^0 at every size) before comparing anything.

10.1 Times

Machine as in Section 2.1. The twelve-thread run started at load average 2.80; the single-threaded run followed it immediately and started at 11.05 decaying from the first run's own threads, which is why its contact time is 28.6 s against the 26.9 s of the baseline profile and 5.1 s on twelve threads. “table” is `farSetup` with the shape table already on disk (`tables/bench.1.txt`, `tables/bench.12.txt`). All times in seconds.

build	thr	before			after				speedup
		contact	far	total	contact	table	far	total	
32^3 , $\lambda/32$	1	28.59	22.80	51.46	28.59	0.68	0.026	29.37	$1.8\times$
64^3 , $\lambda/32$	1	28.59	149.31	178.19	28.59	0.69	0.172	29.72	$6.0\times$
128^3 , $\lambda/32$	1	28.59	1157.46	1189.69	28.59	0.70	1.491	34.26	$34.7\times$
32^3 , $\lambda/8$	1	27.12	45.40	72.60	27.12	0.85	0.034	28.08	$2.6\times$
32^3 , $\lambda/4$	1	27.17	91.84	119.09	27.17	0.90	0.047	28.19	$4.2\times$
32^3 , $\lambda/32$	12	5.14	4.29	9.49	5.14	0.71	0.005	5.92	$1.6\times$
64^3 , $\lambda/32$	12	5.14	25.91	31.15	5.14	0.76	0.034	6.03	$5.2\times$
128^3 , $\lambda/32$	12	5.14	200.76	206.54	5.14	0.73	0.200	6.73	$30.7\times$
32^3 , $\lambda/8$	12	3.57	9.82	13.44	3.57	0.89	0.006	4.52	$3.0\times$
32^3 , $\lambda/4$	12	3.59	14.87	18.52	3.59	0.93	0.008	4.59	$4.0\times$

The far-field fill itself:

build	1 thread			12 threads		
	before	after	factor	before	after	factor
32^3 , $\lambda/32$	22.80 s	0.026 s	$882\times$	4.29 s	0.005 s	$880\times$
64^3 , $\lambda/32$	149.31 s	0.172 s	$866\times$	25.91 s	0.034 s	$752\times$
128^3 , $\lambda/32$	1157.46 s	1.491 s	$776\times$	200.76 s	0.200 s	$1002\times$
32^3 , $\lambda/8$	45.40 s	0.034 s	$1348\times$	9.82 s	0.006 s	$1607\times$
32^3 , $\lambda/4$	91.84 s	0.047 s	$1947\times$	14.87 s	0.008 s	$1963\times$

The headline the task statement asked for: a 128^3 self-volume Green function at $\lambda/32$ whose far field takes 1.491 s single-threaded and 0.200 s on twelve threads, against 1157 s and 201 s — seconds, not minutes — and a whole build of 34.26 s and 6.73 s against 1189.69 s and 206.54 s. The far-field speedup rises with the cell size, from $882\times$ at $\lambda/32$ to $1947\times$ at $\lambda/4$, because `quadOrd` raises the quadrature order with ks while the multipole order L is set by ρ and barely moves (Section 8.3).

10.2 Where the time goes now

The build is now the contact integrals and nothing else. At 128^3 , $\lambda/32$ on twelve threads, 5.14 s of the 6.73 s total is `wekTrp` at `intOrd = 48`; the far field is 0.200 s, the geometry-table setup 0.73 s (and zero on the second frequency of the same shape), the touching shell 0.060 s, and the

embedding, gather and FFT together 0.60 s. Single-threaded the same build is $28.59 + 0.70 + 1.49 + 0.05 + 3.43$ s. The *after* total is nearly flat in N — 29.37, 29.72, 34.26 s single-threaded and 5.92, 6.03, 6.73 s on twelve threads at 32^3 , 64^3 , 128^3 — because everything that grows with N has been removed except the FFT, which is 2.91 s of the 128^3 single-threaded total.

This is the attribution the task statement asked for, and it points at the next task rather than at this one. **wekTrp** evaluates **wekS**, **wekE** and **wekV** by an order-48 tensor rule; the companion document *The Moments of Gila* gives closed forms for every one of those integrals and a k -series built from them that reproduces the order-48 values to 6×10^{-11} and is about seven times faster. Section 9.5 adds the accuracy argument: on a slender cell that same order-9 shell rule is wrong by 117% and is the sole cause of the operator’s indefiniteness. *The contact integrals are now both the whole cost and the whole error of the build, and the tool that fixes them already exists.*

10.3 Accuracy of the assembled operator, before and after

Three checks, printed for every configuration. The first is a consistency check on the harness; the second and third are the ones that matter.

	4^3	16^3	32^3	128^3
staged <i>before</i> vs GlaVacOprMem ’s own egoFur	0	0	0	0
<i>after</i> vs <i>before</i> , $\lambda/32$	5.32×10^{-14}	2.06×10^{-14}	1.66×10^{-14}	9.89×10^{-15}
both operators on one random vector	2.57×10^{-14}	2.55×10^{-14}	1.32×10^{-14}	—

and at the coarser cells, 32^3 : *after* vs *before* is 8.71×10^{-15} at $\lambda/8$ and 5.88×10^{-15} at $\lambda/4$, with the operator applied to a random vector differing by 1.09×10^{-14} and 1.15×10^{-14} .

That difference is Gila’s error, not this library’s. The worst separated offset of the 128^3 block at $\lambda/32$ is 5.579×10^{-13} at $(0, 5, 5)$, and Section 9.1 puts **farTensor** at 4.2×10^{-15} against the 220-bit reference there; at $\lambda/8$ the worst is 1.882×10^{-13} at $(0, 2, 0)$ and at $\lambda/4$ it is 8.736×10^{-14} at $(0, 0, 2)$. The **egoFur** difference *shrinks* as N grows — 5.3×10^{-14} at 4^3 , 9.9×10^{-15} at 128^3 — because the transform averages a fixed per-offset error over more offsets while $\max|\mathbf{egoFur}|$ is set by the self term. So the honest statement is:

The assembled operator changes by 6×10^{-15} to 5×10^{-14} relative, and every part of that change is Gila’s fixed Gauss–Legendre rule being replaced by something 30 to 10^5 times more accurate.

The per-entry accuracy of the far tensor itself, before and after, is Sections 2.3 and 9.1:

worst tensor error per entry vs the 220-bit reference	before	after
cube $\lambda/32$, separations 2–64, three directions	7.9×10^{-13}	4.4×10^{-15}
cube $\lambda/8$	3.2×10^{-13}	5.5×10^{-15}
cube $\lambda/4$	2.8×10^{-13}	9.3×10^{-15}
slender, long axes	3.4×10^{-11}	8.7×10^{-15}
slender, short axis $(0, 0, n)$	1.60	8.7×10^{-15}

10.4 Per-offset cost

Part (i), table **i_cost.txt**: one offset of route (i) at fixed L , single-threaded, zero allocations, with the h_l and Y_{lm} recursions and Miller’s downward j_l inside the timing. “terms” is the number of complex multiply–adds the route issues.

shape	L	terms	ns/offset	ns/term
c32	4	51	300.0	5.88
c32	8	152	412.5	2.71
c32	12	307	604.2	1.97
c32	16	516	866.6	1.68
c32	20	779	1229.2	1.58
c32	30	1673	2564.6	1.53
c32	40	2904	4891.6	1.68
c32	56	5576	9216.6	1.65

so $t(L) \approx 1.6$ ns per term above $L = 12$, and the whole-box evaluation costs under a microsecond at every L the far plateau uses. Route (ii), the eight octants, costs 53–84 μ s at the offsets that need it (67 256 to 104 384 terms at 0.79–0.81 ns each) — and it is used at twelve offsets of two million.

The mean over a whole 32^3 octant is 360 terms per offset at $\lambda/32$, 473 at $\lambda/8$, 687 at $\lambda/4$ and 644 on the slender cell (the last including its 67 k -series and 104 octant offsets), against the task statement’s target of “a few hundred floating-point operations per offset”. *The target is met in the currency of terms and exceeded by a factor of a few in flops*, and the resulting fill is 706 to 1410 ns per offset single-threaded (0.185 s for a 64^3 block at $\lambda/32$, 0.046 s for a 32^3 block at $\lambda/4$, 0.040 s for a 32^3 slender block including its 67 k -series offsets read from the cache) — against 498.1 μ s for the order-4 Gauss–Legendre rule it replaces, a factor of 350 to 700 per offset before threading. The 128^3 fill of Section 10.1 is 1.491 s over 2 097 144 offsets, i.e. 711 ns each.

11 Unequal cells: the far field of a composite volume

Everything before this section assumed that target and source cell have the same edges. Gila’s composite volumes (`GlaCmpVol`) put a coarse region next to a fine one, and the external block between two regions of different scale is a family of pairs with $\mathbf{s}_t \neq \mathbf{s}_s$. Touching unequal pairs are already handled by the gcd-cell averaging of `egoCntOut!`; this section is about the *separated* ones. The mathematics changes in exactly one place, the per-axis weight, and the rest of the method carries over with its proofs. What does not carry over is the near band: the sub-box split the task statement proposed was implemented, measured, and found not to converge where it was needed, and the near band is served instead by an identity that reduces every unequal pair on Gila’s lattice to a weighted box sum over the equal-cell tensors of the fine cell. The section names every number’s source; the five sub-agent reports are `scratch/reports/xtheory.md`, `xtable.md`, `xgeom.md`, `xref.md`, `xnear.md`, the integration report `xunify.md`, and the verification report `xverify.md`.

11.1 The trapezoid weight

For a target cell of edges $\mathbf{s}_t$ centred at $\mathbf{R}$ and a source cell of edges $\mathbf{s}_s$ at the origin, put per axis

$$a_i = \frac{1}{2}|s_{t,i} - s_{s,i}|, \quad b_i = \frac{1}{2}(s_{t,i} + s_{s,i}), \quad b_i - a_i = \min(s_{t,i}, s_{s,i}), \quad b_i^2 - a_i^2 = s_{t,i}s_{s,i}. \quad (57)$$

The convolution of the two intervals is a trapezoid rather than a triangle: flat at height $b_i - a_i$ on $|t| \leq a_i$, falling as $b_i - |t|$ on $a_i \leq |t| \leq b_i$, zero beyond, with kinks at $\pm a_i$ and $\pm b_i$ and none at 0. The volume form of Theorem 3.5 is otherwise unchanged, $T_{ab} = V_t^{-1} \int_D w [(\partial_a \partial_b + \delta_{ab} k^2)g](\mathbf{R} + \boldsymbol{\delta}) d\boldsymbol{\delta}$ on $D = \prod [-b_i, b_i]$ with $V_t = \prod s_{t,i}$ the *target* volume (`srfMat` still divides by V_t). The one fact that does all the work is that

$$w_i(t) = (b_i - |t|)_+ - (a_i - |t|)_+, \quad (58)$$

the trapezoid is the triangle of half-width b_i minus the triangle of half-width a_i . Every linear functional of w_i is therefore a difference of two triangle functionals, and the three-term moment formula of the task statement collapses to

$$\mu_n(a, b) = \int_{-b}^b w t^n dt = \frac{2(b^{n+2} - a^{n+2})}{(n+1)(n+2)} = \mu_n(b) - \mu_n(a) \quad (n \text{ even; } 0 \text{ for odd } n), \quad (59)$$

with $\mu_n(s)$ the triangle moment (7). Four consequences follow at once. The weight is even on every axis, so the parity argument of Section 4.5 holds verbatim: only even l survive and the same four (l, m) classes are stored. The geometry table is the equal-cell table `tabWhl` with `wgtTrpX(a, b)` in place of `wgtTrpT(s)` and nothing else changed; at $a = 0$ the subtraction of an exact zero leaves the same `BigFloat` operations, which is why the equal-cell tables are bit-identical (Section 11.6). The table is stored divided by $\prod b_i$, and since the raw integral is even under $(\mathbf{R}, \mathbf{s}_t, \mathbf{s}_s) \rightarrow (-\mathbf{R}, \mathbf{s}_s, \mathbf{s}_t)$,

$$G(\mathbf{R}; \mathbf{s}_t, \mathbf{s}_s) = \frac{V_s}{V_t} G(-\mathbf{R}; \mathbf{s}_s, \mathbf{s}_t), \quad (60)$$

so one table serves the *unordered* pair, keyed on $(\mathbf{a}, \mathbf{b})$; the caller supplies the factor $\prod b_i/V_t$ (folded into the frequency prefactor) and the sign of $\mathbf{R}$. Finally $V_t^{-1} \int_D w = \prod (b_i^2 - a_i^2)/V_t = V_s$, so the a priori scale $\text{est}(\mathbf{R})$ of Section 6.7, which for equal cells multiplied the pointwise dyadic scale by $\int w/V_t = V_t$, must multiply by the *source* volume V_s ; consistently, $\text{est}(\mathbf{R}; \mathbf{s}_t, \mathbf{s}_s)/\text{est}(-\mathbf{R}; \mathbf{s}_s, \mathbf{s}_t) = V_s/V_t$ as in (60). The difference form (59) loses at most $\log_{10}(1/(1 - (a/b)^{n+2})) \leq 0.66$ digits in `Float64` at ratio 16 and is exact in the rational arithmetic the table is built in.¹

11.2 The bounds: Theorem A', and why Theorem B needs no change

For the bounds the two triangles are *not* subtracted, because Theorem A works with positive measures. Write $w'_i = -\text{sign}(t) \mathbf{1}_{a_i < |t| < b_i}$ and $w''_i = \delta_{-b_i} + \delta_{b_i} - \delta_{-a_i} - \delta_{a_i}$ distributionally; the positive measures are $|w'_i| dt$, the flat measure on the two ramps only, with moments $\lambda_n(a, b) = 2(b^{n+1} - a^{n+1})/(n+1)$, and $|w''_i|$, four unit point masses with moments $2b_i^{2i} + 2a_i^{2i}$. At $a_i = 0$ the two masses at $\pm a_i$ merge into the triangle's $2\delta_0$, whose moment vector $[2, 0, 0, \dots]$ is the equal-cell code's `zro`. The analyticity radius is $r_d = |\mathbf{b}| = \max_D |\delta|$ and $\rho = |\mathbf{b}|/|\mathbf{R}|$.

Lemma 11.1 (integration by parts onto the trapezoid weight). *Extend w by zero. For $\phi \in C^2$ near $\overline{D}$, $a \neq b$, c the third axis, and Q_a the rectangle in the axes b, c ,*

$$\int_D w \partial_a \partial_b \phi = \int_D \text{sign}(\delta_a) \text{sign}(\delta_b) \mathbf{1}_{a_a < |\delta_b| < b_a} \mathbf{1}_{a_b < |\delta_b| < b_b} w_c(\delta_c) \phi d\delta, \quad (61)$$

$$\int_D w \partial_a^2 \phi = \int_{Q_a} w_b w_c [\phi|_{\delta_a=b_a} + \phi|_{\delta_a=-b_a} - \phi|_{\delta_a=a_a} - \phi|_{\delta_a=-a_a}] d\delta_b d\delta_c. \quad (62)$$

Proof. Verbatim Lemma 6.1: w is Lipschitz and vanishes on ∂D , so both boundary terms vanish; $w'_a w'_b$ is sign sign on the ramps and 0 on a plateau, which is (61); and $-\int w'_a \partial_a \phi = -\int_{-b_a}^{-a_a} \partial_a \phi + \int_{a_a}^{b_a} \partial_a \phi = \langle w''_a, \phi \rangle$, which is (62). Only the support of w'_a and the location of the point masses changed. When $a_a = 0$ the last two evaluations coincide and give $-2\phi(0)$: the triangle lemma. $\square$

Theorem 11.2 (Theorem A for the trapezoid box). *Let $\rho = |\mathbf{b}|/|\mathbf{R}| < 1$. Then (46) holds with $r_d = |\mathbf{b}|$, V_t the target volume in the prefactor, and, for $l = 2p$,*

$$V_l^{ab} = \sum_{i+j+q=p} \frac{p!}{i!j!q!} \lambda_{2i}(a_a, b_a) \lambda_{2j}(a_b, b_b) \mu_{2q}(a_c, b_c) \quad (a \neq b), \quad (63)$$

¹`scratch/reports/xtheory.md`, Section 1.1; the ratio-16 figure is $a/b = 15/17$ at $n = 0$.

$$V_l^{aa} = 2 \sum_{i+j+q=p} \frac{p!}{i!j!q!} (b_a^{2i} + a_a^{2i}) \mu_{2j}(a_b, b_b) \mu_{2q}(a_c, b_c) + |k|^2 W_l, \quad W_l = \sum_{i+j+q=p} \frac{p!}{i!j!q!} \mu_{2i} \mu_{2j} \mu_{2q}. \quad (64)$$

Every term is positive; only even l contribute.

Proof. Each step of the proof of Theorem 6.4 was checked against the new weight. Steps that do not see w are unchanged: $\partial_{\mathbf{R}} = \partial_{\boldsymbol{\delta}}$ on $g(\mathbf{R} + \boldsymbol{\delta})$, the representation $T_{ab} = \frac{ik}{f^2 V_t} \sum (-1)^l \psi_{lm}(\mathbf{R}) \mathcal{M}_{lm}^{ab}$ (it is the definition of the volume form with the new w), the Cauchy–Schwarz over m and Lemma 6.2. Three steps see w . (i) Uniform convergence of (14) with all $\boldsymbol{\delta}$ -derivatives on $\overline{D}$ needs $\max_{\overline{D}} |\boldsymbol{\delta}| = |\mathbf{b}| < |\mathbf{R}|$. (ii) Lemma 11.1 replaces Lemma 6.1, and Minkowski’s integral inequality applies because $|w'_a| dt$, $|w''_a|$ and $w dt$ are positive measures; this is the only place $w \geq 0$ is used. (iii) Theorem 5.1 is applied with $x \leq r_d = |\mathbf{b}|$, which holds on the ramps and at $\pm a_i, \pm b_i$, all inside $\overline{D}$, and the multinomial expansion of $|\boldsymbol{\delta}|^{2p}$ against the product measures gives (63)–(64) with the new one-dimensional moments. Odd l vanish because the parity argument of Section 4.5 uses only that w is even in every coordinate. Nothing else changes; at $a_i = 0$ on every axis the statement is Theorem 6.4 term by term, and the code’s `farMomW(1Top, a, b)` at $\mathbf{a} = 0$ reproduces `farMomW(1Top, s)` bitwise in `Float64`, `BigFloat` and `Rational{BigInt}` (359 comparisons, 0 mismatches).² $\square$

Remark 11.3 (sub-boxes and the gcd average). Theorem 6.6 needs no change. Its hypotheses are an affine weight $u_i = \alpha_i + \beta_i t \geq 0$ on $[-h_i, h_i]$ and $\rho_j = |\mathbf{h}|/|\mathbf{R} + \mathbf{c}_j| < 1$. A ramp $[a_i, b_i]$ is the octant piece with $h_i = (b_i - a_i)/2$, $\alpha_i = h_i$, $\beta_i = \mp 1$; a plateau $[-a_i, a_i]$ has $h_i = a_i$, $\alpha_i = b_i - a_i$, $\beta_i = 0$, for which the one-dimensional identities of the proof hold with their β terms absent ($\nu_i = \pi_i = (b_i - a_i)(\delta_{h_i} + \delta_{-h_i})$, no β_i -measure); the proof never uses $\beta_i \neq 0$ or $\alpha_i = h_i$. The gcd average

$$G(\mathbf{R}; \mathbf{s}_t, \mathbf{s}_s) = \frac{1}{N_t} \sum_{j=1}^{N_t} \sum_{j'=1}^{N_s} G(\mathbf{R} + \mathbf{c}_j - \mathbf{c}'_{j'}; \mathbf{g}, \mathbf{g}), \quad g_i = \min(s_{t,i}, s_{s,i}), \quad (65)$$

with $\mathbf{c}_j, \mathbf{c}'_{j'}$ the centres of the N_t target and N_s source sub-cells of edges $\mathbf{g}$, is the decomposition of the measure $w d\boldsymbol{\delta}$ into $N_t N_s$ translated triangle measures (a sum of indicator convolutions), so each piece is Theorem 6.4 at the real offset $\mathbf{R} + \mathbf{c}_j - \mathbf{c}'_{j'}$ with $r_d = |\mathbf{g}|$, and the errors add with the weights $1/N_t$: a budget of $\text{tol} \cdot \text{est}/N_s$ per piece certifies $\text{tol} \cdot \text{est}$ for the sum. The factor $1/N_t$ and not $1/(N_t N_s)$ is because G already carries the source volume.

Remark 11.4 (the scale, the lower bound, the n -tail). The proven lower scale `estLo` keeps its two hypotheses with $r_d = |\mathbf{b}|$: $|k| |\mathbf{b}| \leq \pi$, true for all twelve test shapes of Section 11.3 at $g = \lambda/32$, $f = 1$ (0.405 at ratio 2 on one axis to 2.89 at ratio 16 on three), and the $l \geq 2$ tail below the leading shell, whose factor becomes $|k| V_s j_0(|k| |\mathbf{b}|)/(4\pi |f|^2)$. Measured, the second fails (`estLo` = 0) at the nearest offsets for ratio ≥ 4 on two or three axes and at every offset up to 32 fine cells for ratio 16 on two or three axes; where it holds, `estLo/est` is 0.005–0.03 at one cell of gap and 0.02–0.42 at thirty-two, so `est` remains the scale, as for equal cells.³ The n -tail certificate of the k -series order integrates the same pointwise remainder majorant against the same measures, so it is Theorem 11.2 with $|k|^l e^{\dots}/(2l+1)!! \rightarrow j_{\text{tail}}(l, N, |k| |\mathbf{b}|)/|\mathbf{b}|^{l+2N+2}$ and $V_l^{ab} \rightarrow V_{l+2N+2}^{ab}$; in the code `nCutVec` now takes the prefactor volume V_t , the scale volume V_s and r_d as separate arguments. Route (iii)’s $x = |k| D_{\max}$ uses $D_{\max} = (\sum_i (|R_i| + b_i)^2)^{1/2}$, the largest distance between a point of the target and a point of the source, which is what `panSpan` already returns for unequal panels.

²`scratch/xwork/xtheory/out_xbit.txt`.

³`scratch/xwork/xtheory/out_xlow.md`.

The exact-rational checks. Everything in this subsection that is an identity was checked in `Rational{BigInt}` arithmetic, on polynomials in the six coordinates or the three difference coordinates, with box integrals by exact antiderivatives and no floating-point number anywhere.⁴ Every one of the 132 rows is exactly 0:

identity	cases	what was compared
$\mu_n(a, b)$ (59)	7 shapes $\times$ 3 axes $\times$ $n \leq 20$	the two-term form, the task statement's three-term form, and the direct piecewise integral (plateau plus two ramps)
convolution	7 shapes $\times$ 125 monomials	$\int_T \int_S (\mathbf{x} - \mathbf{y} - \mathbf{R})^\alpha$ against $\prod_i \mu_{\alpha_i}(a_i, b_i)$, $\alpha_i \leq 4$, $\mathbf{R} = (7/3, -2/5, 11/7)$
Lemma 11.1	7 shapes $\times$ 343 monomials $\times$ 6 (p, q)	$\int w \partial_p \partial_q \phi$ against (61) and (62), $\phi = \delta^e$, $e_i \leq 6$
V_l^{ab} , W_l	7 shapes $\times$ $l \leq 20 \times 7$ entries	the multinomial code <code>farMomW(a, b)</code> against the expansion of $ \delta ^l$ integrated against the piecewise one-dimensional moments
swap (60)	6 shapes $\times$ 2 $\mathbf{R} \times j \leq 3$	the raw integral of $ \mathbf{x} - \mathbf{y} ^{2j}$ and its six second derivatives, both orientations
gcd (65)	ratios 2, 3; 1-3 axes; both orientations and mixed	the raw integral against the sum over $N_t N_s \leq 27$ equal-cell pieces, and G against the $1/N_t$ average
weight form	7 shapes $\times$ $j \leq 3$	$V_t^{-1} \int_D w \mathbf{R} + \delta ^{2j}$ with trapezoid moments against the face-pair form

11.3 Where Gila produces unequal separated pairs, and how it computes them today

Reading `src/glaCmpOpr.jl`, `glaCmpVol.jl` and `vacuum/glaVacOprMemGen.jl`, and confirming each statement by logging the calls of a real `GlaVacOprMem` build, gives a picture that differs from the task statement's in one architectural respect.⁵ A composite region pair of different scale whose bounding boxes *touch* (face, edge or corner; the test is a closed-box test) is not built with unequal cells at all: `_sndBlk` remeshes the coarse region at the fine scale and builds an *equal-cell* external operator with the contact correction, wrapped in `GlaSndOprVac`. For a 64^3 fine region against a 4^3 neighbour of $16g$ that is a $64^3 \times 64^3$ equal-cell block at g with 2.1×10^6 offsets, all of them the equal-cell library's. The unequal-cell `genEgoCrcExt!` runs in two situations: (a) the direct constructor `GlaOprVac(trg, src)` on two `GlaVols` of different scale, touching or not, which is the route the test suite exercises (`test/crsSclTest.jl`); and (b) composite region pairs of different scale that do *not* touch, i.e. nested refinement (a fine core inside a ring inside a coarse shell, core against shell) and two-body operators. In (b) the ring cannot be one coarse cell thick: `chkCmpVol` demands even cell counts per region and `chkParCmpVol` an even per-partition sum for every region pair, and a one-coarse-cell ring of any finer scale gives an odd sum and is rejected, so the thinnest legal gap is *two* coarse cells.

The lattice. Every non-contact offset $\mathbf{R}$ of both routes has $2\mathbf{R}/g$ integer, and R_i/g is a half-integer on every coarsened axis of even ratio: $R_i = (r_i + 1)g/2 + \kappa g$ for a face gap of κ fine cells, and the lateral positions $(r_i - 1)g/2 - jg$ are half-integers too. An unequal-cell table would therefore be keyed on the half-lattice $2\mathbf{R}/g$, which is what the block's circulant index already is. The gcd average (65) is where the half-integers cancel: the source sub-cell centres $c'_j = (-(r - 1)/2 + j)g$ are

⁴`scratch/xwork/xttheory/xexact.jl` $\rightarrow$ `out_xexact.txt`; shapes $(g, g, g)/(g, g, g)$, $(g, g, g)/(2g, g, g)$, $(g, g, g)/(16g)^3$ ($a/b = 15/17$), $(g, g, g)/(100g, g, 4g)$ ($a/b = 99/101$), $(3g, 5g, g)/(5g, 3g, 7g)$, $(g, g, 16g)/(g, g, 2g)$ and the slender $(g, g, 1/512)/(1/8, 1/8, 1/512)$.

⁵`scratch/reports/xgeom.md`, Sections 0, 1, 1b, 5; scripts `scratch/xwork/xgeom/enum.jl`, `instr.jl`, `nest.jl`.

half-integers for even r , so $R_x - c'_j = (r + \kappa - j)g$ runs over the *integer* fine offsets $\kappa + 1, \dots, \kappa + r$, and laterally $(r - 1)/2 - j' + \text{lat}/g$ is an integer for the same reason. The nearest sub-offset of the $\kappa = 1$ layer has max-norm 2, the one-fine-cell-gap offset that the equal-cell library routes to the octants. Since $g = 1/32$ and r is a power of two, every offset is a dyadic rational, the `Float64` values Gila’s `grdSel` produces are exact, and no classification depends on rounding.

Offset counts and classes. Per cell pair Gila classifies: contact if $|R_i| < \text{cntTol} + (s_{t,i} + s_{s,i})/2$ on every axis (`egoCntOut!`, gcd-cell self-tensor averaging); else adaptive (`egoSrfAdp!`, `hcubature` at relative tolerance 10^{-6} , absolute 10^{-9}) if $\text{sep} = \max_i \text{round}(|R_i| / \max(s_{t,i}, s_{s,i})) \leq 1$; else the fixed rule `egoSrfFxd!` with `quadOrd(sep, kSc1)`. On the test-suite geometry (a coarse 2^3 block of cells rg on the axes of the set A sharing its full face with a fine block of g , the fine cell the target) the exact-rational enumeration reproduces the real build’s logged $(\mathbf{R}, \text{class}, \text{sep}, \text{ord})$ set with 0 missing, 0 extra and 0 class disagreements on five builds, and obeys closed forms: $27r^{|A|}$ distinct offsets, $\prod_{\text{lateral}}(r_i + 2)$ contact offsets, and the adaptive class is every offset with $|R_i| < 1.5 \max(s_{t,i}, s_{s,i})$ on all axes minus the contact ones.⁶

coarse cell ($g = 1/32$)	offsets	contact	adaptive	fixed	nearest non-contact $ \mathbf{R} /g$
$r = 2, x$	54	9	9	36	2.5
$r = 2, xyz$	216	16	56	144	2.5
$r = 4, x$	108	9	27	72	3.5
$r = 4, xyz$	1728	36	540	1152	3.5
$r = 8, x$	216	9	63	144	5.5
$r = 8, xyz$	13 824	100	4508	9216	5.5
$r = 16, x$	432	9	135	288	9.5
$r = 16, xyz$	110 592	324	36 540	73 728	9.5

The nearest non-contact offset $((r + 3)/2)g$ is 1.25, 0.875, 0.6875, 0.594 coarse cells for $r = 2, 4, 8, 16$ and always rounds to $\text{sep} = 1$: the whole first coarse cell of separation is adaptive, and the fixed rule starts at $\kappa = r$. For the two realistic face-sharing blocks of route (a), and for the nested-refinement blocks of route (b) with the two-cell ring around a 1λ core:⁷

block	offsets	contact	adaptive	fixed	$\rho \geq 0.6$	$\rho \geq 1$
(a) fine 64^3 of g vs coarse 4^3 of $16g$ ($\lambda/2$), face	1 404 928	324	36 540	1 368 064	15 936	1456
(a) fine 64^3 of g vs coarse 8^3 of $8g$ ($\lambda/4$), face	1 728 000	100	4508	1 723 392	2476	196
(b) core 32^3 of g vs shell of $2g$, two-cell ring	1 118 192	0	0	1 118 192	0	0
(b) core vs shell of $4g$	1 528 704	0	0	1 528 704	0	0
(b) core vs shell of $8g$	2 616 320	0	0	2 616 320	0	0

In route (b) the nearest offset is $((5r + 1)/2)g$ with $\text{sep} = 3$ for every r and the largest whole-box ratio is $\sqrt{3}(r + 1)/2$ over $\sqrt{((5r + 1)/2)^2 + 1/2}$, i.e. 0.4685, 0.4115, 0.3800 for $r = 2, 4, 8$: the whole trapezoid box alone serves every composite non-touching cross-scale block. The near band exists only through route (a), and there it is 1.1 % of the $\lambda/2$ block. Gila’s cost for that block today is about 1.8 h single-threaded (1.37×10^6 fixed-rule calls at 13/4.7/2.5 ms for orders 9/7/6 plus 36 540 adaptive calls at 1–54 ms).

Gila’s accuracy today. Both of Gila’s separated rules were evaluated with `genEgoCrcExt!`’s own arguments at every key of the 220-bit reference cache of Section 11.6 (356 distinct keys at three

⁶`scratch/xwork/xgeom/enum_out.md, instr_out.md.`

⁷`scratch/reports/xgeom.md, Sections 3 and 5; scratch/xwork/xgeom/nest_out.md.`

frequencies, twelve coarse shapes) and compared entry by entry.⁸ The metric is the max-entry-normalised error $\max_{ab} |G - G_{\text{ref}}| / \max_{ab} |G_{\text{ref}}|$; the worst per-entry relative error over entries above $10^{-30} \max |G_{\text{ref}}|$ is in brackets where it differs by more than a factor two.

rule	where	error against the 220-bit reference, $f = 1$
adaptive (sep ≤ 1)	111 offsets, all shapes	2.7×10^{-9} to 2.28×10^{-4} , median 1.2×10^{-6} ; 106 of 111 above 10^{-8} , 13 above 10^{-5} ; worst two are slabs (8, 8, 1) at (8.5, 0, 0) g and (16, 16, 1) at (12.5, 0, 0) g
fixed, order 9 (sep = 2), cubes and slabs	$\lambda/16$ to $\lambda/2$ cells	1.5×10^{-14} to 8.3×10^{-12} ; the $\lambda/2$ cube 1.5×10^{-14} to 5.6×10^{-12} , the same as the $\lambda/16$ cube
fixed, orders 7/6/5/4, cubes and slabs	sep ≥ 3	$\leq 6.3 \times 10^{-13}$ except order 6 across the $\lambda/2$ cell at 4λ : 1.7×10^{-11} (2.1×10^{-10})
fixed, rods ($r, 1, 1$), fine cell displaced sideways	sep reached through the thin axis	8.8×10^{-13} , 2.6×10^{-9} , 1.42×10^{-6} , 1.77×10^{-4} (2.28×10^{-4}) for $r = 2, 4, 8, 16$
other frequencies $1 + 0.1i$, 0.37	80 offsets, rods and cubes	adaptive 2.0×10^{-8} to 6.3×10^{-6} , fixed 3.2×10^{-14} to 1.5×10^{-11}

The adaptive rule delivers its tolerance of 10^{-6} and nothing more, at every face gap and not only in the touching layer; it is also faster than the order-9 fixed rule from $\kappa = 2$ on (1–17 ms), the signature of `hcubature` stopping after one or two Genz–Malik steps. The fixed rule is 10^{-12} on cubes and slabs at every order and every cell size, and fails on rods displaced sideways because the classification $\text{sep} = \max_i \text{round}(|R_i| / \max(s_{t,i}, s_{s,i}))$ reaches 2 through the thin axis while the fine cell sits $1.5g$ from a $16g$ long side face: nine Gauss–Legendre points along a $\lambda/2$ panel whose singularity is $1.5g$ away. The equal-cell `quad0rd` schedule was measured on cubes; the aspect ratio of the rod is the untested dimension. The sign convention was verified on both sides: flipping R_y or R_z flips G_{ab} for $a \neq b$ involving that axis by -1.000000 in Gila and in the reference alike, with $\mathbf{R}$ the target centre minus the source centre in both, so no orientation or transposition correction is needed anywhere.⁹

11.4 Three routes, and the one that was dropped

Route 1: the whole trapezoid box. Wherever Theorem 11.2 certifies $\text{tol} \cdot \text{est}$ within the table ($L \leq 56$), one expansion per offset at 0.9–12.5 μs for $L = 12$ –56, the same code as the equal-cell whole box with the trapezoid table. On the realistic $\lambda/2$ block that is 1 387 696 of the 1 404 604 non-contact offsets, 98.8 %, and 4.15 s single-threaded summed over the L histogram, with $L \leq 24$ for 872 468 of them.¹⁰ The band edge sits at $\rho_{\text{whl}} = 0.58$ –0.60; below it the whole box converges, above it (16 908 offsets here) it does not within $L = 56$.

Route 2: the gcd average as a box sum. On Gila’s lattice every sub-offset of (65) is an integer offset of the fine cell, so the average is a weighted sum of entries of the fine equal-cell `egoToe` that the equal-cell library already fills:

$$G(\mathbf{R}) = \frac{1}{N_t} \sum_{\mathbf{t}} \left(\prod_i m_i(t_i) \right) \text{ego}[\lfloor \mathbf{m} + \mathbf{t} \rfloor] \text{ (reflected)}, \quad m_i(t) = \#\{(j, j') : j - j' = t\}, \quad (66)$$

with $\mathbf{m} = \mathbf{R}/g + (\mathbf{n}_s - \mathbf{n}_t)/2$ the integer lattice coordinate, $\mathbf{n}_t, \mathbf{n}_s$ the sub-cell counts per axis and $\prod_i (n_{t,i} + n_{s,i} - 1)$ weighted adds per offset, i.e. a convolution of the fine block with the coarse cell’s

⁸`scratch/work/xgeom/acc2.jl` $\rightarrow$ `cmp.all.md`; `scratch/reports/xgeom.md`, Sections 2b and 4.

⁹`scratch/work/xgeom/cmp.out.md`, last block.

¹⁰`scratch/reports/xtable.md`, Section 4; `scratch/work/xtable/census.txt`, `costsum.out`.

indicator, all weights $+1/N_t$. The cancellation of the sum, $\Lambda = \sum_{jj'} \max |G_{jj'}| / (N_t \max |\sum G_{jj'}|)$, was measured at 1.0–2.3 over the whole test matrix at all three frequencies (1.0 at ratio 2, 1.1–1.3 at ratio 8, 1.5–2.3 at ratio 16 on three axes),¹¹ so the average loses at most one bit to interference at these cell sizes; the a priori fear that it would cancel at $\lambda/4$ – $\lambda/2$ coarse cells was wrong, and the disagreements that raised it are Section 11.5. The form matters more than the formula: evaluated naively as $N_t N_s$ separate equal-cell tensors per offset (4096 at ratio 16 on three axes) it costs 38.9 ms per offset and 658 s for the 16 908 offsets of the near band, 160 times the whole far field; as the box sum it is one equal-cell **farBlock!** on the 33^3 fine block the band needs (0.95 s, 35 929 offsets) plus 28.6 μ s of adds per offset (0.48 s), identical to the naive sum to the last bit at the offset checked.¹² The equal-cell tolerance of the pieces is the pair’s tolerance; the a posteriori certificate of the sum is the $1/N_t$ -weighted sum of the pieces’ certificates.

Route 3: the k -series on unequal panels. Where the pair is off the gcd lattice (a lateral coordinate 0 against an axis whose source cell is an even multiple of g puts m_i at a half-integer; Gila never produces it, the reference matrix does), or where a sub-offset would touch, the moment series of the companion document is applied to the thirty-six face pairs of the two cells, which **pairMoments** takes as arbitrary axis-aligned rectangles, at 128 bits with the order N from the k -series bound at tol/Λ . It is a correctness fallback and not a cost: 6–21 s per offset at $N = 13$ –44, reached by 103 of the 438 reference records and by 0 offsets of either block (Section 11.6).

The twenty-seven-piece split: implemented, measured, dropped. The task statement expected the near band to be served as for equal cells, by sub-boxes: with kinks at $\pm a_i$ the minimal legal split per axis is three pieces, two ramps $[\pm a_i, \pm b_i]$ of affine weight and a plateau $[-a_i, a_i]$ of constant weight, twenty-seven sub-boxes with the reflection rule as before, covered by Theorem 6.6 with $\beta = 0$ on a plateau axis (Remark 11.3). The split was built in full: eight piece-type tables per pair (plateau bits per axis), selection by Theorem B with the budget $\text{tol} \cdot \text{est}$ over the pieces, and the reflection rule with the plateau piece its own mirror.¹³ It converges where the whole box nearly converges, and it does not converge where the whole box fails. The reason is geometric and was predicted before the measurement: the all-plateau piece has radius $|\mathbf{a}| = (r - 1)\sqrt{3}g/2$ on a cubic coarse cell, against a nearest separated offset of $|\mathbf{R}| \approx (r + 3)g/2$, so its ratio is 0.35, 0.87, 1.17, 1.24 at $r = 2, 4, 8, 16$, and the ramp-plateau pieces next to the target are worse. Measured, the worst piece ratio at the nearest offset $\kappa = 1$ and the piece order the bound asks for, with the fine cell centred on the coarse face (a position of the reference matrix that Gila’s lattice does not produce on an even axis) and over the corner sub-cell (a lattice position):¹⁴

¹¹`scratch/work/htable/estratio_f1.txt, estratio_f2.md, estratio_f3.md; htable.md`, Sections 2d and 3e.

¹²`scratch/reports/htable.md`, Section 4, and Section 4b: box sum against the per-offset average on 340 offsets, worst 9.3×10^{-16} , summation order.

¹³`scratch/work/htable/farx.jl: tabPce, tnsPceX!, boundLpce; htable.md`, Sections 1, 1b, 2.

¹⁴`scratch/reports/htable.md`, Section 2, rows `x1c`, `x2c`, `d1c`, `x1e`, `x1v`.

coarse cell, position	$ \mathbf{R} /g$	ρ_{whl}	L_{whl}	worst piece ρ_j	L_{split}	split time / whole time
$r = 2, x, \text{centred}$	2.5	0.825	> 56	0.522	51	123 μs / —
$r = 2, x, \text{centred}, \kappa = 2$	3.5	0.589	48	0.333	30	56.5 μs / 9.0 μs
$r = 2, x, \text{diagonal } \kappa = 1$	3.2	0.644	56	0.397	36	67.5 μs / 12.5 μs
$r = 2, xyz, \text{centred}$	2.5	1.039	> 56	0.577	> 56	—
$r = 2, xyz, \text{corner}$	2.6	1.000	> 56	0.522	52	185 μs / —
$r = 4, xy, \text{centred}$	3.5	1.050	> 56	1.049	diverges	—
$r = 4, xy, \text{corner}$	3.8	0.965	> 56	0.761	> 56	—
$r = 4, xyz, \text{centred}$	3.5	1.237	> 56	1.453	diverges	—
$r = 4, xyz, \text{corner}$	4.1	1.058	> 56	0.839	> 56	—
$r = 8, xyz, \text{centred}$	5.5	1.417	> 56	3.317	diverges	—
$r = 8, xyz, \text{corner}$	7.4	1.053	> 56	0.962	> 56	—
$r = 16, xyz, \text{centred}$	9.5	1.550	> 56	7.079	diverges	—
$r = 16, xyz, \text{corner}$	14.2	1.034	> 56	0.991	> 56	—

On the lattice positions the worst piece ratio stays below 1 but not below the 0.58–0.60 at which fifty-six shells suffice, so the split fails there as well. On the near band of the realistic $\lambda/2$ block the split converges at 0 of the 2213 offsets of a representative sector (all-plateau piece radius $13.0g$ against $|\mathbf{R}| \geq 12.7g$), and where it does converge it is 5.4 to 8.9 times slower than the whole box (56.5 against 9.0 μs , 67.5 against 12.5, 40.3 against 4.6, 39.0 against 4.5) and 30 to 100 times slower than a box-sum add. Where a reference exists it is as accurate as the other routes (1.5×10^{-15} worst over 152 offsets at $f = 1$).¹⁵ Subdividing the plateau would restore convergence, and the gcd decomposition is exactly such a subdivision (a constant weight may be cut anywhere), with no new tables, no new bound and no cancellation. The split is therefore not in `farfield.jl`; it stays in `scratch/xwork/htable/farx.jl` as the cross-check it was useful as, and the pieces carry no k -series remainder certificate of their own, which a production use would have needed.

Routing. `farRouteX` takes an exact rational offset: touching or overlapping pairs ($|R_i| \leq b_i$ on every axis, tested in exact arithmetic behind a `Float64` screen) raise an error; route 1 if Theorem 11.2 is met within the table; else route 2 if $\mathbf{m}$ of (66) is integer and every sub-offset has max-norm ≥ 2 ; else route 3. The selection runs in `Float64` as for equal cells, with the thresholds of Section 8.2 recomputed from the trapezoid moments and est with V_s .

11.5 Three defects found on the way

D11: the n -cut masked its own failure. The first whole-box results at the $\lambda/4$ and $\lambda/2$ coarse cells disagreed with the gcd average and with the reference by 5.9×10^{-14} to 6.8×10^{-10} while the gcd average sat at 4×10^{-16} to 2×10^{-15} from the reference. Two causes, in one pair of lines that the equal-cell `farSetup` shares.¹⁶ First, the per-shell k -series order N_l is certified at two radii, the nearest whole-box radius and the farthest, and `nCutVec` returns -1 for “no $N \leq n_{\text{max}}$ meets the budget”; the two vectors were combined with `max.`, which turns an uncertified -1 at one radius into the other radius’ value. Second, with the split enabled the near radius was taken from the smallest *piece* centre (1.66 g for a ramp piece at $\kappa = 1$), where the whole-box n -tail with $r_d = |\mathbf{b}|$ is vacuous ($\rho = 8$) and returns -1 at every $l \gtrsim 10$; the far-end value, $N = 0$ at high l where the far radius is 3300–7500 g , was then used at the near offsets of 11–27 g , whose high- l columns need $N = 1$ –4, and `frqBox` contracted those columns with the $n = 0$ term alone. The fix is `cutMax` (elementwise maximum with -1 absorbing, so an uncertified slot propagates and triggers the recomputation at

¹⁵`scratch/xwork/htable/recheck2.f1.md`; times from `htable.md`, Section 2, at load 8.5–12.6.

¹⁶`scratch/reports/htable.md`, Sections 2d and 2e; `scratch/xwork/htable/attrib.txt`, `ncutfull.txt`.

N_{cap}) and the whole-box n -cut certified at whole-box radii only. The four disputed offsets before and after, $f = 1$, against the 220-bit reference:

coarse cell	$\mathbf{R}/g$	ρ_{whl}	L	whole box, old n -cut	whole box, fixed	split	gcd average
$r = 8, xy$	(10.5, 3.5, 0)	0.582	54	5.9×10^{-14}	1.1×10^{-15}	9.6×10^{-16}	4.6×10^{-16}
$r = 8, xyz$	(12.5, 3.5, 3.5)	0.580	50	7.7×10^{-13}	8.2×10^{-16}	3.9×10^{-16}	1.0×10^{-15}
$r = 8, xyz$	(8.5, 8.5, 3.5)	0.623	56	3.2×10^{-12}	3.4×10^{-16}	—	8.7×10^{-16}
$r = 16, xyz$	(24.5, 7.5, 7.5)	0.551	52	6.8×10^{-10}	2.8×10^{-16}	—	1.8×10^{-15}

The n -cut vectors themselves show the mechanism. For the $\lambda/2$ cube the old vector was $N_l = 12, 12, 12, 10, 9, 8, 7, 6, 4, 3, 2$ at $l = 0, 2, \dots, 20$ and 0 from $l = 22$ on; the fixed one is $13, 12, 11, 11, 10, 10, 9, 9, 8, 8, 7$ and decreases to 2 at $l = 56$, monotone in l as $c_{ln} \sim k^{2n}/(2l+2n+1)!!$ predicts, and its $N_0 = 13$ raised the table’s n_{max} from 12 to 13 (the N_{cap} recomputation of Remark 5.6 ran for the first time on a real shape). A uniform $N = 8$ gives 1×10^{-10} on that pair, $N = 10$ gives 2×10^{-14} and $N \geq 11$ gives 3×10^{-16} . The same `max.` idiom was in the equal-cell `farSetup`; at the four shipped shapes both radii are certified at $n_{\text{max}} = 12$ ($N \leq 6$ at $\lambda/32$), so it never fired and the equal-cell results are bit-identical before and after the fix, but a cell with $|k|r_d \gtrsim 2.9$ (edge $\geq \lambda/2$ at $f = 1$) would have been silently under-truncated. A constant that is right on the tested domain is again not a certificate.

D12: the table cache was not multi-process safe. Two Julia processes extending the same shape-table file each appended the missing l -segment (the append lock is per process and nothing re-reads before appending), and a later reader concatenated the duplicated (l, m) columns and summed the shells $l = 40\text{--}54$ twice: 4.68×10^{-13} at $(6.5, 1.5, 1.5)g$ on the ratio-4 cube at $L = 54$ against a certificate of 3.4×10^{-14} , a violation by $14\times$, where the same offset from a clean table is 5.12×10^{-16} .¹⁷ The file held 602 columns of which 406 were distinct. The fix (`mrGeo`) drops a segment’s columns whose (l, m) index is already present when merging, so a duplicated file reads correctly (406 columns, 5.12×10^{-16}) and a well-formed file reads exactly as before (bit-identity re-proved below). The writer still appends without a cross-process lock; a duplicated segment now costs only disk (2.80 against 1.88 MB for one pair table).

D13: a frequency set reused at a radius where its n -cut was never certified. Found by the adversarial agent of Section 11.6 and the one defect it had to fix.¹⁸ `farSetupX` certifies the k -series cut of a pair’s whole-box table at two radii: the far end $r_{\text{hi}} = 4n_{\text{blk}}r_d$ and the near end r_{nc} , the smallest $|\mathbf{R}|$ among the offsets it was given that takes route 1. The n -tail relative to `est` decreases with $|\mathbf{R}|$ for $l \geq 2$, so the set is certified on $[r_{\text{nc}}, r_{\text{hi}}]$ and nowhere else, and nothing checked the radius of a later request. Holding a `FrqSetX` across blocks is exactly what the interface recommends, to skip the two- to three-second selection pass; a held set asked for an offset *nearer* than any it was built from silently received the far end’s N . A set built for the pair g against $(4g, g, g)$ at $f = 1$ from the single offset $(3.5, 0, 0)g$, which takes route 2, and then asked for $(6.5, 0, 0)g$, which takes route 1 at $L = 36$, returned a tensor wrong by 1.59×10^{-11} against both the 220-bit reference and a freshly built set, under a certificate of 1.7×10^{-14} of $\max |G|$: a violation by 10^3 . The same set at $(34.5, 0, 0)g$ was correct (4.4×10^{-16}), and at $(102.5, 0, 0)g$ — past r_{hi} altogether — correct in fact (3.5×10^{-16}) and certified by nothing. The fix stores r_{nc} and r_{hi} in the set, has `farRouteX` raise on

¹⁷`scratch/reports/xunify.md`, Section D.0; `scratch/xwork/xunify/attrib*.jl`; the corrupted file is kept in `xwork/xunify/tab.corrupt/`.

¹⁸`scratch/reports/xverify.md`, Sections 5 and 2b (8); `scratch/xwork/xverify/k_adv_prefix.txt` is the pre-fix table, `farfield_prefix.jl` the pre-fix source.

a route-1 request outside the interval, and certifies from the first route-1 threshold when the given offsets place no route-1 offset at all — without that third line a set built only from near offsets has a certified interval collapsed to the single point r_{hi} and refuses everything. Afterwards the reused set reproduces the fresh one bitwise at $(6.5, 0, 0)g$ (1.21×10^{-15} against the reference), and $(102.5, 0, 0)g$ is refused by name. `farTensorX` and `farBlockX!` pass their own offsets when they build the set, so the check cannot fire on the documented call paths, and it did not change a single value in the block runs; it fires only for a hand-built or reused set, which is the case that was wrong. *An interval of validity that is computed and then not carried on the object is not a certificate either.*

11.6 Verification

The reference cache. `scratch/xwork/xref/reftensors_x.txt` holds 441 records (438 distinct keys, three written twice by two agents racing with identical values) of 220-bit volume tensors for unequal pairs, keyed on the exact rationals $(\mathbf{R}, \mathbf{s}_t, \mathbf{s}_s, f)$: the twelve coarse shapes (ratios 2, 4, 8, 16 on one, two and three axes against the fine cell $g = 1/32$), face gaps $\kappa \in \{1, 2, 3, 4, 6, 8, 16, 32\}$ along x with the fine cell centred, over a corner sub-cell and over the corner-corner sub-cell, a diagonal family past the coarse edge at $\kappa \in \{1, 2, 4, 8, 16\}$, $f \in \{1, 1 + 0.1i, 0.37\}$, and two slender pairs, $(1/32, 1/32, 1/512)$ against $(1/32, 1/32, 1/64)$ and against $(1/8, 1/8, 1/512)$, along both their thin and their long axis; all 416 keys of the planned matrix are present.¹⁹ Two rules were written. Rule A is the face-pair reference of Section 9, `pairKer` on the thirty-six pairs of the two cells with the `srfSum!` signs, divided by V_t ; its own order convergence (`ordN` 44 against 64) is $1-7 \times 10^{-63}$ on all four cubic ratios, and it reproduces the equal-cell record at $(5, 1, 0)$ to 0.0. Rule B is a graded tensor Gauss–Legendre volume rule with the trapezoid weight, cut at $\pm a_i$ and $\pm b_i$ and graded by factors of two toward the singularity, sharing no code with rule A; it is 10–20 times cheaper on these geometries ($16.5 \mu\text{s}$ per point, 0.4 to 20 million points, 5 to 330 s per record) and is the production rule of the cache, with rule A the independent check. The two agree to 1.6×10^{-58} – 3.9×10^{-57} at a face gap of one fine cell and to $1-2 \times 10^{-63}$ (the 220-bit floor of a nine-entry sum is $1-4 \times 10^{-64}$) from two cells on; the near-gap discrepancy is rule B’s order-32 truncation on the two panels of width $g/2$ nearest the singularity (order 40 brings it to 1.4×10^{-64}), and its largest value, 6.8×10^{-53} at ratio 16 on three axes and $\kappa = 1$, is the same 32-point error on the same outermost panel in both rules. Every record of the twelve coarse shapes is therefore good to at least 52 digits and most to 58–63; the least accurate record of the cache is the slender pair’s thin-axis $\kappa = 1$ record at 3.0×10^{-50} , measured against both a higher order and rule A. The gcd identity (65) holds between cache records and the equal-cell reference cache to 3.2×10^{-64} when whole and pieces share a rule, and the swap identity (60) to 7.0×10^{-65} (rule A) and 1.4×10^{-64} (rule B, the 2^3 sum form). The machine time of the cache was about 4.3 h.

Against the reference: every record, four ways. `farfield.jl` evaluated all 438 records by `farBlockX!` on each group’s offsets with certificates, by `farTensorX` per offset, and both again in the swapped orientation (coarse target, fine source, offset $-\mathbf{R}$) for the identity (60).²⁰ Errors are against the 220-bit reference as in Section 9.1; “digits lost” is $\log_{10}$ of the per-entry error over ε .

¹⁹`scratch/reports/xref.md`, Sections 3, 4, 6.

²⁰`scratch/xwork/xunify/acc.jl` $\rightarrow$ `acc.txt`, `acc.rows.md`; `scratch/reports/xunify.md`, Section C. Twelve threads, load 2.4–54 (other agents), 5.56 GB peak RSS.

f	records	routes 1/2/3	worst $e_{\max}$	worst per entry	digits lost	swap identity	violations
0.37	64	36/11/17	2.7×10^{-15}	2.7×10^{-15}	1.1	3.1×10^{-16}	0
1	310	173/69/68	3.4×10^{-15}	4.7×10^{-15}	1.3	4.9×10^{-15}	0
1 + 0.1i	64	35/11/18	1.4×10^{-15}	1.6×10^{-15}	0.9	2.7×10^{-16}	0

route	records	$e_{\max}$ median / worst	per entry median / worst	certificate / max $ G $ min / median / max	certificate / actual error
1 whole box	244	4.9×10^{-16} / 2.4×10^{-15}	6.9×10^{-16} / 4.7×10^{-15}	4.6×10^{-15} / 1.3×10^{-14} / 3.5×10^{-14}	2.4 / 23.9 / 135
2 gcd box sum	91	5.8×10^{-16} / 3.4×10^{-15}	7.8×10^{-16} / 4.7×10^{-15}	4.9×10^{-15} / 1.6×10^{-14} / 3.2×10^{-14}	4.0 / 29.3 / 673
3 k -series	103	4.2×10^{-17} / 9.7×10^{-17}	6.0×10^{-17} / 9.7×10^{-17}	4.6×10^{-15} / 4.8×10^{-15} / 8.1×10^{-13}	48.7 / 115 / 8930

Over 438 tensors at fourteen unequal pairs, three frequencies and separations from one to thirty-two fine cells the worst max-norm error is 3.4×10^{-15} (a 4096-term box sum on the $\lambda/2$ cube at (12.5, 12.5, 7.5) g) and the worst per-entry error 4.7×10^{-15} , 1.3 digits; the certificate was never below the actual error (0 violations). `farTensorX` is bitwise equal to `farBlockX!` on 438 of 438 and returns the same certificate; the swapped orientation takes the same route on 438 of 438. The swap identity is exactly 0 on all 244 whole-box records, because the table is the unordered pair's and (V_s/V_t) enters as a single exact factor; it is up to 4.9×10^{-15} on the box sums (15 of 91 above 10^{-15} , the 4096 pieces visited in the reflected order) and 10^{-38} – 2.6×10^{-36} on route 3 (the 128-bit value assembled in the other face order). Route 3 was reached by 103 records, every one because the pair is off the gcd lattice, and delivers the reference to 10^{-17} ; its certificate is $\text{tol} \cdot \text{est}$ at 4.6 – 5.0×10^{-15} of $\max |G|$ on 99 of them, and 1.5×10^{-14} to 8.1×10^{-13} on the four thin-axis records of the slender pair, where est over-states $\max |G|$ by 100 to 8100 (Section 11.8). These numbers agree with the earlier prototype's recheck of the same cache (272 offsets at $f = 1$: whole box worst 1.4×10^{-15} over 147, split 1.5×10^{-15} over 152, gcd average 5.1×10^{-15} over 156; 64 offsets each at 1 + 0.1i and 0.37: 2.6×10^{-15} and 2.7×10^{-15}).²¹

The equal-cell paths did not move. The integration touched five equal-cell function bodies (`tabWhl`: two lines; `farShape`: key and file name; `farMomW`: four moment vectors; `nCutVec`: the volumes separated; `farSetup`: the `cutMax` line). The old and new files were loaded as two modules in one process, each with its own copy of the shipped tables and k -series cache, and compared: every field of fresh `tabWhl` and `tabOct` tables at 256 bits on the four shipped shapes (0 mismatches in value, sign and precision), the `farMomW`, `nCutVec` arrays, a fresh table file written by each (34 974 803 bytes, byte identical), `farTensor` at 30 offsets $\times$ 4 shapes as `UInt64` bit patterns of all nine entries and the route kind (120 of 120 identical, routes 108/12/0), and `farBlock!` on 12^3 blocks of the $\lambda/32$ cube and the slender cell (13 894 and 13 952 nonzero entries, 0 differing).²² Why it cannot move: $\mu_n(0, s) = 2(s^{n+2} - 0)/\dots$ and $x - 0 = x$ in every type, $0^0 = 1$ and $0^{2i} = 0$ is the old `zro` vector, the forwarding `nCutVec` passes the very expressions the old body computed, and `cutMax` equals `max`. whenever neither slot is -1 , which at the shipped shapes is always. `verify.jl` part (a) runs unmodified on the integrated file and its four table files are byte-identical to the staged ones, except `a_rho.txt`, whose staged copy was found truncated at four lines and is now the full 81-line table it announces.

²¹ `scratch/xwork/xtable/recheck2_f1.md`, `recheck2_f2.md`, `recheck2_f3.md`.

²² `scratch/xwork/xunify/bitid.jl` $\rightarrow$ `bitid.txt`, re-run after the D12 fix; `scratch/xwork/xtable/bitid2.out` for the prototype.

Blocks. `farBlockX!` on the two blocks of Section 11.3, $f = 1$, against the reference records that fall inside them (9 for the ratio-4 block: the corner-corner lateral $(1.5, 1.5)g$ family; 13 for the $\lambda/2$ block: the $(7.5, 7.5)g$ family and the diagonal family).²³ “Cold” is a process that has loaded the file and builds or reads the pair’s table and the fine set; “warm” has both in memory and refills the fine block per call; a caller holding the `FrqSetX` skips the selection pass as well (2.7 s warm on the large block).

block	offsets	routes 1/2/3	threads	cold	warm	warm, certified
ratio 4 cube, 2^3 vs 8^3	1692	1352/340/0	1	8.91 s	1.21 s	1.29 s
			12	4.86 s	0.99 s	1.09 s
$\lambda/2$ cube, 4^3 vs 64^3	1 404 604	1 387 696/16 908/0	1	17.92 s	9.81 s	17.63 s
			12	10.71 s	5.69 s	7.13 s

The references inside the blocks are reproduced to 8.7×10^{-16} (9 records, small block) and 3.4×10^{-15} (13 records, large block), identically at one and twelve threads. Gila today: 3.5 s adaptive plus 12.6 s fixed rule plus the contact self build for the small block, about 1.8 h for the large one, both single-threaded, at the accuracy of Section 11.3. The large block’s phases on twelve threads, warm: selection 2.94 s (serial: 1.4×10^6 exact-rational offsets converted, thresholded and n -cut certified at $2.1 \mu\text{s}$ each), route decisions 0.67 s, the fine 33^3 block 1.08 s (of which 0.8–0.95 s is the serial `farSetup` of the equal-cell set, redone because the driver passes no set; the threaded fill itself is 0.01–0.04 s), and the fill 1.00 s wall for 10.99 CPU-s of whole-box expansions ($3.2 \mu\text{s}$ each) and 0.26 CPU-s of box sums ($9.5 \mu\text{s}$ each, 4096 adds). The fill parallelises $4.6\times$ (it is memory-bound on the four (l, m) classes of the table), the certified fill $5.2\times$; what does not shrink is serial. The whole-box L histogram of the large block, 20:13 636, 22:539 332, 24:319 500, $\dots$, 56:1544, the route census and the certificate statistics are identical between the one- and twelve-thread runs and identical to the prototype’s independent count. Certificates: route 1 min / median / max 4.7×10^{-15} / 8.0×10^{-15} / 4.1×10^{-14} of $\max|G|$ with 516 872 of the 1 387 696 above tol (the rule certifies $\text{tol} \cdot \text{est}$ and $\text{est} / \max|G|$ is 1–4, as for equal cells); route 2 9.0×10^{-15} / 1.2×10^{-14} / 3.0×10^{-14} ; against the 13 references the box-sum certificate is 4–9 times the actual error and the whole-box one 15–50 times; 0 violations. The pair tables themselves cost what the equal-cell whole box costs at the same L : 3.3–5.7 s cold per pair for $L_{\text{whl}} = 40$ –56 against 4.0 s for the equal-cell whole box to 56, 0.4–2.5 MB on disk against 19.5 MB for an equal-cell file that also carries the octant table, and +0.2–0.3 GB over the loaded process; the only pair whose n -cut exceeds the floor $n_{\text{max}} = 12$ is the $\lambda/2$ cube (13).²⁴

The adversarial verification. A separate agent, `xverify`, was given the integrated library with the instruction to break it, in the manner of the audit of Section 12.10, in parallel with the writing of this section. It re-derived the routing and lattice arithmetic from the definitions before running anything, re-evaluated the whole reference cache with its own script, its own reference reader and its own error metrics, built new 220-bit references for geometries this work had not tried, and compared Gila against the library on two blocks.²⁵

²³`scratch/work/xunify/block.jl` $\rightarrow$ `block_r4_t1.txt`, `block_r4_t12.txt`, `block_big_t1.txt`, `block_big_t12.txt`; `scratch/reports/xunify.md`, Section D. Single-thread runs at load 6.2–9.4 with another twelve-thread job alongside (upper bounds); the twelve-thread runs were the only Julia process, one-minute load 4.9–5.8 including themselves.

²⁴`scratch/work/xunify/build.txt`; `xunify.md`, Section E.

²⁵`scratch/reports/xverify.md`; the deliverable is `notes/farfield/verify.jl` parts (j)–(m), writing `tables/j_xref.txt`, `k_adv.txt`, `l_route.txt`, `l_thread_t<n>.txt`, `m_gila.txt`, `m_block_t<n>.txt`. Parts (a)–(i) of `verify.jl` are byte-identical before and after (md5 163e37ce... of their body) and were re-run at the end: no numerical output changed, the four files that differ do so in timing columns and in the known truncation of `a_rho.txt`.

The cache re-evaluation is the independent check of the table above. All 439 records (the 438 plus one added meanwhile) were evaluated four ways — **farBlockX!** with certificates, **farTensorX** per offset, and both again swapped — in 4.9 h single-threaded at 1-minute load 3.2–11.5, and reproduce the earlier table to every digit that can be compared: routes 244/92/103, worst $e_{\max}$ 3.42×10^{-15} at the same record, worst per entry 4.71×10^{-15} , 1.3 digits lost, the same certificate ratios (2.4 / 23.9 / 135 on route 1, 4.0 / 29.2 / 673 on route 2, 48.7 / 115 / 8930 on route 3), 0 violations in 878 certified evaluations, and **farTensorX** bitwise equal to **farBlockX!** with the same certificate on 439 of 439. One test is new: the reflection identity $G_{ab}(\sigma \mathbf{R}) = \sigma_a \sigma_b G_{ab}(\mathbf{R})$ over the three axis flips and $-\mathbf{R}$, which exercises the sign logic of the box sum and the route selection at mirrored offsets. It holds exactly on every whole-box record, to 5.17×10^{-15} on the box sums (the mirrored pieces are visited in another order) and to 10^{-36} on the k -series. The loosest certificate in the cache is confirmed to be the slender pair’s thin-axis record at $(0, 0, 2)g_z$, 8930 times the actual error, which is the est looseness of Section 11.8 and not an accuracy failure.

Forty-seven new 220-bit records were then computed for geometries chosen to break the routing rather than to exercise it: 66 offsets, 50 of them referenced. Worst max-entry error 5.1×10^{-15} , worst per entry 5.3×10^{-14} , swap identity 3.5×10^{-15} , reflection identity 8.4×10^{-16} , 0 certificate violations.

tried	result
odd ratios 3 and 6, on one axis and on all three	routes 1, 2 and 3 all reached; worst 3.0×10^{-15} . Ratio 3 keeps every offset on the gcd lattice, where the even ratios’ lateral-0 offsets fall off it
a pair coarse on x in the target and on y in the source, $(4, 1, 1)g$ against $(1, 2, 1)g$, and its swap coarse cells of $\lambda/2$ and λ : $ k r_d$ to 5.6, per-shell n -cut to 18, the cap $N = 40$ never reached	worst 1.4×10^{-15} ; the swapped group reproduces every value to 2×10^{-16} , route 1 exactly worst max-entry 5.1×10^{-15} ; worst per entry 5.3×10^{-14} , one entry of a 32 768-piece box sum at $(17.5, 15.5, 15.5)g$, 2.4 digits, certificate $7\times$ the error and holding
complex $f = 1 + i$ and $0.5 + 2i$, $\Im k$ to 12.6 mixed-sign offsets, including a fine cell beside the long face of a 16:1:1 rod	every route works; worst 2.0×10^{-15} , certificates 6–106 $\times$ worst 4.3×10^{-15} ; the reflection signs and the negative lattice coordinates are right
non-integer ratios, 3/64 against 1/32 and 1/32 against 1/48	refused by farSetupX , as GlaExtInf refuses them; nothing returns a value
$\mathbf{R} = 0$, face, edge and corner contact, and a gap of $g/1000$	the four contacts raise; the hairline gap routes to the k -series and is correct
Float32	NaN on every near offset (D14 below)

The routing and lattice logic was then checked exhaustively rather than by sampling. The $27r^3$ offsets of the ratio- r blocks, $r = 2, 4, 8, 16$, were re-enumerated from the closed form, with an independent contact test and an independent derivation of the lattice coordinate $\mathbf{m}$ and of the nearest sub-offset. Over 125 256 separated offsets: every one of the 16, 36, 100 and 324 touching offsets is refused, every separated one takes route 1 or route 2 and route 3 is never reached, nothing is off the lattice, **xLat** and **xNear** agree with the independent derivation everywhere, every route-2 offset’s nearest piece has max-norm exactly 2 — the one-fine-cell face gap — and the route-1/route-2 boundary sits at $\rho = 0.59$ – 0.68 , the Theorem A’ thresholds at $L = 56$. **farTensorX** is bitwise equal to **farBlockX!** on all 15 616 offsets of the $r = 2, 4, 8$ blocks, and the $r = 4$ block’s 15 228 complex entries are bitwise identical filled on 1, 4 and 12 threads: 0 entries differ. The independent comparison against Gila on the whole ratio-4 block (1692 offsets) and a 2172-offset sample of the ratio-16 block reproduces Section 11.3: the adaptive rule 10^{-7} to 5×10^{-4} , the fixed rule 2×10^{-14} to 1.8×10^{-12} , the library within 3.4×10^{-15} of every reference, at 6–8 μ s per offset against 5–13 ms.

Three findings are defects rather than confirmations. One is real and was fixed: D13, above, a **FrqSetX** reused at a radius where its n -cut had never been certified, which is the one code change

this round made and which the equal-cell bit-identity check was re-run against (all four parts clean: 120 of 120 **farTensor** bit patterns, both 12^3 **farBlock!** blocks with 0 differing entries, fresh shape files byte-identical). One is pre-existing, equal-cell, and *not* fixed: in **Float32** the upward recursion for $h_l(k|\mathbf{R}|)$ overflows at $l \approx 27$, so every offset whose selected L reaches 28 returns NaN in all nine entries — at $\lambda/32$ and $f = 1$ that is every offset with $|\mathbf{D}| \leq 4$, and cross-scale route 2 inherits it across the whole near band, where the certificate is still returned as a number. Part (g) of the equal-cell verification had tested **Float32** at one far offset only and missed it. It is listed as D14 and is the root’s call, because fixing it means touching the equal-cell path that everything else here is proved bit-identical to; a **Float32** caller today must keep $L < 28$, which the routing does not know. The third is cosmetic: on a non-integer-ratio pair, **farTensorX** and **farBlockX!** without a prepared set fail with **InexactError** from the gcd sub-cell count before **farSetupX**’s clear message is reached. The agent also showed, in a patched copy it did not ship, that replacing $\min(s_t, s_s)$ by $\gcd(s_t, s_s)$ per axis — three lines — makes all three routes work on non-integer rational ratios to 1.4×10^{-15} , which is the natural way to lift that limitation if Gila’s lattice ever does.

Its verdict, which is also this section’s: the library is correct at the level this round can test, and the reader should still read **xverify.md** before relying on a statement about a geometry outside the ratios, frequencies and cell sizes listed here.

11.7 The near field of unequal cells (item 5)

The task statement asked whether the thin-indicator regime of **moments.jl**, the perpendicular box with one axis narrow relative to its offset ($2(\text{hi} - \text{lo}) < \text{hi}$) that is integrated by a precision-adaptive Gauss–Legendre rule against the two-dimensional closed forms, loses digits on the geometries unequal cells produce, and whether the three-dimensional analogue of the centred Taylor series that removed the same problem in two dimensions exists.²⁶ Both were answered by measurement.

The branch is not an off-lattice exotic. A census of every three-dimensional box of every face pair over the twelve coarse shapes, the equal-cell control, all touching and separated positions up to thirty-two fine cells (234 position classes) shows that it fires on *every* separated pair, equal cells included: the fine-width indicator against the near source face is $[\kappa, \kappa + 1]$ in units of g , which sits exactly on the strict gate at $\kappa = 1$ and is thin for $\kappa \geq 2$, so the equal-cell lattice at $|\mathbf{D}| = 2$ escapes and $|\mathbf{D}| \geq 3$ does not; on touching pairs it fires once a ratio is ≥ 2 (the far source face at $[r, r + 1]$). The statement “never reached on the lattice” in the companion document is about the fifteen touching sub-geometries of equal cells only. The k -series route (iii) of Section 8.6 therefore already runs 99 % of its perpendicular boxes at $|\mathbf{D}| \geq 3$ through this branch. At 128 bits, where route (iii) runs, the branch loses 0 to 0.99 of 38.2 digits against 320 bits on the seven face pairs measured (5.8×10^{-38} relative at worst), twenty-two orders below route (iii)’s target; nothing in the k -series route changes.

In **Float64**, over 2772 face pairs ($8 \text{ shapes} \times 11 \text{ positions} \times 36 \text{ pairs}$) against the same formulas at 320 bits, and confirmed identical on 504 of them against the independent graded quadrature of **notes/gen/ref/mom.jl**: the worst loss is 0.99 digits, on a *parallel* pair through the two-dimensional integrator; the worst thin-branch pair loses 0.93 (two thin axes, $m = 0$); the worst closed-form three-dimensional pair 0.90 (the $m = -1$ seed of a corner box, identical for equal and unequal cells); no pair loses a digit. Per box, over the 2228 thin-branch boxes of a 2734-box sweep, the Gauss–Legendre rule loses 0.51 / 0.71 at the median and 0.95 / 0.96 at worst with one / two thin axes, the same level as the divergence-identity closed forms it bypasses (1.15 per box). The

²⁶ **scratch/reports/xnear.md**; scripts **scratch/xwork/xnear/geom.jl**, **facedig.jl**, **box3tay.jl**, **boxval.jl**, **hybrid.m.jl**, **ksr128.jl**, outputs in **xwork/xnear/out/**.

cross-scale geometries, ratios 2–16, gaps 1–32 fine cells, add nothing to the loss of the equal-cell lattice (0.91 / 0.87 in the companion document).

The three-dimensional centred Taylor series exists and was written (`box3Tay`, generic in the number type): r^m expanded about the box centre in the three shifted coordinates with a three-index recurrence from $g \partial_p f = (m/2) f \partial_p g$, the level sums against the exact affine moments, the same five controls as the two-dimensional one, and a polydisc gate $\lambda > 1/2$ with λ the positive root of $|\mathbf{h}|^2 \lambda^2 + 2(\mathbf{u}_m \cdot \mathbf{h}) \lambda - \rho^2 = 0$. It has a trap the two-dimensional series cannot have: at $m = -1$ the function $1/r$ is harmonic, so on a diagonal centre $\mathbf{u}_m = (a, a, a)$ every pure second derivative vanishes and level 2 is exactly zero; the two-dimensional stopping rule of two consecutive small levels then stops at $n = 3$ with the seed wrong by 3×10^{-5} to 3×10^{-3} (11–13 digits lost on three boxes of the sweep). Three consecutive small levels and $n \geq 4$ fix it. With the fix the series converges on 94.6% / 99.8% of the one- / two-thin-axis boxes at all fourteen m , is rejected on 1.0% (all at $\lambda = 0.463$ – 0.497 , a thin axis beside a long corner axis on the ratio-16 shapes), and where it converges is at one ε (median 0.00, worst 0.35 digits, condition number ≤ 3.4). It costs 2–3 times the Gauss–Legendre rule per call at the median, 6.3 times in aggregate over the sweep, and 2126 times at worst (115 levels at $\lambda = 0.67$); gated at $\lambda \geq 1$ it caps the levels at 64 for 1.9 times the aggregate cost. At face-pair level the worst case does not follow the box: 558 of the 1595 thin pairs also hold closed-form corner boxes at $\lambda = 0.41$ – 0.50 where no series converges, pinning them at 0.71, and the closed-only pairs stay at 0.90, so the global worst moves from 0.93 to 0.90 (series on thin boxes) or 0.71 (series attempted on every box, at 2.4 / 33 times the median / 90th-percentile cost). Less than one binary digit, for a second code path, in a fill that does not run in Float64. **The recommendation, followed, is to keep `box3Slv`; `moments.jl` is unchanged.** If a Float64 near-field fill is ever wired and its budget is counted in ε , the drop-in is `scratch/xwork/xnear/box3tay.jl` gated at $\lambda \geq 1$, with the three-level stopping rule.

11.8 What is not covered, and what is open

est is a scale, not a bound, in both directions. Across the thin axis of the slender pair (1/32, 1/32, 1/512) against (1/8, 1/8, 1/512) at 2–9 thin cells of separation the tensor is 10^2 to 10^4 below the pointwise scale, so $\text{est}/\max |G|$ is 100, 550, 2500, 8100 and the route-3 certificates there are 1.5×10^{-14} to 8.1×10^{-13} of $\max |G|$ against actual errors of 2.3×10^{-17} to 9.0×10^{-17} : valid and loose by 2600–8900 $\times$. In the other direction *est* *under*-states $\max |G|$ by 2–4 $\times$ on the ratio-16 rod at $\kappa = 1$, $f = 1$ and 0.37 ($\text{est}/\max |G| = 0.50$ and 0.25), so the a priori certificate there is 2–4 times tighter than *tol*.²⁷ The a posteriori certificate (53) discharges both, as for equal cells.

0.13% of the realistic block is certified only to 1–4 tol. 1800 offsets at $\rho_{\text{whl}} = 0.55$ – 0.60 meet the bound at $L = 56$ with 1–4 times $\text{tol} \cdot \max |G|$ and the table has no more shells. Where a reference exists they are correct to 3×10^{-15} ((24.5, 7.5, 7.5) g at $L = 54$: 2.9×10^{-16}), but their proven bound is what it is. $L_{\text{max}} > 56$ or routing that band to the box sum (the fine block would grow from 33^3 to about 41^3) would tighten it.²⁸

The table writer is unlocked. D12 made a duplicated segment harmless on read; the append itself still has no cross-process lock, so two processes extending one table file still write the duplicate and pay its disk space.

²⁷`scratch/reports/xunify.md`, Section C, route-3 table; `xtable.md`, Sections 3b and 3d.

²⁸`scratch/reports/xtable.md`, Section 4b; `xunify.md`, Section D.2 and open item 6.

The exact-rational offsets cost as much as the far field. On the large block the selection pass of `farSetupX` ($2.1 \mu\text{s}$ per offset) and the route pass ($1.0 \mu\text{s}$) are 4.2 s single-threaded against 4.4 s of whole-box expansions, and on twelve threads the serial selection (2.9 s) is half the total; the offset vector as `NTuple{3,Rational{BigInt}}` is about 1 GB for 1.4×10^6 offsets, most of the 2.95 GB peak. A caller holding the `FrqSetX` skips the selection pass; an integer-lattice interface (`m` and the gcd cell instead of `R`) would remove both passes, and is the natural interface for Gila’s circulant, whose index already is that lattice.

Route 3 costs 6–21 s per offset at $N = 13$ –44 and 128 bits on the thirty-six unequal panels, is reached only off the gcd lattice, which Gila’s block geometry does not produce (0 of 1.4×10^6 offsets), and was verified only through the 103 off-lattice records of the reference matrix, never through a block. Its results are cached to disk per (pair, orientation, frequency).

Other limits. The fine equal-cell set of the near band costs 0.8–0.95 s of serial `farSetup` per `farBlockX!` call that starts without a `FrqSetX` (the threaded fill of the 33^3 block is 0.01–0.04 s); `fineSet` caches it in the pair set. The verification covers ratios 2, 4, 8, 16 on one to three axes against a $\lambda/32$ fine cell and two slender pairs, at three frequencies; odd ratios, which `GlaExtInf` allows and Gila’s power-of-two refinements never use, put `R` on the integer lattice and were exercised only by the exact-rational checks (ratio 3). No number in this section was measured on a GPU. And, as for equal cells, nothing under `src/` was touched: `farBlockX!` fills a $3 \times 3 \times n$ array for a list of exact offsets, and wiring it into `genEgoCrcExt!` (whose `egoFunOut!` loop is what it replaces, with `egoCntOut!` left in place for the contact class) is the next task, together with the equal-cell wiring it shares a table directory with.

12 The registry of approach families

The search was run as a portfolio, several mechanisms alive at once and kept independent in the early rounds so that they would not converge on the same attractive reduction, with a written registry of what each had achieved and where it was blocked. A route was marked blocked for a band only together with the measurement that blocked it. Both the successes and the failures are recorded here, because the failures are what justify the shape of the final method: they are the reason there is a subdivided band and a k -series band at all.

A caveat on every timing in this section. The per-offset and per-block times quoted below were measured in the first round, on a machine carrying four to ten other agents’ Julia processes. They are sound for ordering the families against each other, which is what they are used for here, and they are *not* the deliverable’s timings: those were re-measured on a quiet machine and are in Sections 10 and 10.4.

	family	status	the number that decided it
(a)	Cartesian Taylor about the centre	alive, cross-check	3×10^{-16} – 1.5×10^{-14} from 3 cells; $\sqrt{2}\rho < 1$ fails at 2
(b)	spherical addition theorem	won , whole box	8.6×10^{-15} max-norm over 427 references
(b')	the same on eight octants	won , near band	the 12 two-cell offsets at 4.2×10^{-15}
(c)	one exact axis, two expanded	alive, not used	$\tau_\xi = 1$ <i>exactly</i> at two cells
(d)	subdivision of the box	won as (b'); reference	ρ_j falls like $1/(m(n-1)+1)$; the needle needs ~ 200 boxes
(e)	spectral / Ewald	blocked, all bands	$P \approx 1651$ against 256, i.e. 768 GB, at $f = 1 + 0.1i$
(f)	interpolation of the far tensor	alive, marginal	pays only above 2.2×10^3 flops/offset; ceiling $9.6\times$
(g)	derivative-free coefficients	alive, cross-check	4.8×10^{-15} over 290 rows; agrees with (b) to 1.1×10^{-14}
(k)	the moment k -series	won , slender needle	67 offsets; $\varepsilon\Lambda = 2.3 \times 10^{-13}$ forces 128 bits
<i>unequal cells</i> (Section 11; measured in the cross-scale round, load 3–12)			
(u ₁)	the whole trapezoid box, Theorem A'	won , far field	98.8% of the $\lambda/2$ block; 2.4×10^{-15} over 244 records
(u ₂)	the gcd average as a box sum over the fine block	won , near band	$\Lambda = 1.0$ – 2.3 ; 658 s naive against 1.4 s as a box sum
(u ₃)	the twenty-seven-piece split	built, measured, dropped	0 of 2213 near-band offsets converge at ratio 16; 5.4–8.9 $\times$ slower where it does
(u ₄)	the k -series on thirty-six unequal panels	fallback, off the lattice	103 records at 10^{-17} ; 6–21 s per offset; 0 offsets of either block

12.1 (a) Cartesian Taylor expansion about the centre separation

Mechanism. $\partial_a \partial_b g(\mathbf{R} + \boldsymbol{\delta}) = \sum_{\boldsymbol{\alpha}} (\partial^{\boldsymbol{\alpha} + \mathbf{e}_a + \mathbf{e}_b} g)(\mathbf{R}) \boldsymbol{\delta}^{\boldsymbol{\alpha}} / \boldsymbol{\alpha}!$, integrated term by term against the exact moments (7); only even $\boldsymbol{\alpha}$ survive.

Outcome: works from three cells outward, with a proven bound. The sign and normalization were pinned against the 220-bit reference to 8×10^{-26} at $(5, 1, 0)$ and 10^{-40} at $(8, 8, 0)$. Over 185 offsets the measured accuracy is 3×10^{-16} to 1.5×10^{-14} with -0.11 to $+1.84$ digits lost, and the truncation degree saturates beyond about sixteen cells at $p = 8, 12, 16$ – 18 at $\lambda/32, \lambda/8, \lambda/4$, giving a constant far cost of 6408 flops and $7.7 \mu\text{s}$ per offset, 34 to 185 times faster than the rule it replaces. The remainder bound was proved with explicit constants — the exact ball min-modulus $|\mathbf{R}| - \rho$ for $\rho \leq |\mathbf{R}|/2$ and $\sqrt{|\mathbf{R}|^2/2 - \rho^2}$ beyond, a Helmholtz alternative for the diagonal, and a tail sharpened by the moment weights — and is 2.0 to 7.7 times pessimistic in p .

Where it stopped. Its convergence criterion is the *complex-ball* one, $\sqrt{2}r_d < |\mathbf{R}|$, which fails at the six nearest non-touching axis offsets of a cubic grid. Remark 5.9 settles what that means: the series itself converges for $\rho < 1$ and it is only the Cauchy bound that is unavailable, but the practical consequence stands, since at $(2, 0, 0)$ the best attainable is 7.7×10^{-14} in **Float64** at $p = 100$ (1.4×10^6 derivatives, 9.1×10^5 sum terms) and the derivatives overflow beyond $p \approx 100$. A 2^3 subdivision restores it, at 5.5×10^{-14} with $p = 36$ and 1.7×10^7 flops. The slender cell breaks it along the short axis and not marginally: the criterion demands $n \geq 33$ there, and at $n = 16$ the ratio is $\tau = 2.00$ and the expansion diverges. An attempt to recover the anisotropy by replacing the ball with a Cartesian polydisc was made and recorded as not useful: the tail needs $s_i/r_i < 1$ on every axis, so $r_\perp > s$ always, and for a cubic cell at $n = 4$ the best achievable $\max_i s_i/r_i$ is 0.69–0.77 against the isotropic $\tau = 0.61$.

12.2 (b) Spherical addition theorem, whole box and octant split

Mechanism. Sections 4.1–4.8.

Outcome: this is the method. Sign + with no extra factor, bit exact against the reference at (5, 1, 0). Against the 220-bit reference over 163 tensors from three cells outward, all four shapes, all three directions and both frequencies, the whole-box form is accurate to 4.2×10^{-15} with at most 1.3 digits lost and is rounding limited (at the same L in **BigFloat** the error is 10^{-19} to 10^{-22}); symmetries and f -homogeneity are bit exact. Cost 391/1250/2731/4972 ns at $L = 10/20/30/40$ with no allocation, 96.5 % of the offsets of a 128^3 block at $\lambda/32$ needing $L \leq 8$, and 2.37 s for that whole far field single-threaded with the Miller repair (0.59 s without it). The conditioning is confined to the geometry table: the l -sum cancels by at most 8.4 at $\lambda/32$ and 67 at $\lambda/4$, the n -sum by 3 to 4, while the exact-rational contraction cancels by 5.8×10^8 and is therefore not done in floating point at all. The octant split closes the near band: the twelve two-cell offsets reach 4.2×10^{-15} (worst per entry 4.3×10^{-15}) at both frequencies with sub-box cancellation ≤ 1.55 .

Where it stopped, and what closed it. Two gaps, both truncation and neither conditioning. The twelve offsets with $|\mathbf{n}| \leq 2.56$ need $L = 94$ for 6.6×10^{-14} and do not reach 10^{-15} by $L = 100$; the octant split solves them (Section 8.5). The slender needle $(0, 0, n)$ with $n < 23$ has $\rho > 1$ and *diverges*; at $n = 24$ the error is 1.3×10^{-4} at $L = 40$, at $n = 32$ it is 1.5×10^{-9} and only at $n = 64$ does it reach 3.4×10^{-16} . Neither the whole box nor the octant split has a certificate on the 67 offsets of Section 8.6, and those go to the k -series.

Added here. For a cubic cell the entire $l = 2$ shell vanishes identically (Remark 4.3), so the surviving shells are $\{0, 4, 6, 8, \dots\}$; this was not noticed in the round that produced the term-count tables and is an unclaimed saving on the shapes Gila uses most.

12.3 (c) One exact axis, the other two expanded

Mechanism as proposed. Do the integral along one axis in closed form with the other two coordinates frozen, expand only in the remaining two, so that the ratio becomes two dimensional — the natural tool for a slender cell whose isotropic ρ is dominated by its long edges.

Outcome: the proposal does not exist; an inversion of it works. With the transverse coordinates frozen, the line integral of $e^{ik\varrho}\varrho^{-l}$ is an incomplete-Hankel object for every $l \geq 1$, so there is no closed form to have. Reversing the roles does work: $r^2 = \varrho(t)^2 + \xi$ with $\xi = 2R_2y + y^2 + 2R_3z + z^2$ a polynomial in the two transverse coordinates, so $F(r) = H(r^2)$ is expanded in ξ (the operator $D = (1/2r)d/dr$ acts in closed form on $e^{ikr}r^{-l}$), the transverse moments are exact per level, and the along-**R** axis carries a Taylor series whose coefficients come from the closed system $G'_l = ik(R_1 + t)G_{l+1} - l(R_1 + t)G_{l+2}$. Accuracy $\leq 1.3 \times 10^{-13}$ per entry from three cells outward, all directions, $\lambda/32$ and $\lambda/4$, $f = 1$ and $1 + 0.1i$, with ≤ 1.07 digits lost; $(M^*, Q^*, \mu s) = (25, 28, 142)$ at three cells, (13, 22, 53) at four, (7, 14, 18) at eight — comparable with the Taylor route. On the slender cell it is the best of the Cartesian family: 9 μs at 9.4×10^{-14} for (16, 0, 0) and 150 μs for the needle at (0, 0, 32) against 1.2 ms.

Where it stopped. At two cells, by an identity rather than by slow convergence: $\tau_\xi = 1$ *exactly*, even with two exact axes, because the nearest point of the box is at distance s and the transverse excursion is also s . Only subdivision moves that. The needle at $n \leq 16$ is blocked as well ($\tau_\xi = 1.13$ at $n = 16$). And its Q rule is not proven: the bound with $e^{1.5kR_0\theta}$ is valid but 1.8 times loose at $kR = 22$, while the M rule was never violated in 110 cases.

12.4 (d) Subdivision of the difference box

Mechanism. Section 4.7, with any of (a), (b), (g) as the per-sub-box engine.

Outcome: it supplies the near band, and a third independent reference. An independent `BigFloat` Gauss–Legendre *volume* rule written for this family matches the face-pair `pairKer` reference to 10^{-63} – 10^{-65} on the cubes and 1.4×10^{-62} on the slender cell — the third confirmation of Theorem 3.5 — and it is the only usable reference on the slender cell, where `pairKer` stalls beyond fifteen minutes per offset. Its own remainder bound locates the singularities exactly on the complex line at $|t| = |\mathbf{X}|/|\boldsymbol{\delta}'|$ (the same fact as Proposition 5.8) and is never violated over 364 rows, with a median looseness of 2.29 in degree (1.75 to 7.74). The per-sub-box ratio falls like $1/(m(n-1)+1)$ and not like $1/m$: at $n = 2$, $m = 1, 2, 3, 4, 6, 8$ give 0.866, 0.522, 0.433, 0.333, 0.243, 0.190. The optimum is $m = 2$ or 3 at two cells and $m = 1$ from three cells outwards; $m = 8$ at two cells costs 3.5 times more than $m = 2$, and non-uniform splits gain only 1.00–1.09 on cubes (1.5 on the slender cell). Cancellation between sub-boxes is 1.0 to 4.0.

Where it stopped. Two things. Its bound is a certificate and not a selector: a degree factor of 2.29 is a flop factor of about twelve, since work grows like p^3 . And it does not rescue the needle at an acceptable price: $(0, 0, 2)$ needs about two hundred boxes for $\rho_j < 0.5$ ($10 \times 10 \times 2$ graded, $p = 25$, 2.7×10^7 flops) and $(0, 0, 8)$ needs $8 \times 8 \times 1$ at $p = 21$, 3.1×10^6 flops. It converges there; the cost is the objection. One trap is worth recording because it is invisible in the ratio tables: *odd m is unsafe for an offset with a zero component*, because a sub-box centred on the zero axis has $\rho_j = 5.66$ against 0.99 for the even split at the slender $(0, 0, 2)$ — the same conclusion Lemma 4.4 reaches from the weight.

12.5 (e) Spectral and Ewald splitting

Mechanism. The Fourier transform of g is $1/(|\boldsymbol{\xi}|^2 - k^2)$ and cell averaging is a product of sinc^2 factors, so the whole Toeplitz kernel might be obtained for all offsets at once by one FFT of an analytic spectral kernel, with the near field corrected in real space.

Outcome: blocked in every band, on structure and not on effort. The spectrum $\hat{T}_{ab} = V_t \prod_i \text{sinc}^2(\xi_i s_i/2) (\delta_{ab} k^2 - \xi_a \xi_b)/(f^2(|\boldsymbol{\xi}|^2 - k^2))$ was verified against real space to 3.4×10^{-13} , so the transform is right. The plain alias sum needs $M \approx 4 \times 10^{11}$ images per axis. Ewald damping fixes the aliasing ($M = 2$ at $\eta s \leq 2.5$), but the pole at $|\boldsymbol{\xi}| = k$ is fatal for a different reason: the inverse DFT of the periodized spectrum is the P -periodization of a kernel that does not decay, with error $e^{-\text{Im}(k)Ps}$. At real frequency that is an $O(1)$ error at every P ; at $f = 1 + 0.1i$ and $N = 128$ it needs $P \approx 1651$ against 256, i.e. 768 GB. And even if the pole were handled, the cost is about 10^{11} flops against 5×10^9 for the direct route. Reopen only for a mechanism that removes the pole, not for a better damping.

12.6 (f) Interpolation of the smooth far tensor

Mechanism. The tensor entries are analytic in $\mathbf{R}$ away from the origin; evaluate on a coarse sub-lattice and interpolate.

Outcome: alive, marginal, and not needed. The analyticity radius is exactly the cell gap. At $\lambda/32$ the Chebyshev degree for 10^{-13} is 18 for a patch of $H = 32$ cells and 26 for $H = 64$, and the fraction of offsets that must still be computed directly is 0.10 over a 128^3 grid, so the ceiling is a factor 9.6. But the interpolation arithmetic is about 2.1×10^3 flops per offset, so it pays only if the direct method costs more than 2.2×10^3 flops per offset — which the whole-box multipole sum at $L \leq 8$ does not. Equispaced nodes never reach 10^{-13} (Runge), and there is nothing at $\lambda/8$, $\lambda/4$, or in the two-to-eight-cell band. Kept open as a pure constant-factor option if the per-offset cost of the chosen method ever rises.

12.7 (g) Derivative-free Taylor coefficients

Mechanism. Generate the Taylor coefficients c_α of $g(\mathbf{R} + \boldsymbol{\delta})$ directly from the differential equation the kernel satisfies, instead of differentiating it.

Outcome: the strongest of the Cartesian routes, and the independent cross-check of (b). The working route is the pair recurrence on $G = g(\mathbf{R} + \boldsymbol{\delta})$ and $S = |\mathbf{R} + \boldsymbol{\delta}| G$: $\nabla S = ik(\mathbf{R} + \boldsymbol{\delta})G$ and $P\nabla G = (\mathbf{R} + \boldsymbol{\delta})(ikS - G)$ with $P = |\mathbf{R} + \boldsymbol{\delta}|^2$, a two-level recurrence seeded by $a_0 = e^{ik\varrho}/(4\pi f^2 \varrho)$, $b_0 = e^{ik\varrho}/(4\pi f^2)$, and the assembly

$$T_{ab} = \frac{1}{V_t} \sum_{\boldsymbol{\beta} \text{ even}} \mu_{\boldsymbol{\beta}} \left[(\beta_a + 1 + \delta_{ab})(\beta_b + 1) c_{\boldsymbol{\beta} + \mathbf{e}_a + \mathbf{e}_b} + \delta_{ab} k^2 c_{\boldsymbol{\beta}} \right],$$

which equals `srfSum!`'s output with no extra sign or factor. Re-derived independently and confirmed to 10^{-76} at 256 bits, with the trace identity $\text{tr } T_q = 2k^2 \langle g \rangle_q$ holding to 2×10^{-58} . Over 290 (case, frequency) rows against the 220-bit reference, and 200 of them inside Gila's operating range, the worst tensor error is 4.8×10^{-15} and the worst per entry 8.5×10^{-15} , with $q^* \leq 14$ far out. Cost 17.6–18.6 ns per coefficient and a 32^3 block in 1.85 s against 23.3 s, a 12.6-fold block speedup.

Where it stopped, with numbers. Three failure modes, all shared with (a) because it is the same expansion: $\rho \geq 1$ on the slender needle diverges, with the boundary at $n = 22.6$ exactly and no $(0, 0, n)$ with $n \leq 39$ reaching 10^{-13} within $q \leq 40$; $k \cdot 2s \geq 10$ (cells larger than $\lambda/2$, outside Gila) costs $10^{0.2k \cdot 2s}$; and $\rho \rightarrow 1$ at $(2, 0, 0)$ needs $q \approx 100$ where the cancellation is 1.8×10^4 , so the error curve turns around at 1.1×10^{-13} . Its imaginary part is limited by the formulation and not by the recurrence: the tensor is one complex sum, so the relative error of $\text{Im } T_{ab}$ is 10^{-13} to 4×10^{-12} whenever $|\text{Im}|/\max |T|$ falls below 10^{-3} — which is exactly the defect the y_l/j_l split of Section 4.9 does not have. Four sub-routes were measured and blocked: the Helmholtz march from exact Cauchy data (3.5 digits lost at order 30); the addition theorem expanded into Cartesian *monomials* (3.0 digits at order 16, amplification $10^{0.33n}$, from the $(1 + \sqrt{2})^l$ growth of the Legendre coefficients in the monomial basis — note this is *not* the (l, m) -organized route of Section 4, which is where the same expansion becomes well conditioned); a one-dimensional ODE with angular recovery (condition $10^{0.33n}$, 65–125 times more expensive); and automatic differentiation (correct, 37–94 times slower). Two further measurements are worth carrying: order 50–60 at $kR = 30$ loses 6–9 digits, so the order must be set by ρ and never pushed; and the flat error floor at large kR is the seed $e^{ik\varrho}$, removable for twelve flops by a double-double phase, which recovers 2.5 of the 3.35 digits lost at $kR = 3000$ — worth doing for the h_0 seed of (b) as well, since kR reaches about 350 on a 128^3 block at $\lambda/4$.

12.8 (k) The moment k -series carried outward

Mechanism. The near-field series of the companion document, $\int_A \int_B g = \frac{1}{4\pi f^2} \sum_n (ik)^n I_{n-1}/n!$, applied to separated pairs.

Outcome: correct everywhere, affordable only where kD and Λ are both small. Its bound (Theorem 5.7) is tight to ± 3 terms over 233 rows, which makes it the sharpest bound in this document. The obstruction is the pair of constants in front of it. The assembly amplification is $\Lambda = 68$ to 1405 at $\lambda/32$, 15 to 24 at $\lambda/4$ and 2.2×10^3 to 8.1×10^4 for the slender cell, so the `Float64` band edges for 10^{-13} on the assembled tensor are four cells at $\lambda/32$, three at $\lambda/8$, two at $\lambda/4$ and *none at all* for the slender cell, where $\varepsilon\Lambda = 4.8 \times 10^{-13}$ already at two cells. In 128-bit arithmetic the error is 10^{-31} to 10^{-28} everywhere. And the cost is prohibitive as a general method: the moment ladder scales as $m_{\max}^{2.6 \dots 5.1}$ and is re-run per m on three-dimensional pairs, giving 32 ms per offset

at two cells at $\lambda/32$ (against 13.9 ms for the rule it would replace) and 671 ms at thirty-two cells (against 0.55 ms).

Where it stopped, and the merge that failed. The idea that would have unified the near and far fields is the shifted exponential, $e^{ikr} = e^{ikR_0}e^{ik(r-R_0)}$ with $R_0 = |\mathbf{R}|$: the outer series then needs only 11 terms at $\lambda/32$ and 19–21 at $\lambda/4$ with cancellation ≤ 1.5 at every separation. It fails on the centred moments $\int \int w(r - R_0)^n$, which cannot be generated stably: the naive binomial loses $n \log_{10}(2.3N)$ digits; the divergence recurrence subtracts nR_0/d per step; and the route through $q = (2\mathbf{R} \cdot \boldsymbol{\delta} + |\boldsymbol{\delta}|^2)/R_0^2$ converges only when $(2N + 3)/N^2 < 1$, that is $N \geq 4$ on an axis — which is family (a)’s own radius, so the merge is circular. Blocked as a merge; the series stays what it is, the tool for the touching shell and for the slender needle, where $kD_{\max} \leq 0.60$ and $N \leq 16$.

Side findings. Two, both about the code being replaced. Gila’s fixed rule on the slender cell at $(0, 0, 8)$ has a relative tensor error of 1.79×10^{-3} (order-5 rule; 2.8×10^{-5} on the worst face pair), and at $(0, 0, 20)$ and $(0, 0, 24)$ it is 4.5×10^{-6} and 1.9×10^{-6} , because `quadOrd` keys the order on the separation in cells and not on the physical geometry. And the `Float64` moment ladder itself loses 5.5 digits at $m_{\max} = 48$.

12.9 Three disagreements between rounds, and how they were resolved

1. *Does the Cartesian series converge at the first axis offset?* One round recorded $(2, 0, 0)$ as divergent and asymptotic; another recorded it as convergent with an effective ratio of 0.73. Proposition 5.8 and the measurement in Remark 5.9 settle it: convergent, with an observed asymptotic ratio of 0.887 tending to $\rho = 0.866$, and no Cauchy *bound* available. The practical conclusion — hand the offset to the octant split — is unchanged, but the reason is different, and the difference matters for the slender needle, where $\rho > 1$ and the divergence is real.
2. *The sub-box term count.* The bound (37) selects $L_j \leq 56$ at the twelve two-cell offsets when evaluated by `bands.jl` and $L_j \leq 48$ when evaluated in the round that introduced it, from the same written formula and the same geometry. The discrepancy is about a factor 100 in the tail, does not move any band edge, and is recorded unresolved in Remark 8.1; the larger value is used here because it is what the stated formula produces.
3. *The truncation rule for the Taylor routes.* One round proposed $q^* \approx 17/\log_{10}(1/\rho)$ and the audit measured $q^* \approx 7 + 7.1/\log_{10}(1/\rho)$, the first being 3.35 times too costly and a pure $9.7/\log_{10}(1/\rho)$ undershooting by two at $\rho \leq 0.06$, where ninety per cent of a block’s offsets sit. The audit’s rule is the one to use; it is a fit and not a bound, which is a reason to prefer the multipole form, whose L comes from Theorem 6.4.

12.10 The second round: what was rebuilt after the families closed

The first round settled the mathematics and produced no library. The second round produced one and then attacked it. Six passes, in order, each with the number that made it necessary.

The theory pass. Wrote Sections 3–8 from the round-1 reports and checked every bound in `BigFloat` against the quantity it bounds. It found that the constant 17 of Theorem 5.3 is loose by 32 to 34 000; that the end-to-end bound over-states by 1.3×10^3 to 3.0×10^7 ; that for a cubic cell the entire $l = 2$ shell vanishes (Remark 4.3); and that the band edges follow from the bounds alone (Section 8.3) rather than from experiment.

The unification pass. Built `farfield.jl`: the three routes, the exact geometry table, the selector, the caches, threading. Accuracy 8.6×10^{-15} max-norm over the 403 references then available; the 128^3 far field in 2.05 s single-threaded. It also found four defects in the round-1 sources it

superseded — a tail bound that was a partial sum and therefore vacuous at its own top shell, a geometry table that accepted `Float64` edge lengths and became noise beyond $L \approx 26$, a Bessel seed losing 7.7×10^{-14} of phase at $k\mathbf{R} = 348$, and a per-thread workspace sized by `nthreads()` instead of `maxthreadid()`.

The sharper bound. Replaced the two lossy steps of Theorem 5.3 by exact identities (Section 6): a median gain of 3.2×10^3 in the bound, of 1.47 in per-offset cost, and — the point — a bound on the truncation the code actually performs rather than on a different reorganization of the same series.

The preparation pass. Integrated Theorem A/B into the selector, moved the tolerance to 10^{-14} for the reason in Section 8.2, rebuilt the geometry table at 256 bits (bit-for-bit identical to the 1024-bit table over all 321 373 stored numbers of four shapes) and sized it lazily in L , cutting a cold 32^3 build from 828 s to 202 s and one cold far offset from 828 s to 1.3 s. It also found that the n -truncation cut was being evaluated at a radius where the expansion does not converge, that the exact-zero threshold inside the contraction did not track the build precision, and that the selection arithmetic must be done in `Float64` whatever the working type, because $(2l + 1)!!$ overflows a `Float32` at $l = 29$.

The adversarial audit. Twelve cell shapes with aspect ratios 1 to 1020, six shape-adapted offsets each, four frequencies: 288 tensors against an independent 220-bit graded volume rule whose own convergence is a median 6.0×10^{-50} ; 60 further random cases for the bound; 40 boundary offsets; 21 frequency cases including $f = -1$, $f = 1 - 0.1i$, $f = 10$ and $f = 1 + 3i$; determinism at 1, 4 and 12 threads. It produced the ten defects of Section 13, of which D9 — the k -series of j_l capped at a compile-time constant — destroyed the answer completely (9.6×10^{-3} per entry) as soon as $|k|r_d$ exceeded about 3.2.

The merge. Fixed D9 by Remark 5.6, confirmed the audit’s other two accuracy defects fixed, removed a cost model that was computed on every offset and read by nobody (28–31 % of the whole far-field fill), threaded the k -series route (633 s $\rightarrow$ 128 s on a cold slender block, bitwise identical), and closed the five API gaps. It also found, while testing positivity, that the indefiniteness two earlier passes had attributed to the discretization is Gila’s order-9 contact rule (Section 9.5).

12.11 The cross-scale round: unequal cells

The third round extended the library to pairs of cells with different edges (Section 11). Five agents ran in parallel from one brief — the theory, the table and routes, Gila’s geometry and accuracy, the 220-bit references, and the near-field item — followed by an integration pass and an adversarial one. Four families were alive; two won, one is the fallback, one was dropped.

(u₁) The whole trapezoid box. Mechanism. The equal-cell table with the trapezoid moments $\mu_n(a, b) = \mu_n(b) - \mu_n(a)$ in place of the triangle’s; Theorem A with the ramp-only flat measure and point masses at $\pm a_i$, $\pm b_i$ (Theorem 11.2). **Outcome: the far route.** One table per unordered pair, 3.3–5.7 s cold; 98.8 % of the realistic $\lambda/2$ block at 0.9–12.5 μs per offset; 2.4×10^{-15} worst over 244 reference records; the equal-cell tables bit-identical at $a = 0$. **Where it stops.** At $\rho_{\text{whl}} = 0.58\text{--}0.60$, the same edge as the equal-cell whole box; 16 908 offsets of the block lie above it. Its first results at the $\lambda/4$ and $\lambda/2$ coarse cells were wrong by up to 6.8×10^{-10} , which was the n -cut defect D11 and not the table.

- (u₂) **The gcd average as a box sum. Mechanism.** Every Gila pair lives on the common gcd lattice, so $G(\mathbf{R}; \mathbf{s}_t, \mathbf{s}_s) = N_t^{-1} \sum_{jj'} G(\mathbf{R} + \mathbf{c}_j - \mathbf{c}_{j'}; \mathbf{g}, \mathbf{g})$ over integer fine offsets, evaluated as a weighted sum of entries of the fine equal-cell `egoToe` (Section 11.4). **Outcome: the near band.** No new table, no new bound, all weights $+1/N_t$; cancellation $\Lambda = 1.0\text{--}2.3$ measured over the matrix at three frequencies (the a priori fear that it would cancel at $\lambda/2$ was wrong); 3.4×10^{-15} worst on a 4096-term sum. **Where it stopped, once.** Evaluated naively as $N_t N_s$ separate tensors per offset it costs 38.9 ms per offset and 658 s for the band, 160 times the far field; as a box sum 1.4 s. The formula was right and the first implementation of it was the wrong one.
- (u₃) **The twenty-seven-piece split. Mechanism.** The sub-box route of Section 4.7 with three pieces per axis, two ramps and a plateau, eight piece-type tables and Theorem B with $\beta = 0$ on a plateau axis. **Outcome: built in full, dropped.** The all-plateau piece has radius $|\mathbf{a}| = (r-1)\sqrt{3}g/2$ against a nearest offset of $(r+3)g/2$, a ratio of 0.87, 1.17, 1.24 at $r = 4, 8, 16$: it diverges exactly where the whole box fails. Where it converges it is 5.4–8.9 times slower than the whole box and as accurate (1.5×10^{-15} over 152 records). It served as the third route in every cross-check and is kept in `scratch/xwork/htable/farx.jl`.
- (u₄) **The k -series on unequal panels. Mechanism.** Route (iii) with `pairMoments` on the thirty-six face pairs of two different rectangles. **Outcome: the correctness fallback.** 10^{-17} against 103 off-lattice records, 6–21 s per offset at $N = 13\text{--}44$; reached by 0 offsets of either block, because Gila’s geometry never leaves the gcd lattice.

Two disagreements were resolved on the way. The root’s prediction that the gcd sum would cancel at $\lambda/4\text{--}\lambda/2$ coarse cells, made to explain whole-box against gcd disagreements of $5.9 \times 10^{-14}\text{--}6.8 \times 10^{-10}$, was wrong: Λ was 1.1–1.6 at those offsets and the whole box was the inaccurate side (D11). The root’s prediction that the plateau piece of the split would not converge for $r \geq 8$ held, and the measurement extended it to $r = 4$ on two and three axes.

13 Defects found on the way

Two kinds: defects in the code being replaced or in the round-1 sources, and defects in this work’s own library, found by the adversarial audit and by the merge pass. Both are recorded because both cost real accuracy and because the second list is what the reader should look at first if a number here ever fails to reproduce.

13.1 In Gila and in the round-1 sources

1. **quadOrd keys the rule order on the separation in cells, not on the physical geometry.** On the slender cell $(1/32, 1/32, 1/512)$ the assembled tensor along the short axis is wrong by 0.789, 0.227, 3.68×10^{-2} , 4.50×10^{-3} , 1.79×10^{-3} , 2.97×10^{-6} at $n = 2, 3, 4, 6, 8, 16$ — a $1/512$ boundary layer across a $1/32$ face is invisible to a rule indexed by cell count (Section 2.3).
2. **The order-9 touching-shell and contact rule is wrong by 117% on a slender cell.** Measured against the exact moment series (Section 9.5): 1.17 relative at $(0, 0, 1)$, 0.127 at the contact cell, 0.024–0.037 at the four singly-shifted offsets. It is the sole cause of the indefiniteness of that block’s anti-Hermitian part, which two passes of this work had attributed to the discretization.
3. **The face-pair assembly amplifies. `srfSum!` subtracts nearly equal numbers; the amplification is 36 to 1300 at $\lambda/32$ and 4.3×10^3 to 1.67×10^5 on the slender cell’s long axes, turning per-pair**

errors of 10^{-15} into tensor errors of 10^{-11} . It is a *fine*-cell effect, growing like $(|\mathbf{R}|/s)^2$. The volume form removes it.

4. **A tail bound that is a partial sum.** One round-1 routine returned $\text{cum}[i] = \sum_{L < l \leq l_{\text{top}}}$, so $\text{cum}[\text{end}] = 0$ and the bound certified $L = l_{\text{top}}$ for free. Fixed by summing to $L + 16$ and offering cuts only at $l \leq L$; the discarded $l > L + 16$ part is below 10^{-40} of the retained tail at every shape and frequency used.
5. **A geometry table that accepted Float64 edge lengths.** The contraction cancels by 2.1×10^5 at $L = 20$, 5.8×10^8 at $L = 40$ and 7.31×10^{11} at $L = 56$: a Float64 table is noise beyond $L \approx 26$. The shipped code takes Rational{BigInt} only.
6. **A Bessel seed normalised on $\sin z/z$ recomputed from the Float64 argument.** At $\lambda/4$ and 128 cells $k\mathbf{R}$ reaches 348 and the rounding of z alone injects 7.7×10^{-14} of phase — at the target. Replaced by a double-double phase, twelve flops.
7. **The $(-1)^l$ of the addition theorem.** It is invisible on the whole box, where the triangle weight kills every odd l , and mandatory on a sub-box, where the affine weight does not: omitting it is a 3.7×10^{-2} error. The library folds it into the frequency contraction for both tables, so the two paths cannot diverge.
8. **A per-thread workspace sized by nthreads() and indexed by threadid().** In Julia 1.12 the first counts the :default pool and the second is a global id that also numbers the interactive threads; with JULIA_NUM_THREADS=8 this raises BoundsError at index 9. maxthreadid() is the correct size. Found by the bit-identity check between the threaded and serial fills.
9. **pairMoments fails on Rational{BigInt} panels (sqrt at moments.jl:442).** Convert to BigFloat first, which is exact for these lengths.
10. **Gila’s two separated-pair rules on unequal cells** (Section 11.3). The adaptive rule `egoSrfAdp!` runs `hcubature` at relative tolerance 10^{-6} and delivers 2.7×10^{-9} to 2.28×10^{-4} , median 1.2×10^{-6} , on every offset of the first coarse cell of separation (36 540 offsets of a $\lambda/2$ block) and not only in the touching layer. The fixed rule `egoSrfFxd!` is 10^{-12} on cubes and slabs but 1.77×10^{-4} on a 16:1:1 rod approached from the side, because `sep` is reached through the thin axis while the fine cell is $1.5g$ from a $\lambda/2$ panel; and 1.7×10^{-11} with order 6 across a $\lambda/2$ cell at 4λ .

13.2 In this work’s own library

The audit ran twelve shapes, six offsets each, four frequencies against an independent 220-bit reference, plus sixty random cases for the bound, forty routing boundaries and twenty-one frequencies. It produced ten numbered defects. Their state now:

	defect	measured	state
D1	Float32 : the bound overflows	$(2l + 1)!!$ overflows at $l = 29$ and $ k ^l$ at $l = 49$; one NaN poisons every cumulative tail, so every offset fell through to the k -series: 20.9 s for one tensor instead of 1 μ s	fixed: the selection arithmetic is Float64 whatever the working type
D2	the n -cut evaluated at a vacuous radius	over-truncation 4.65×10^{-13} ($\lambda/128$ cube), 4.25×10^{-14} (aspect 4)	fixed
D9	the k -series of j_l capped at a compile-time constant	error $\propto (k r_d)^{25.6}$ above $ k r_d \approx 3.2$, reaching 9.6×10^{-3} per entry with a reported bound 7.2×10^8 times smaller	fixed: N adaptive, and the call errors above $N = 40$
D10	L under-selected at large separation	1.23×10^{-13} per entry at (48, 48, 48)	fixed
D3	the k -series route has no cost bound and no fallback	0.05 s to 76.3 s per offset; 99.9 % of a cold slender build	<i>not</i> fixed; threaded (633 $\rightarrow$ 128 s)
D4	a cost model computed on every offset and read by nobody	309–312 ns of a 988–1095 ns fill, 28.5–31.3 %	fixed
D5	BigFloat output above 192 bits is no more accurate	128 vs 256 bits 3.15×10^{-38} ; 192 vs 256 3.95×10^{-58}	<i>not</i> fixed
D6	unguarded global caches in the single-offset path	5 trials $\times$ 6 concurrent route-(iii) offsets: 0 exceptions, results bitwise correct — the hazard is in the source and would not fire	fixed for the caches; the process-global setprecision remains fixed
D7	a Julia 1.12 world-age warning from the lazy include	printed on the first k -series call	
D8	shape-table build time and memory	298–1770 s and 18.9 GB at 1024 bits	fixed: 14.3–18.3 s and 0.66–0.74 GB, by in-place GMP/MPFR arithmetic in all three inner loops, bit-identical tables

The cross-scale round (Section 11.5) added two more and the adversarial verification of the integrated library (Section 11.6) a third; that same round also found a pre-existing **Float32** failure in the equal-cell *value* path, distinct from D1 and listed here as D14. A later audit asked whether D13 had a twin on the equal-cell side, where the same interval is computed and the same comparison was missing. It does, and it is D15.

	defect	measured	state
D11	the n -cut masked its own failure: the two nCutVec evaluations combined with max. , which turns an uncertified -1 into the other radius' N , and the near radius taken from sub-box centres where the whole-box tail is vacuous	whole box wrong by 5.9×10^{-14} to 6.8×10^{-10} at the $\lambda/4$ and $\lambda/2$ coarse cells; latent in the equal-cell farSetup , never firing at the shipped shapes ($N \leq 6$ at $n_{\max} = 12$)	fixed: cutMax (-1 absorbing, triggers the N_{cap} recomputation) and whole-box radii only; 3×10^{-16} – 1.2×10^{-15} at the same offsets; equal cells bit-identical
D12	the table cache was not multi-process safe: two processes extending one file each appended the missing segment and a reader summed the duplicated shells	4.68×10^{-13} against a certificate of 3.4×10^{-14} at $(6.5, 1.5, 1.5)g$, $L = 54$: a $14\times$ violation	fixed on read (mrgGeo drops columns already present; the file reads as 406 columns, 5.12×10^{-16}); the append itself is still unlocked
D13	a FrqSetX reused at a radius where its n -cut was never certified: farSetupX certifies the whole-box table on $[r_{\text{nc}}, r_{\text{hi}}]$ and no caller compared $ \mathbf{R} $ against it	1.59×10^{-11} at $(6.5, 0, 0)g$ from a set built for $(3.5, 0, 0)g$ only, under a certificate of 1.7×10^{-14} of $\max G $: a violation by 10^3 ; a fresh set gives 1.2×10^{-15}	fixed: the interval is stored in the set and farRouteX raises outside it; the reused set is now bitwise equal to a fresh one; the documented call paths never fire it and no block value changed; equal cells bit-identical
D14	Float32 returns NaN at every offset whose selected L reaches 28: the upward $h_l(k \mathbf{R})$ recursion overflows. Pre-existing, equal-cell, inherited by cross-scale route 2	all nine entries NaN for $ \mathbf{D} \leq 4$ at $\lambda/32$, $f = 1$ ($ h_{27}(0.79) > 3.4 \times 10^{38}$; forced L : fine to 24, Inf at 28, NaN from 32); $ \mathbf{D} \geq 5$ is fine, 1.4 – 4.4×10^{-7}	<i>not</i> fixed (equal-cell path, under the bit-identity mandate): a Float32 caller must keep $L < 28$, which the routing does not know
D15	D13's equal-cell twin: farSetup certifies the k -series cut of the whole-box and octant tables on $[r_{\text{nc}}, r_{\text{hi}}]$, an interval fixed by the offsets the set was <i>built</i> for, and farRoute compared no later request against it	3.2×10^{-8} at $\mathbf{D} = (2, 2, 6)$ on the $\lambda/32 \times \lambda/32 \times \lambda/512$ cell, from a set built with an explicit empty offs , under a certificate of 6.5×10^{-15} of $\max G $: a violation by 5.0×10^6 , where a fresh set gives 1.1×10^{-15} . The realistic form — a set built for a far block and reused at every nearer offset its table still admits — gives 9.4×10^{-9} against 9.1×10^{-15}	fixed: the interval is stored in the set and farRoute raises outside it, on the whole-box radius and on the eight octant vertex radii. Every in-interval request is now bitwise equal to a fresh set, 120/120 tensors and two 12^3 blocks are bit-identical, and the default offs = nothing path never fires it over the whole 128^3 block it claims. The one caller in the verification suite that did fire it, part (b), had cached one set per shape and frequency but sized it on the first offset

Four of these deserve a sentence, because the mechanism generalises.

D9 is the one that mattered, and it was invisible from inside Gila's range. The geometry table stores the first $N + 1$ coefficients of the k -power series of $j_l(k|\boldsymbol{\delta}|)$, and the order that series needs is set by $|k|r_a$, a shape *and frequency* quantity. The code carried a constant 12 that a per-shell

cut could only lower, so a shape needing more got 12 and no warning. Every cell whose longest edge is at most $\lambda/4$ has $|k|r_d \leq 2.72$ and needs $N \leq 12$: the constant was exactly right for the stated range, and the failure appeared only when the audit deliberately went outside it. *A constant that is correct on the whole tested domain is not a certificate.* The fix is three lines and turns a silent 9.6×10^{-3} into a refusal.

D2 and D10 are one bug with two faces. The same budget — an absolute tolerance against the a priori scale est — was evaluated at a radius at which the expansion does not converge, so the cut it returned was meaningless in both directions: sometimes far too small (the $\lambda/128$ cube, wrong by 4.65×10^{-13} at every separation), sometimes saturated at the cap with no certificate (the extreme-aspect shapes, wrong by up to 9.6×10^{-3}). Two symptoms that look like opposite bugs were one line.

D11 is D2 and D9 again, one layer up. D2 evaluated the n -cut at a vacuous radius; D9 capped N at a constant that was right on the tested domain. D11 did both at once and then hid the result: the near-radius evaluation was vacuous because it used a sub-box centre with the whole-box r_d , it returned the sentinel -1 as designed, and `max.` replaced the sentinel by the far radius’ N , which was 0 at high l because the far radius is thousands of cells. The equal-cell library carried the same two lines and was saved by its cells being at most $\lambda/4$, where both radii are certified at $N \leq 6$. The first coarse cell outside that range ($\lambda/2$) exposed it within a day of use. A sentinel that can be combined away by an arithmetic operator is not a sentinel.

D4 is the cheapest lesson in the list. `farRoute` computed a cost model on every offset that all three of its call sites destructured as `_`. It was 360 ns of a 936 ns per-offset fill; making it a keyword recovered 28.5% of the 32^3 far-field build and 31.3% of the 64^3 one, and changed no value.

14 What remains uncertain

Everything below is a limitation of the deliverable as it stands, with the number that measures it. None of it is a reason not to use the method; all of it is what a reader should know before relying on a particular number.

14.1 Two things that are not proven

est(R**) $\geq \max_{ab} |T_{ab}|$ is not a theorem.** The selection rule compares the proved bound against $\text{tol} \cdot \text{est}(\mathbf{R})$, and the statement “relative error tol” needs $\text{est} \leq \max_{ab} |T_{ab}|$, which is false: measured over the 427 references, $\text{est}/\max |G_{\text{ref}}|$ runs over $[1.10, 6.34]$ on the cubes and reaches 135 on the slender needle. It over-states, which is the safe direction — the rule certifies $\text{tol} \cdot \text{est}/\max |T|$, i.e. 1.1 to 6.3 times tol — but it is a measurement and not a proof, and if a shape existed for which est under-stated the true magnitude the selected L would be too small. The cross-scale round found such a case: on the 16:1:1 rod at one fine cell of gap $\text{est}/\max |G|$ is 0.50 at $f = 1$ and 0.25 at $f = 0.37$ (Section 11.8). The whole box does not converge at that offset, so no L was selected from the under-stated scale, but the hypothesis is now known to fail on a shape Gila produces. The a posteriori certificate (53) discharges the hypothesis entirely at no cost and evaluates to 4.5×10^{-14} over the same references; the proven lower scale est_{l_0} exists but is vacuous at $\lambda/4$ (Section 7.3) and fails at the nearest unequal offsets for ratios ≥ 4 on two or three axes (Remark 11.4).

The rounding error is measured, not bounded. Section 7 states the number: $\leq 16.4\varepsilon \max_{ab} |T_{ab}|$ over sixty random cases spanning aspect ratios 1 to 1020, with an l -sum amplification of at most 3.14×10^3 . Nothing in this document bounds it. Related, and also measurements rather than theorems: the upward recurrence for h_l is accurate to 3×10^{-15} for $l \leq 40$, $|z| \in [0.3, 300]$, $\text{Im } z \in [0, 30]$, but that is 72 measured rows and not a proof — which is why the shipped bound uses the *certified* majorant $\text{hb}(l, z)$ of Theorem 5.2 rather than $|h_l|$ itself, at a cost of 1.001–1.008 in per-offset work on cubes and 1.2 on the slender cell. Miller’s downward recurrence returns j_l to 6.8×10^{-15} ; the odd- l radial moments use $W_l \leq \sqrt{W_{l-1}W_{l+1}}$, exact in the even case and a genuine bound in the odd; and the monotonicity in $|\mathbf{R}|$ that the threshold table assumes was checked against a direct evaluation at 4000 radii per (shape, frequency), which is a check and not a proof.

14.2 Where the accuracy statement has to be qualified

The small part of a complex entry. For real f the tensor splits into two independent real sums, $\text{Re } T$ from y_l and $\text{Im } T$ from j_l , and each part keeps its own relative accuracy. For complex f it is one complex sum, and then a part carrying less than about 10^{-3} of the entry’s own modulus is accurate only to $\varepsilon \max_{ab} |T_{ab}|$ in absolute terms, i.e. to a *relative* error of about 10^{-16} divided by that fraction. This is visible in the Re/Im columns of every verification table at 10^{-13} to 9×10^{-12} , and it is a property of the formulation, not of the expansion: the independent Cartesian route has it too, and none of the seven families in the registry avoids it. It is not fixable in `Float64` without splitting the complex-frequency sum by magnitude. **Entries themselves — and, at real f , parts of entries — meet 10^{-14} ; parts of entries at complex f that carry less than 1 % of their entry do not.**

Aspect ratios were verified only where references exist. The library is generic in three independent rational edge lengths and so is the bound, but a reference is an expensive object. What exists: the four production shapes at five frequencies (427 tensors), the twelve audit shapes at aspect ratios 1 to 1020 at four frequencies (288 tensors, of which four rows are limited by the reference’s own 3.6×10^{-15} floor at an aspect-678 needle two cells apart), and sixty random cases. *No claim is made about a shape outside that set beyond what the shape-generic bound gives.*

Cells above $\lambda/4$ are refused or expensive, not wrong. Above $|k|r_d \approx 3.2$ the k -series inside j_l needs more terms than a $\lambda/4$ cell does; the library now computes how many and builds the table to match, and above $N_{\text{cap}} = 40$ ($|k|r_d \approx 22$) it raises an error rather than truncating. But N_{cap} is a policy, not a theorem: nothing says the expansion is the right tool there, and a $\lambda/2$ cube needs $N = 16$ and 171 s of table, a λ cube $N = 24$ and more. There is also a reversal worth knowing: on a $\lambda/2$ cell the two-cell offset does not converge on either expansion route and falls through to the k -series, which answers it to 5.6×10^{-17} — *on a coarse cell the near field is the accurate part and the far field is the expensive one*, the opposite of the usual expectation.

14.3 Cost and engineering

The memory of the table build is measured on four shapes. The in-place rewrite of Section 1.3 brought the cold build from 21–25 GB to 0.66–0.74 GB on the four production shapes and was checked bit-identical on 24 tables. The twelve audit shapes of aspect ratio up to 1020 were not rebuilt after the rewrite; their tables are the same code path at the same L and n_{max} , so nothing suggests a different outcome, but it is inferred and not measured.

The cost discontinuity at the k -series boundary. One offset of the slender needle costs 1.9 to 9.5 s and its lattice neighbour one lattice step further out costs 0.13 ms: a factor of 10^5 across one step. It is a property of the routing rule, not of the machine, and the route has no cost cap — it raises only when 200 series terms are not enough. A $\lambda/8$ plate at $N = 28$ still takes tens of seconds per offset. It is paid once per (shape, frequency) and cached, and it is 67 offsets of half a million, but a user meeting it cold on a new shape will wait two minutes on eight threads.

Precision above 192 bits buys nothing. The geometry table is stored at `TABPRC = 192`, so a `BigFloat(256)` result advertises 77 digits and carries 58. The type is generic; the accuracy is capped.

No GPU path. `farBlock!` is CPU-threaded and bitwise deterministic in the thread count. Nothing here was written for or tested on a GPU, and the machine used had no CUDA device.

Wiring into `GlaVacOprMem` is a later task. `farBlock!` fills a Gila-layout `egoToe` and `bench.jl` drives the real build stage by stage through `GilaElectromagnetics.GilaVacuum`, verifying that the staged “before” path is bit-identical to `GlaVacOprMem`’s own output and that the “after” path differs from it only by Gila’s own quadrature error. But nothing under `src/` was touched, by instruction. Replacing the `egoFunInn!` loop in `genEgoCrcSlf!` is the next task, and it needs a decision on where the shape tables live and who builds them.

14.4 Unequal cells

Section 11.8 lists what the cross-scale extension leaves open; the items that qualify a number in this document are these.

The certificates of unequal pairs are loose where `est` is far off, and 0.13 % of a realistic block is certified only to 1–4 tol. Across the thin axis of the slender pair $(1/32, 1/32, 1/512)$ against $(1/8, 1/8, 1/512)$ the tensor is 10^2 – 10^4 below the pointwise scale, so the route-3 certificate is 2600–8900 times the actual error there (1.5×10^{-14} to 8.1×10^{-13} of $\max |G|$ against errors of 10^{-17}). On the $\lambda/2$ block 1800 offsets at $\rho_{\text{whl}} = 0.55$ – 0.60 meet the bound at $L = 56$ only to 1–4 times $\text{tol} \cdot \max |G|$; where a reference exists they are correct to 3×10^{-15} , but that is what is proven.

The verification of unequal cells covers ratios 2, 4, 8, 16 on one to three axes against a $\lambda/32$ fine cell, two slender pairs, and three frequencies, and the adversarial round added ratios 3 and 6, a pair coarse on a different axis in each cell, coarse cells of $\lambda/2$ and λ , two complex frequencies and mixed-sign offsets (Section 11.6). What that round did not turn into a verified case: non-integer rational ratios, which the library refuses rather than computes — a three-line change makes all three routes work on them and was measured but not shipped; coarse cells above λ and ratios above 32; and `Float32` anywhere in the near band, which returns `NaN` rather than a poor answer (D14). Route 3 on unequal panels was verified through 103 off-lattice reference records and never through a block, since no block offset reaches it. The routing itself is no longer a sample: it is exhaustive over the $125 \cdot 256$ separated offsets of the four test blocks, and `farBlockX!` is bitwise deterministic in the thread count.

Two costs are engineering, not mathematics. The exact-rational offset vector costs as much to classify as the far field costs to compute (4.2 s against 4.4 s single-threaded on the $\lambda/2$ block, and about 1 GB of `BigInt`); an integer-lattice interface would remove it. The table writer still appends without a cross-process lock (D12 made the duplicate harmless, not absent). And, as for equal cells, nothing under `src/` was touched: `farBlockX!` replaces the `egoFunOut!` loop of `genEgoCrcExt!` only once someone wires it there.

The contact path is the remaining bottleneck, and it is now also the remaining accuracy problem. After this work a 128^3 build at $\lambda/32$ is 6.73 s on twelve threads of which 5.14 s is `wekTrp` at `intOrd` 48 (Section 10); and Section 9.5 shows that the same integrals are wrong by 117% on a slender touching shell. The moment series of the companion document computes them exactly. *Both problems have the same fix, and it is not in this deliverable.*

15 Acknowledgement of method

This document, the library it describes and every measurement in it were produced by Claude (Anthropic, Claude Fable 5.1) over two working days in September 2026, run as a portfolio of independent sub-agents coordinated by a root agent, in the same way as the two companion documents.

The first round launched eight sub-agents on the seven approach families of the task statement plus the groundwork, with no agent told which family looked promising and no agent reading another’s transcript — only its report. That independence was the point: three of the families converged on the same object from different directions (the Cartesian Taylor series, the derivative-free coefficient recurrence and the addition theorem are one expansion in three organizations), and the fact that they agreed to 10^{-14} per entry over 16 340 offsets, having been written without knowledge of each other, is a stronger statement than any one of them verifying itself. Two families were killed by measurement in that round — the spectral route by the non-decaying periodization it implies (768 GB at the frequency Gila uses), the interpolation route by an arithmetic cost above the direct method’s — and both are recorded in Section 12 with the number that killed them, so that they are not reopened by intuition.

The second round was a chain rather than a portfolio: a theory pass that wrote and checked every bound, a unification pass that built the library, a pass that sharpened the bound by a median factor of 3.2×10^3 , a preparation pass that integrated it, an *adversarial* audit whose only instruction was to break the library, and a merge pass that fixed what the audit found. The audit is the reason this document can distinguish what is certified from what is measured. It found ten defects, one of which (Section 13, D9) destroyed the answer completely outside the tested range while the library’s own bound reported success, and it found it by refusing to stay inside the range the task statement named. Two conclusions this work had already written down twice — that the slender operator’s indefiniteness is a property of the discretization, and that the $\lambda/128$ cube’s error was rounding — were both wrong, and both were overturned by later passes with a measurement.

Paul Virally ran the independent Mathematica cross-check of the volume formulation himself (Section 3.6): an `NIntegrate` of $(1/V_t) \int_D w [(\partial_a \partial_b + \delta_{ab} k^2)g](\mathbf{R} + \boldsymbol{\delta}) d\boldsymbol{\delta}$ at $\mathbf{n} = (5, 1, 0)$, $\mathbf{s} = (1/32)^3$, $f = 1$, at working precision 30–40 with the box split at the kink of the weight, in his own installation, sharing no code with anything here. The off-diagonal entry agreed with the 220-bit face-pair reference to 3.4×10^{-14} , which is Mathematica’s own error estimate; the diagonal entry, recomputed with a fixed 30-point Gauss–Kronrod rule per sub-box after the global adaptive strategy stalled, reproduced `0.001570738221266370299274 + 0.000360155079207178606379i` in every printed digit.

That is the third of the three routes that pin the sign and the normalization, and it is the only one with no author in common with the other two.

The machine was an Apple M3 Pro, 12 cores, 36 GB, Julia 1.12.6, with no CUDA device. A hard limit of two concurrent Julia processes per agent was imposed after ten agents running seventy-seven processes crashed the machine and lost two hours of work; the load average is recorded next to every timing in the sub-agent reports, and every absolute timing quoted in Section 10 was re-measured on a quiet machine at the end, because the round-1 timings were taken with four to ten other processes running and are only good for ordering methods against each other.